

15. **SAE International.** SAE International is a global body of scientists, engineers, and practitioners that advances self-propelled vehicle and system knowledge in a neutral forum for the benefit of society. The mission involves standards-setting activities mostly related to aerospace, automotive, and commercial vehicles.
16. **Hytorc Company.** Manufacturer of hydraulic and pneumatic automatic torque tensioners, mechanical tensioners, and *Stretch-to-Load* technology.
17. **Loctite Adhesives.** Provider of adhesives for industry, construction, commercial, and household use, including epoxies, cyanoacrylates, contact cements, construction adhesives, and the Loctite threadlocker. Part of Henkel Corporation that offers several dozen additional adhesive brands serving a wide range of uses. Part of Henkel Corporation.
18. **International Organization for Standardization (ISO).** An independent, non-governmental international organization that brings together experts to develop voluntary, consensus-based International Standards that support innovation and provide solutions to global challenges. Based in Geneva, Switzerland.
7. What would be the nearest standard metric screw thread size to the American Standard 7/8–14 thread? By how much do their major diameters differ?
8. A machine screw is found with no information given as to its size. The following data are found by using a standard micrometer caliper: The major diameter is 0.196 in; the axial length for 20 full threads is 0.630 in. Identify the thread.
9. A threaded fastener is made from nylon 6/6 with an $M10 \times 1.5$ thread. Compute the maximum tensile force that can be permitted in the fastener if it is to be stressed to 75% of the tensile strength of the nylon 66 dry. See Appendix 13.
10. Compare the tensile force that can be carried by a 1/4–20 screw if it is to be stressed to 50% of its tensile strength and if it is made from each of the following materials:
 - (a) Steel, SAE Grade 2
 - (b) Steel, SAE Grade 5
 - (c) Steel, SAE Grade 8
 - (d) Steel, ASTM Grade A307
 - (e) Steel, ASTM Grade A574
 - (f) Steel, metric Grade 8.8
 - (g) Aluminum 2024-T4
 - (h) SAE 430 annealed
 - (i) Ti–6Al–4V annealed
 - (j) Nylon 66 dry
 - (k) Polycarbonate
 - (l) High-impact ABS

PROBLEMS

1. Describe the difference between a screw and a bolt.
2. Define the term *proof strength*.
3. Define the term *clamping load*.
4. Specify suitable machine screws to be installed in a pattern of four, equally spaced around a flange, if the clamping force between the flange and the mating structure is to be 6000 lb. Then recommend a suitable tightening torque for each screw.
5. What would be the tensile force in a machine screw having an 8–32 thread if it is made from SAE Grade 5 steel and is stressed to its proof strength?
6. What would be the tensile proof force in newtons (N) in a machine screw having a major diameter of 4 mm with standard fine threads if it is made from steel having a metric strength grade of 8.6?
11. Describe the differences among welding, brazing, and soldering.
12. What types of metals are typically brazed?
13. What are some common brazing alloys?
14. What materials make up commonly used solders?
15. Name five common adhesives, and give the typical properties of each.
16. The label of a common household adhesive describes it as a *cyanoacrylate*. What would you expect its properties to be?
17. Find three commercially available adhesives from your home, a laboratory, a machine shop, or your workplace. Try to identify the generic nature of the adhesive, and compare it with the list presented in this chapter.

MACHINE FRAMES, BOLTED CONNECTIONS, AND WELDED JOINTS

The Big Picture

You Are the Designer

20–1 Objectives of This Chapter

20–2 Machine Frames and Structures

20–3 Eccentrically Loaded Bolted Joints

20–4 Welded Joints

THE BIG PICTURE

Machine Frames, Bolted Connections, and Welded Joints

Discussion Map

- As you develop the design of machine elements (as you have throughout this book), you must also design the housing, frame, or structure that supports them and protects them from the elements.
- You need knowledge of both fabrication methods and design approaches that provide safe, rigid frames and structures that can be reliably manufactured.

Discover

Select a variety of products, machines, vehicles, and even toys. Observe how they are built. What is the basic shape of the structure that holds everything together? Why was that shape chosen by the designer? What functions are performed by the frame?

What kinds of forces, bending moments, and torsional moments (torques) are produced when the product is operating? How are they managed and controlled? What is the load path that delivers them to the ultimate structure?

This chapter will help you identify some efficient approaches to the design of structures and frames and to analyze the performance of fasteners and welded joints loaded in many different ways.

Thus far in this book, you have studied individual machine elements while simultaneously considering how those elements must work together in a more comprehensive machine. As the design progresses, there comes a time when *you must put it all together*. But then you are faced with the questions, “What do I put it in? How do I hold all of the functional components safely, allowing assembly and service while providing a secure, rigid structure?”

It is impractical to generate a completely general approach to the design of a structure or a frame for a machine, a vehicle, a consumer product, or even a

toy. Each is different with regard to its functions; the number, size, and type of components in the product; the intended use; and the expected demand for an aesthetic design. For example, toys often exhibit clever design approaches because the manufacturer wants to provide a safe, functional toy while minimizing the material used and the amount of personnel time required to produce the toy.

In this chapter you will explore some basic concepts for creating a satisfactory frame design, considering the shape of structural components, material properties, the use of fasteners such as bolts, and

the fabrication of welded assemblies. You will learn some of the techniques for analyzing and designing bolted assemblies to consider loads on the bolts in several directions. The design of welded joints to be safe and rigid is discussed.

Chapter 19 covered the part of the story dealing with bolts loaded in pure tension, as in a clamping function. This chapter extends that chapter to consider eccentrically loaded joints: those that must resist a combination of direct shear and a bending moment on a bolt pattern.

The ability of a welded joint to carry a variety of loads is discussed with the objective of designing the weld. Here both uniformly loaded and eccentrically loaded joints are treated.

To appreciate the value of such study, select a variety of products, machines, and vehicles, and observe how they are built. What is the basic shape of the structure that holds everything together? Where are forces, bending moments, and torsional moments (torques) generated? What kinds of stress do they create? Consider the *load path* by following a force from the place where it is generated through all of the means by which that force or its effects are passed to

a series of members and to the point where it is supported by a basic frame of the machine or where it is delivered out of the system of interest. Considering the load paths for all forces that exist in a mechanical device should give you an understanding of the desirable characteristics of the structure and should help you know how to produce a design that optimizes the management of the forces exerted on it.

Take apart a variety of mechanical devices to observe their structure. How did shape contribute to the rigidity and safety of the device? Is it rather stiff? Or is it more flexible? Are there ribs or thickened sections to provide special strength or rigidity for certain parts?

This chapter will help you identify some efficient approaches to the design of structures and frames and to analyze the performance of fasteners and welded joints loaded in many ways.

The subject of machine frames and structures is quite complex. It is discussed from the standpoint of general principles and guidelines, rather than specific design techniques. Critical frames are typically designed with computerized finite-element analysis. Also, experimental stress analysis techniques are often used to verify designs.

YOU ARE THE DESIGNER

In Chapter 16, you were the designer of bearings for conveyor systems for a large product distribution center. Now, how do you design the frame and the structure of the conveyor system? What general form is desirable? What materials and shape should be used for the structural elements? Are the elements loaded in tension, compression, bending, shear, torsion, or some combination of these types of stress? How do the manner of loading and the nature of the stress in the structure affect these decisions? Should the frame be fabricated from standard structural shapes and bolted together? Or should it be made from steel plate and welded? How about using aluminum? Or should it be cast from cast iron or cast steel? Can it be molded from plastic? Can composites be used? How would the weight of the structure be affected? How much rigidity is desirable for this kind of structure? What shapes of load-carrying elements

contribute to a stiff, rigid structure, as well as one that is safe in resisting applied stresses?

If the structure involves bolting or welding, how are the joints to be designed? What forces, both magnitude and direction, must be carried by the fasteners or the welds?

The material in this chapter will help you make some of these design decisions. Much of the information is general in nature rather than giving you specific design procedures. You must exercise judgment and creativity not only in the design of the conveyor frame but also in the analysis of its components. Because the frame design could evolve into a form too complex for analysis using traditional techniques of stress analysis, you may have to employ finite-element modeling to determine whether the design is adequate or, perhaps, overdesigned. Perhaps one or more prototypes should be built for testing. ■

20-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Apply the principles of stress and deflection analysis to propose a reasonable and efficient shape for a structure or frame and for the components involved.
2. Specify materials that are well suited to the demands of a given design, given certain conditions of load, environment, fabrication requirements, safety, and esthetics.

3. Analyze eccentrically loaded bolted joints.
4. Design welded joints to carry many types of loading patterns.

20-2 MACHINE FRAMES AND STRUCTURES

The design of machine frames and structures is largely art in that the components of the machine must be accommodated. The designer is often restricted in where supports can be placed in order not to interfere with the

operation of the machine or in order to provide access for assembly or service.

But, of course, technical requirements must be met as well for the structure itself. Some of the more important design parameters include the following:

Strength	Stiffness
Appearance	Cost to manufacture
Corrosion resistance	Weight
Size	Noise reduction
Vibration limitation	Life

Because of the virtually infinite possibilities for design details for frames and structures, this section will concentrate on general guidelines. The implementation of the guidelines would depend on the specific application. Factors to consider in starting a design project for a frame are now summarized:

- Forces exerted by the components of the machine through mounting points such as bearings, pivots, brackets, and feet of other machine elements
- Manner of support of the frame itself
- Precision of the system: allowable deflection of components
- Environment in which the unit will operate
- Quantity of production and facilities available
- Availability of analytical tools such as computerized stress analysis, past experience with similar products, and experimental stress analysis
- Relationship to other machines, walls, and so on

Again, many of these factors require judgment by the designer. The parameters over which the designer has the most control are material selection, the geometry of load-carrying parts of the frame, and manufacturing processes. A review of some possibilities follows.

Materials

As with machine elements discussed throughout this book, the material properties of strength and stiffness are of prime importance. Chapter 2 presented an extensive amount of information about materials, and the Appendices contain much useful information. In general, steel ranks high in strength compared with competing materials for frames. But it is often better to consider more than just yield strength, ultimate tensile strength, or endurance strength alone. The complete design can be executed in several candidate materials to evaluate the overall performance. Considering the *ratio of strength to density*, sometimes referred to as the *strength-to-weight ratio* or *specific strength*, may lead to a different material selection. Indeed, this is one reason for the use of aluminum, titanium, and composite materials in aircraft, aerospace vehicles, and transportation equipment.

Rigidity of a structure or a frame is frequently the determining factor in the design, rather than strength.

In these cases, the stiffness of the material, indicated by its modulus of elasticity, is the most important factor. Again, the *ratio of stiffness to density*, called *specific stiffness*, may need to be evaluated. See Table 2-17 and Figures 2-23, 2-24, 2-31, and 2-32 for data.

Recommended Deflection Limits

Actually, only intimate knowledge of the application of a machine member or a frame can give a value for an acceptable deflection. But some guidelines are available to give you a place to start. (See Reference 7.)

Deflection due to Bending

General machine part: 0.0005 to 0.003 in/in of beam length

Moderate precision: 0.000 01 to 0.0005 in/in

High precision: 0.000 001 to 0.000 01 in/in

Deflection (Rotation) due to Torsion

General machine part: 0.001° to 0.01°/in of length

Moderate precision: 0.000 02° to 0.0004°/in

High precision: 0.000 001° to 0.000 02°/in

Suggestions for Design to Resist Bending

Scrutiny of a table of deflection formulas for beams in bending such as those in Appendix 14 would yield the following form for the deflection:

$$\Delta = \frac{PL^3}{KEI} \quad (20-1)$$

where P = load

L = length between supports

E = modulus of elasticity of the material in the beam

I = moment of inertia of the cross section of the beam

K = a factor depending on the manner of loading and support

Some obvious conclusions from Equation (20-1) are that the load and the length should be kept small, and the values of E and I should be large. Note the cubic function of the length. This means, for example, that reducing the length by a factor of 2.0 would reduce the deflection by a factor of 8.0, obviously a desirable effect.

Figure 20-1 shows the comparison of four types of beam systems to carry a load, P , at a distance, a , from a rigid support. A cantilever beam is taken as the “basic case”, with a load of 1200 lb, 30 in from its fixed support. Using standard beam formulas, we computed the value of the bending moment and the deflection in terms of P and a , and these values were arbitrarily normalized to be 1.0 for the cantilever. Then the values for the three other cases were computed and ratios determined relative to the basic case. The data show that a fixed-end beam gives both the lowest bending moment and the lowest deflection, while the cantilever gives the highest values for both.

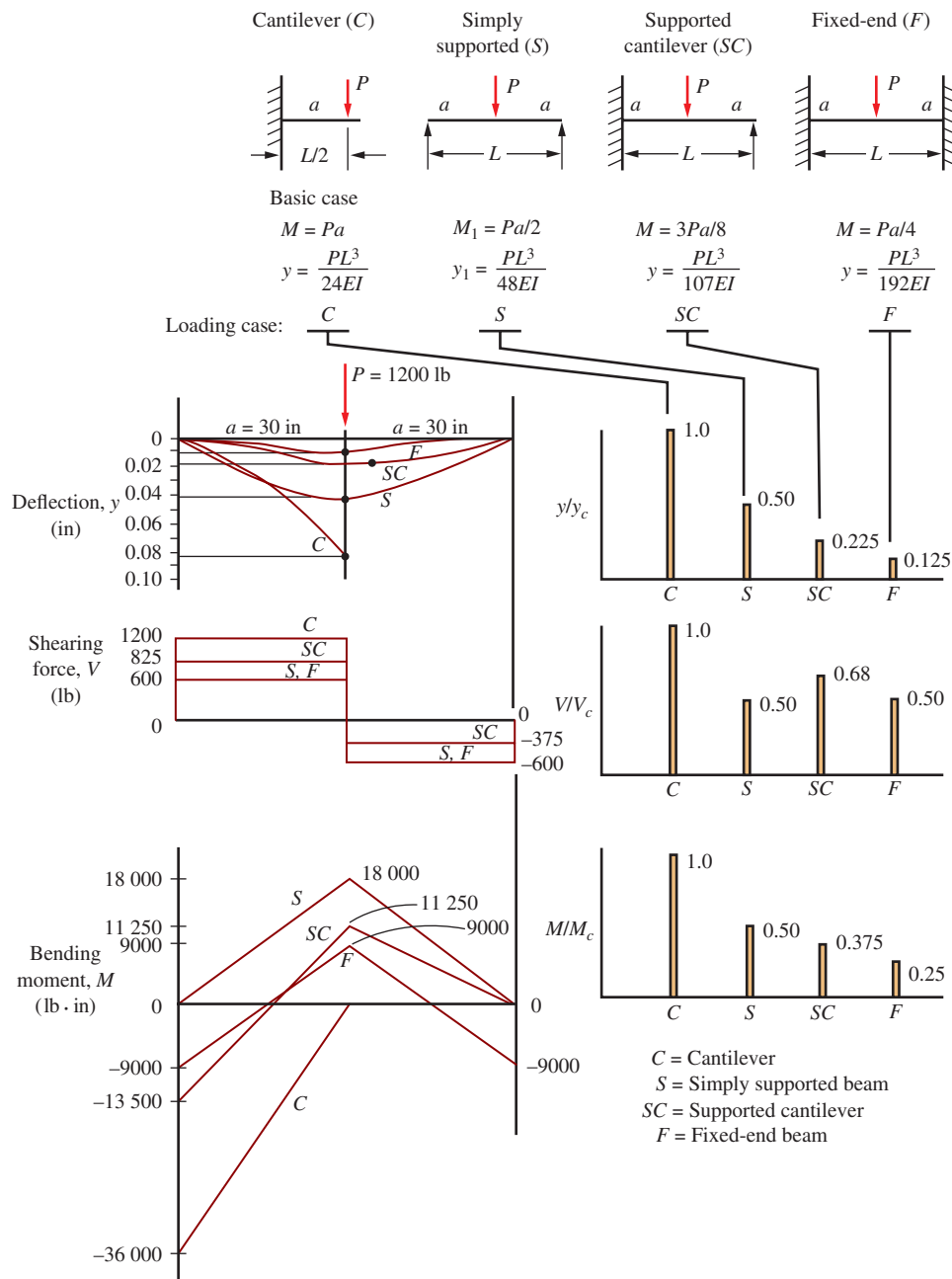


FIGURE 20-1 Comparison of methods of supporting a load on a beam (Reference 17)

In summary, the following suggestions are made for designing to resist bending:

1. Keep the length of the beam as short as possible, and place loads close to the supports.
2. Maximize the moment of inertia of the cross section in the direction of bending. In general, you can do so by placing as much of the material as far away from the neutral axis of bending as possible, as in a wide-flange beam or a hollow rectangular section.
3. Use a material with a high modulus of elasticity.
4. Use fixed ends for the beam where possible.
5. Consider lateral deflection in addition to deflection in the primary load direction. Such loads could be encountered during fabrication, handling, shipping, careless use, or casual bumping.
6. Be sure to evaluate the final design with regard to both strength and rigidity. Some approaches to improving rigidity (increasing I) can actually increase the stress in the beam because the section modulus is decreased.
7. Provide rigid corner bracing in open frames.
8. Cover an open frame section with a sheet material to resist distortion. This process is sometimes called *panel stiffening*.

9. Consider a truss-type construction to obtain structural stiffness with lightweight members.
10. When designing an open space frame, use diagonal bracing to break sections into triangular parts, an inherently rigid shape.
11. Consider stiffeners for large panels to reduce vibration and noise. See Figure 2–29.
12. Add bracing and gussets to areas where loads are applied or at supports to help transfer the forces into adjoining members.
13. Beware of load-carrying members with thin, extended flanges that may be placed in compression. Local buckling, sometimes called *crippling* or *wrinkling*, could occur.
14. Place connections at points of low stress if possible.

See also References 1, 4, 6, 7, and 10 for additional design and analysis techniques.

Suggestions for Design of Members to Resist Torsion

Torsion can be created in a machine frame member in a variety of ways: A support surface may be uneven; a machine or a motor may transmit a reaction torque to the frame; or a load acting to the side of the axis of the beam (or any place away from the flexural center of the beam) would produce twisting.

In general, the torsional deflection of a member is computed from

$$\theta = \frac{TL}{GR} \quad (20-2)$$

where T = applied torque or twisting moment

L = length over which torque acts

G = shear modulus of elasticity of the material

R = torsional rigidity constant

The designer must choose the shape of the torsion member carefully to obtain a rigid structure. The following suggestions are made:

1. Use closed sections wherever possible. Examples are solid bars with large cross section, hollow pipe and tubing, closed or hollow rectangular or square tubing, and special closed shapes that approximate a tube.
2. Conversely, avoid open sections made from thin materials. Figure 20–2 shows a dramatic illustration.
3. For wide frames, brackets, tables, bases, and so on, use diagonal braces placed at 45° to the sides of the frame (see Figure 20–3).
4. Use rigid connections, such as by welding members together.

Most of the suggestions made in this section can be implemented regardless of the specific type of frame

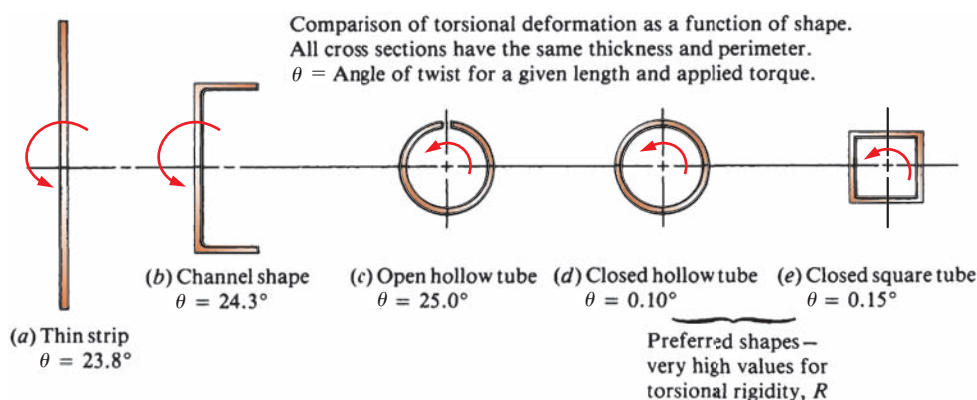


FIGURE 20–2 Comparison of torsional deformation as a function of shape

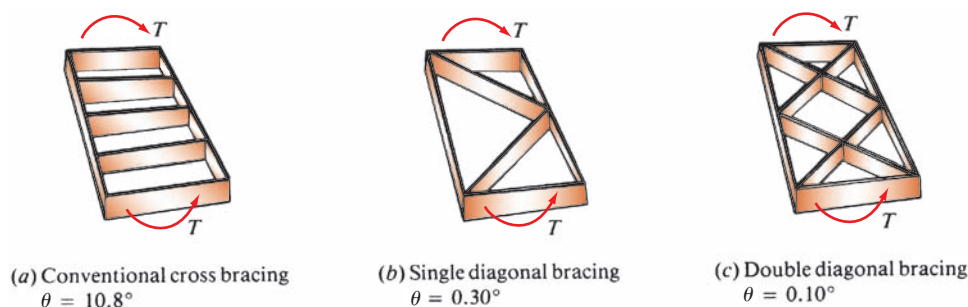


FIGURE 20–3 Comparison of torsional angle of twist, θ , for boxlike frames. Each has the same basic dimensions and applied torque

designed: castings made from cast iron, cast steel, aluminum, zinc, or magnesium; weldments made from steel or aluminum plate; formed housings from sheet metal or plate; or plastic moldings. The references in this chapter provide valuable additional guidance as you complete the design of frames, structures, and housings. See References 1, 4, 6, 7, 10–12, and 18. See also Internet sites 1–12.

20-3 ECCENTRICALLY LOADED BOLTED JOINTS

Figure 20-4 shows an example of an eccentrically loaded bolted joint. The motor on the extended bracket places the bolts in shear because its weight acts directly downward. But there also exists a moment equal to $P \times a$ that must be resisted. The moment tends to rotate the bracket and thus to shear the bolts.

The basic approach to the analysis and design of eccentrically loaded joints is to determine the forces that act on each bolt caused by all the applied loads. Then, by a process of superposition, the loads are combined vectorially to determine which bolt carries the greatest load. That bolt is then sized. The method will be illustrated in Example Problem 20-1.

The American Institute of Steel Construction (AISC) lists allowable stresses for bolts made from ASTM grade steels, as shown in Table 20-1. These data are for bolts used in standard-sized holes, 1/16 in larger than the bolt. Also, a *friction-type connection*, in which the clamping force is sufficiently large that the friction between the mating parts helps carry some of the shear load, is assumed. (See References 1–3, 14, 19, 20, and 22.)

In the design of bolted joints, you should ensure that there are no threads in the plane where shear

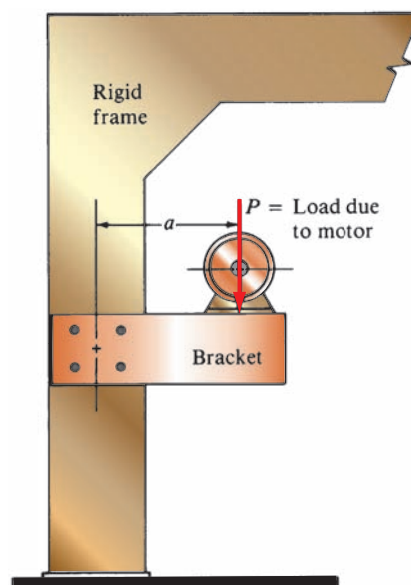


FIGURE 20-4 Eccentrically loaded bolted joint

TABLE 20-1 Allowable Stresses for Bolts

ASTM grade	Allowable shear stress	Allowable tensile stress
A307	10 ksi (69 MPa)	20 ksi (138 MPa)
A325 and A449	17.5 ksi (121 MPa)	44 ksi (303 MPa)
A490	22 ksi (152 MPa)	54 ksi (372 MPa)

occurs. The body of the bolt will then have a diameter equal to the major diameter of the thread. You can use the tables in Chapter 19 to select the standard size for a bolt.

Example Problem 20-1

For the bracket in Figure 20-4, assume that the total force P is 3500 lb and the distance a is 12 in. Design the bolted joint, including the location and number of bolts, the material, and the diameter.

Solution

The solution shown is an outline of a procedure that can be used to analyze similar joints. The data of this problem illustrate the procedure.

Step 1. Propose the number of bolts and the pattern. This is a design decision, based on your judgment and the geometry of the connected parts. In this problem, let's try a pattern of four bolts placed as shown in Figure 20-5.

Step 2. Determine the direct shear force on the bolt pattern and on each individual bolt, assuming that all bolts share the shear load equally:

$$\text{Shear load} = P = 3500 \text{ lb}$$

$$\text{Load per bolt} = F_s = P/4 = 3500 \text{ lb}/4 = 875 \text{ lb/bolt}$$

The shear force acts directly downward on each bolt.

Step 3. Compute the **moment** to be resisted by the bolt pattern: the product of the overhanging load and the distance to the **centroid** of the bolt pattern. In this problem, $M = P \times a = (3500 \text{ lb})(12 \text{ in}) = 42\,000 \text{ lb} \cdot \text{in}$.

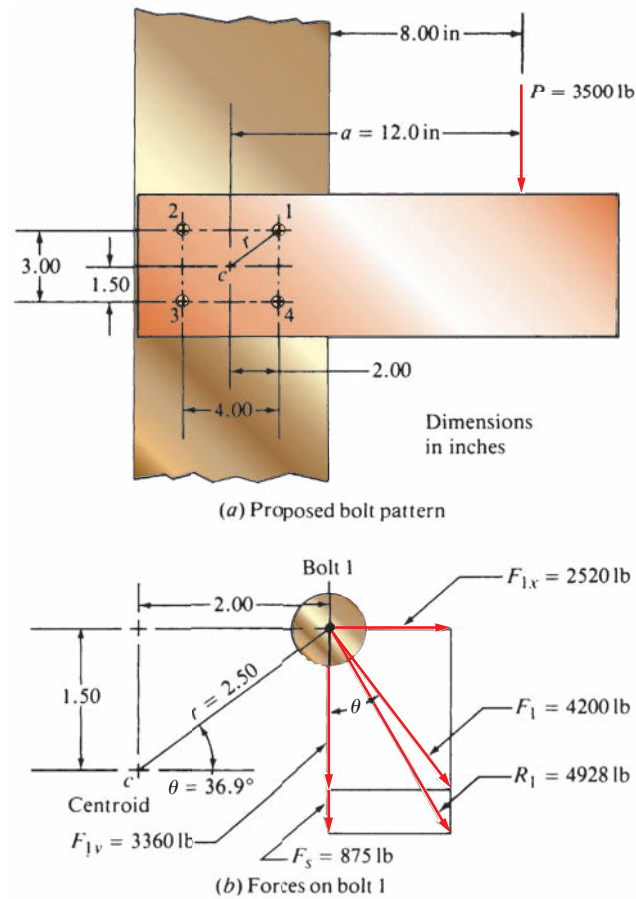


FIGURE 20-5 Geometry of bolted joint and forces on bolt 1

Step 4. Compute the radial distance from the centroid of the bolt pattern to the center of each bolt. In this problem, each bolt has a radial distance of

$$r = \sqrt{(1.50 \text{ in})^2 + (2.00 \text{ in})^2} = 2.50 \text{ in}$$

Step 5. Compute the sum of the *squares* of all radial distances to all bolts. In this problem, all four bolts have the same r . Then

$$\Sigma r^2 = 4(2.50 \text{ in})^2 = 25.0 \text{ in}^2$$

Step 6. Compute the force on each bolt required to resist the bending moment from the relation

$$F_i = \frac{Mr_i}{\Sigma r^2} \quad (20-3)$$

where r_i = radial distance from the centroid of the bolt pattern to the i th bolt

F_i = force on the i th bolt due to the moment. The force acts perpendicular to the radius.

In this problem, all such forces are equal. For example, for bolt 1,

$$F_1 = \frac{Mr_1}{\Sigma r^2} = \frac{(42\,000 \text{ lb} \cdot \text{in})(2.50 \text{ in})}{25.0 \text{ in}^2} = 4200 \text{ lb}$$

Step 7. Determine the resultant of all forces acting on each bolt. A vector summation can be performed either analytically or graphically, or each force can be resolved into horizontal and vertical components. The components can be summed and then the resultant can be computed.

Let's use the latter approach for this problem. The shear force acts directly downward, in the y -direction. The x - and y -components of F_1 are

$$F_{1x} = F_1 \sin \theta = (4200 \text{ lb}) \sin(36.9^\circ) = 2520 \text{ lb}$$

$$F_{1y} = F_1 \cos \theta = (4200 \text{ lb}) \cos(36.9^\circ) = 3360 \text{ lb}$$

The total force in the y -direction is then

$$F_{1y} + F_s = 3360 + 875 = 4235 \text{ lb}$$

Then the resultant force on bolt 1 is

$$R_1 = \sqrt{(2520)^2 + (4235)^2} = 4928 \text{ lb}$$

Step 8. Specify the bolt material; compute the required area for the bolt; and select an appropriate size. For this problem, let's specify ASTM A325 steel for the bolts having an allowable shear stress of 17 500 psi from Table 20–1. Then the required area for the bolt is

$$A_s = \frac{R_1}{\tau_a} = \frac{4928 \text{ lb}}{17\,500 \text{ lb/in}^2} = 0.282 \text{ in}^2$$

The required diameter would be

$$D = \sqrt{\frac{4A_s}{\pi}} = \sqrt{\frac{4(0.282 \text{ in}^2)}{\pi}} = 0.599 \text{ in}$$

Let's specify a 5/8-in bolt having a diameter of 0.625 in.

20-4 WELDED JOINTS

The design of welded joints requires consideration of the manner of loading on the joint, the types of materials in the weld and in the members to be joined, and the geometry of the joint itself. The load may be either uniformly distributed over the weld such that all parts of the weld

are stressed to the same level or eccentrically applied. Both are discussed in this section.

The materials of the weld and the parent members determine the allowable stresses. Table 20–2 lists several examples for steel and aluminum. The allowables listed are for shear on fillet welds. For steel, welded by the

TABLE 20-2 Allowable Shear Stresses on Fillet Welds for Steel and Aluminum

A. Steel

Electrode type	Typical metals joined (ASTM grade)	Allowable shear stress
E60	A36, A500	18 ksi (124 MPa)
E70	A242, A441	21 ksi (145 MPa)
E80	A572, Grade 65	24 ksi (165 MPa)
E90		27 ksi (186 MPa)
E100		30 ksi (207 MPa)
E110		33 ksi (228 MPa)

B. Aluminum

Metal joined	Filler Alloy							
	1100		4043		5356		5556	
	Allowable shear stress							
	ksi	MPa	ksi	MPa	ksi	MPa	ksi	MPa
1100	3.2	22	4.8	33				
3003	3.2	22	5.0	34				
6061			5.0	34	7.0	48	8.5	59
6063			5.0	34	6.5	45	6.5	45

electric arc method, the type of electrode is an indication of the tensile strength of the filler metal. For example, the E70 electrode has a minimum tensile strength of 70 ksi (483 MPa). Additional data are available in publications of the American Welding Society (AWS), the American Institute for Steel Construction (AISC), and the Aluminum Association (AA). See Reference 1 and Internet sites 3, 6–8, and 10.

Types of Joints

Joint type refers to the relationship between mating parts, as illustrated in Figure 20–6. The butt weld allows a joint to be the same nominal thickness as the mating parts and is usually loaded in tension. If the joint is properly made with the appropriate weld metal, the joint will be stronger than the parent metal. Thus, no special analysis of the joint is required if the joined members themselves are shown to be safe. Caution is

advised, however, when the materials to be joined are adversely affected by the heat of the welding process. Heat-treated steels and many aluminum alloys are examples. The other types of joints in Figure 20–6 are assumed to place the weld in shear.

Types of Welds

Figure 20–7 shows several types of welds named for the geometry of the edges of the parts to be joined. Note the special edge preparation required, especially for thick plates, to permit the welding rod to enter the joint and build a continuous weld bead.

Size of Weld

The five types of groove-type welds in Figure 20–7 are made as complete penetration welds. Then, as indicated

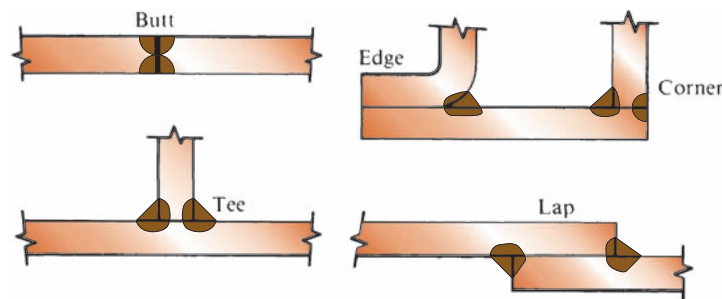


FIGURE 20–6 Types of weld joints

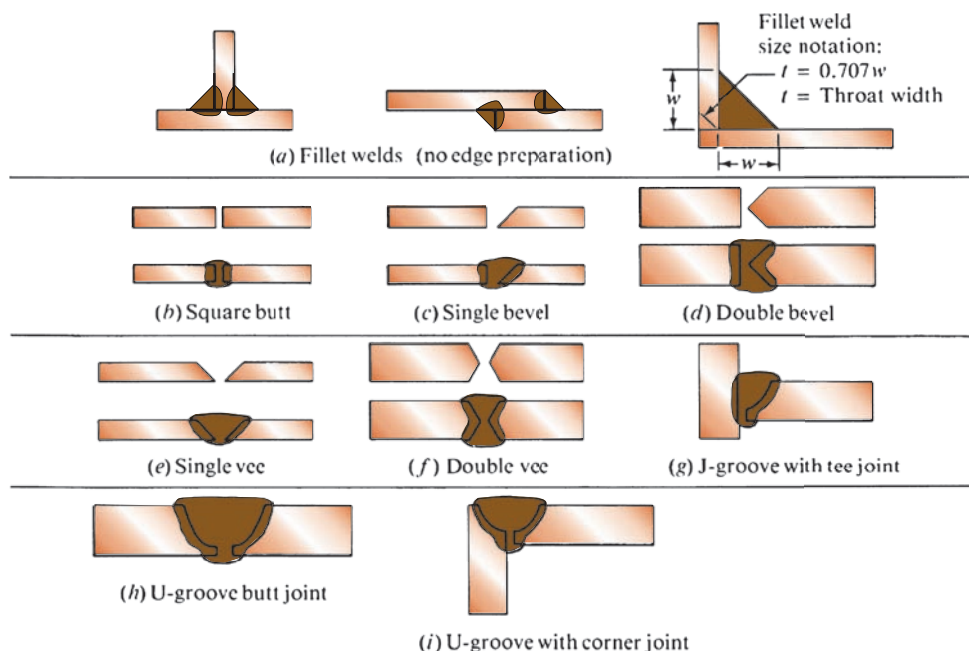


FIGURE 20–7 Some types of welds showing edge preparation

before for butt welds, the weld is stronger than the parent metals, and no further analysis is required.

Fillet welds are typically made as equal-leg right triangles, with the size of the weld indicated by the length of the leg. A fillet weld loaded in shear would tend to fail along the shortest dimension of the weld that is the line from the root of the weld to the theoretical face of the weld and normal to the face. The length of this line is found from simple trigonometry to be $0.707w$, where w is the leg dimension. See Part (a) of Figure 20–7.

The objectives of the design of a fillet-welded joint are to specify the length of the legs of the fillet; the pattern of the weld; and the length of the weld. Presented here is the method that treats the weld as a line having no thickness. The method involves determining the *maximum force per inch* of weld leg length. Comparing the actual force with an allowable force allows the calculation of the required leg length.

Table 20–3 gives data for the allowable shear stress and the allowable force per inch for some combinations of base metal and welding electrode. In general, the allowables for building-type structures are for steady loads. The values for bridge-type loading accounts for the cyclic effects. For true fatigue-type repeated loading, refer to the literature. (See References 1, 6–10, 13, 15, 16, 21, and 23.)

Method of Treating a Weld as a Line

Four different types of loading are considered here: (1) direct tension or compression, (2) direct vertical shear, (3) bending, and (4) twisting. The method allows the designer to perform calculations in a manner very similar to that used to design the load-carrying members themselves. In general, the weld is analyzed separately for each type of loading to determine the force per inch of weld size due to each load. The loads are then combined vectorially to determine the maximum force. This maximum force is compared with the allowables from Table 20–3 to determine the size of the weld required. See References 6 and 7.

The relationships used are summarized as follows:

Type of loading	Formula (and equation number) for force per inch of weld	
Direct tension or compression	$f = P/A_w$	(20–4)
Direct vertical shear	$f = V/A_w$	(20–5)
Bending	$f = M/S_w$	(20–6)
Twisting	$f = Tc/J_w$	(20–7)

In these formulas, the geometry of the weld is used to evaluate the terms A_w , S_w , and J_w , using the relationships shown in Figure 20–8. Note the similarity between these formulas and those used to perform the stress analysis. Note, also, the similarity between the geometry factors for welds and the properties of areas used for the stress analysis. Because the weld is treated as a line having no thickness, the units for the geometry factors are different from those of the area properties, as indicated in Figure 20–8.

The use of this method of weld analysis will be demonstrated with example problems. In general, the method requires the steps in the *General Procedure for Designing Welded Joints*.

GENERAL PROCEDURE FOR DESIGNING WELDED JOINTS ▼

1. Propose the geometry of the joint and the design of the members to be joined.
2. Identify the types of stresses to which the joint is subjected (bending, twisting, vertical shear, direct tension, or compression).
3. Analyze the joint to determine the magnitude and the direction of the force on the weld due to each type of load.
4. Combine the forces vectorially at the point or points of the weld where the forces appear to be maximum.
5. Divide the maximum force on the weld by the allowable force from Table 20–3 to determine the required leg size for the weld. Note that when thick plates are welded, there are minimum acceptable sizes for the welds as listed in Table 20–4.

TABLE 20–3 Allowable Shear Stresses and Forces on Welds

Base metal ASTM grade	Electrode	Allowable shear stress	Allowable force per inch of leg
Building-type structures:			
A36, A441	E60	13 600 psi	9600 lb/in
A36, A441	E70	15 800 psi	11 200 lb/in
Bridge-type structures:			
A36	E60	12 400 psi	8800 lb/in
A441, A242	E70	14 700 psi	10 400 lb/in

TABLE 20–4 Minimum Weld Sizes for Thick Plates

Plate thickness (in)	Minimum leg size for fillet weld (in)
$\leq 1/2$	3/16
$> 1/2$ – $3/4$	1/4
$> 3/4$ – $1\frac{1}{2}$	5/16
$> 1\frac{1}{2}$ – $2\frac{1}{4}$	3/8
$> 2\frac{1}{4}$ –6	1/2
> 6	5/8

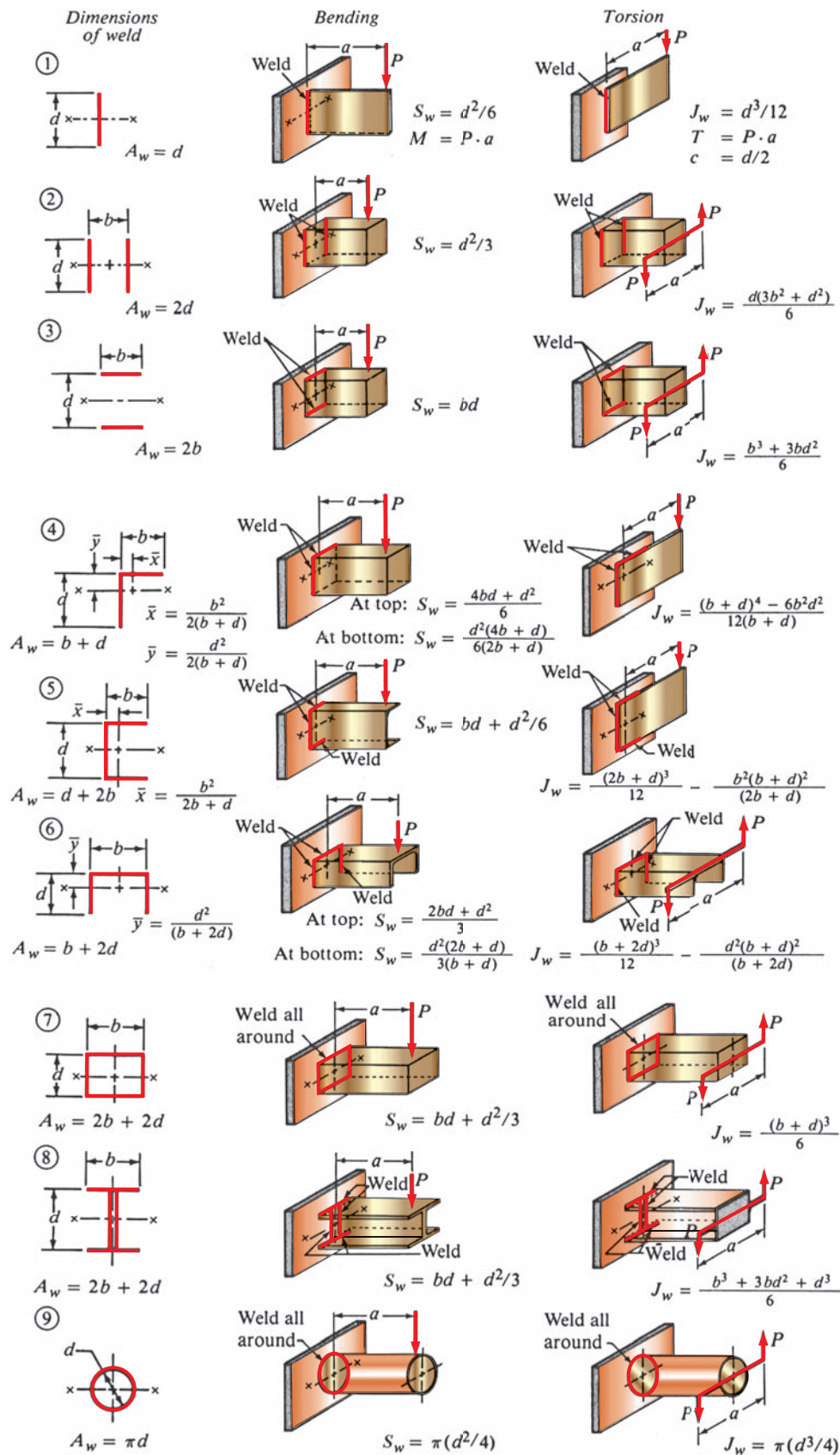


FIGURE 20-8 Geometry factors for weld analysis

**Example Problem
20-2**

Design a bracket similar to that in Figure 20-4, but use welding to attach the bracket to the column. The bracket is 6.00 in high and is made from ASTM A36 steel having a thickness of 1/2 in. The column is also made from A36 steel and is 8.00 in wide.

Solution

Step 1. The proposed geometry is a design decision and may have to be subjected to some iteration to achieve an optimum design. For a first trial, let's use the C-shaped weld pattern shown in Figure 20-9.

Step 2. The weld will be subjected to direct vertical shear and torsion caused by the 3500-lb load on the bracket.

Step 3. To compute the forces on the weld, we must know the geometry factors A_w and J_w . Also, the location of the centroid of the weld pattern must be computed [see Figure 20-9(b)]. Use Case 5 in Figure 20-8.

$$A_w = 2b + d = 2(4) + 6 = 14 \text{ in}$$

$$J_w = \frac{(2b + d)^3}{12} - \frac{b^2(b + d)^2}{(2b + d)} = \frac{(14)^3}{12} - \frac{16(10)^2}{14} = 114.4 \text{ in}^3$$

$$\bar{x} = \frac{b^2}{2b + d} = \frac{16}{14} = 1.14 \text{ in}$$

Force due to Vertical Shear

$$V = P = 3500 \text{ lb}$$

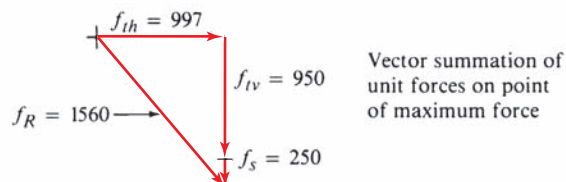
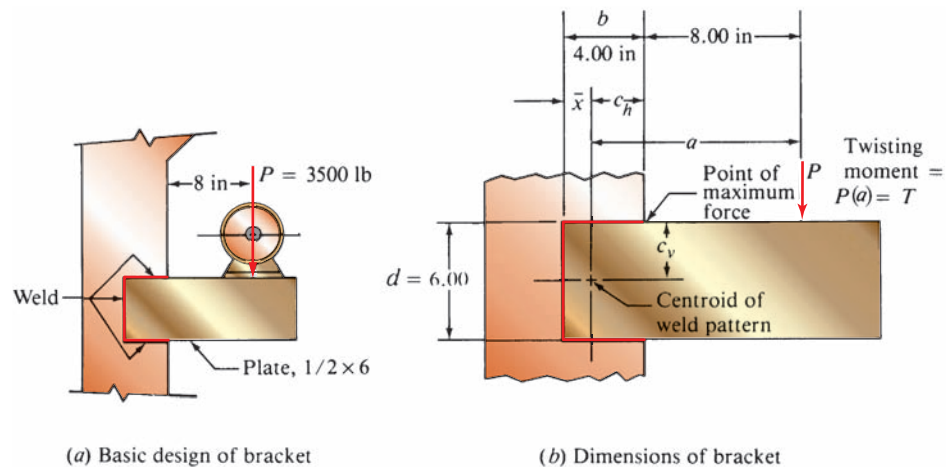
$$f_s = P/A_w = (3500 \text{ lb})/14 \text{ in} = 250 \text{ lb/in}$$

This force acts vertically downward on all parts of the weld.

Forces due to the Twisting Moment

$$T = P[8.00 + (b - \bar{x})] = 3500[8.00 + (4.00 - 1.14)]$$

$$T = 3500(10.86) = 38\,010 \text{ lb} \cdot \text{in}$$



(c) Analysis of forces

FIGURE 20-9 C-shaped weld bracket

The twisting moment causes a force to be exerted on the weld that is perpendicular to a radial line from the centroid of the weld pattern to the point of interest. In this case, the end of the weld to the upper right experiences the greatest force. It is most convenient to break the force down into horizontal and vertical components and then subsequently recombine all such components to compute the resultant force:

$$f_{th} = \frac{T_{C_V}}{J_w} = \frac{(38\,010)(3.00)}{114.4} = 997 \text{ lb/in}$$

$$f_{tv} = \frac{T_{C_h}}{J_w} = \frac{(38\,010)(2.86)}{114.4} = 950 \text{ lb/in}$$

Step 4. The vectorial combination of the forces on the weld is shown in Figure 20–9(c). Thus, the maximum force is 1560 lb/in.

Step 5. Selecting an E60 electrode for the welding, we find that the allowable force per inch of weld leg size is 9600 lb/in (Table 20–3). Then the required weld leg size is

$$w = \frac{1560 \text{ lb/in}}{9600 \text{ lb/in per in of leg}} = 0.163 \text{ in}$$

Table 20–4 shows that the minimum size weld for a 1/2-in plate is 3/16 in (0.188 in). That size should be specified.

Example Problem 20–3

A steel strap, 1/4 in thick, is to be welded to a rigid frame to carry a dead load of 12 500 lb, as shown in Figure 20–10. Design the strap and its weld.

Solution

The basic objectives of the design are to specify a suitable material for the strap, the welding electrode, the size of the weld, and the dimensions W and h , as shown in Figure 20–10. Note that the welds are only on the vertical sides of the strap.

Let's specify that the strap is to be made from ASTM A441 structural steel and that it is to be welded with an E70 electrode, using the minimum size weld, 3/16 in. Appendix 7 gives the yield strength of the A441 steel as 42 000 psi. Using a design factor of 2, we can compute an allowable stress of

$$\sigma_a = 42\,000/2 = 21\,000 \text{ psi}$$

Then the required area of the strap is

$$A = \frac{P}{\sigma_a} = \frac{12\,500 \text{ lb}}{21\,000 \text{ lb/in}^2} = 0.595 \text{ in}^2$$

But the area is $W \times t$, where $t = 0.25$ in. Then the required width W is

$$W = A/t = 0.595 \text{ in}^2/0.25 \text{ in} = 2.38 \text{ in}$$

Let's specify that $W = 2.50$ in

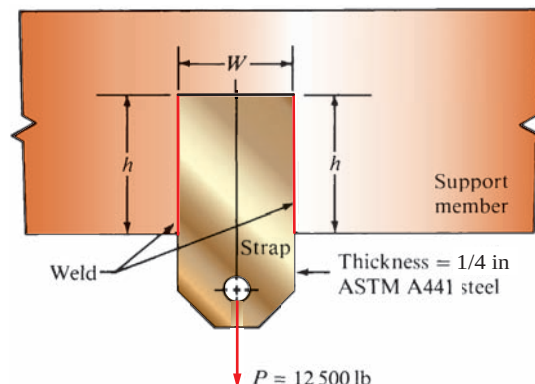


FIGURE 20–10 Steel strap

To compute the required length of the weld h , we need the allowable force on the 3/16-in weld. Table 20-3 indicates the allowable force on the A441 steel welded with an E70 electrode to be 11 200 lb/in per inch of leg size. Then

$$f_a = \frac{11\,200 \text{ lb/in}}{1.0\text{-in leg}} \times 0.188\text{-in leg} = 2100 \text{ lb/in}$$

The actual force on the weld is

$$f_a = P/A_w = P/2h$$

Then solving for h gives

$$h = \frac{P}{2(f_a)} = \frac{12\,500 \text{ lb}}{2(2100 \text{ lb/in})} = 2.98 \text{ in}$$

Let's specify $h = 3.00 \text{ in}$.

Example Problem 20-4

Evaluate the design shown in Figure 20-11 with regard to stress in the welds. All parts of the assembly are made of ASTM A36 structural steel and are welded with an E60 electrode. The 2500-lb load is a dead load.

Solution

The critical point would be the weld at the top of the tube where it is joined to the vertical surface. At this point, there is a three-dimensional force system acting on the weld, as shown in Figure 20-12. The offset location of the load causes a twisting on the weld that produces a force f_t on the weld toward the

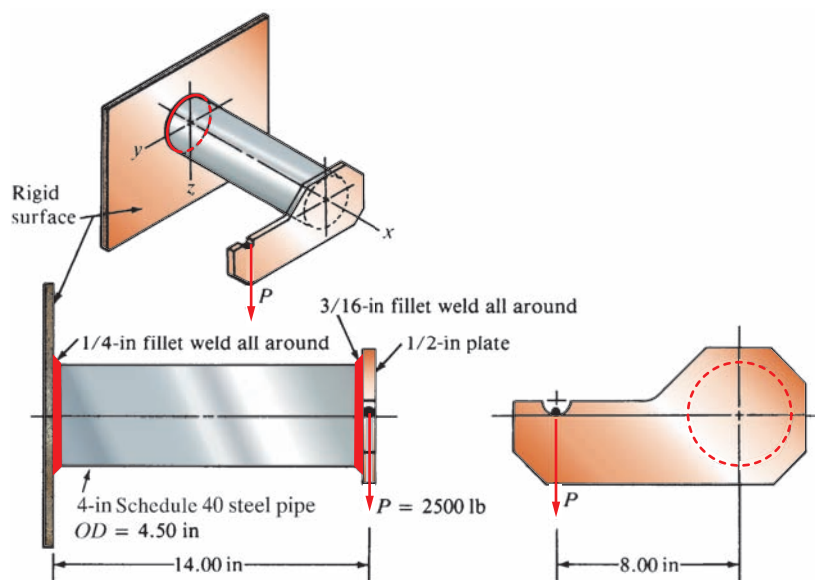


FIGURE 20-11 Bracket assembly

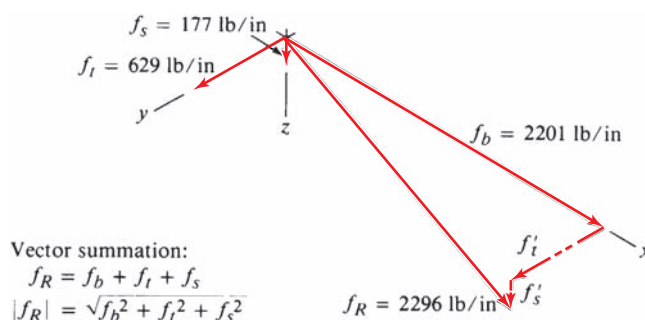


FIGURE 20-12 Force vectors

left in the y -direction. The bending produces a force f_b acting outward along the x -axis. The vertical shear force f_s acts downward along the z -axis.

From statics, the resultant of the three force components would be

$$f_R = \sqrt{f_t^2 + f_b^2 + f_s^2}$$

Now each component force on the weld will be computed.

Twisting Force, f_t

$$f_t = \frac{Tc}{J_w}$$

$$T = (2500 \text{ lb})(8.00 \text{ in}) = 20\,000 \text{ lb} \cdot \text{in}$$

$$c = OD/2 = 4.500 \text{ in}/2 = 2.25 \text{ in}$$

$$J_w = (\pi)(OD)^3/4 = (\pi)(4.500 \text{ in})^3/4 = 71.57 \text{ in}^3$$

Then

$$f_t = \frac{Tc}{J_w} = \frac{(20\,000 \text{ lb} \cdot \text{in})(2.25 \text{ in})}{71.57 \text{ in}^3} = 629 \text{ lb/in}$$

Bending Force, f_b

$$f_b = \frac{M}{S_w}$$

$$M = (2500 \text{ lb})(14.00 \text{ in}) = 35\,000 \text{ lb} \cdot \text{in}$$

$$S_w = (\pi)(OD)^2/4 = (\pi)(4.500 \text{ in})^2/4 = 15.90 \text{ in}^2$$

Then

$$f_b = \frac{M}{S_w} = \frac{35\,000 \text{ lb} \cdot \text{in}}{15.90 \text{ in}^2} = 2201 \text{ lb/in}$$

Vertical Shear Force, f_s

$$f_s = \frac{P}{A_w}$$

$$A_w = (\pi)(OD) = (\pi)(4.500 \text{ in}) = 14.14 \text{ in}$$

$$f_s = \frac{P}{A_w} = \frac{2500 \text{ lb}}{14.14 \text{ in}} = 177 \text{ lb/in}$$

Now the resultant can be computed: (Note: All terms have the unit of lb/in.)

$$f_R = \sqrt{f_t^2 + f_b^2 + f_s^2}$$

$$f_R = \sqrt{629^2 + 2201^2 + 177^2} = 2296 \text{ lb/in}$$

Comparing this with the allowable force on a 1.0-in weld gives minimum size for the weld leg to be

$$w = \frac{2296 \text{ lb/in}}{9600 \text{ lb/in per inch of leg size}} = 0.239 \text{ in}$$

The 1/4-in fillet specified in Figure 20–11 is satisfactory.

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3. **American Welding Society.** A professional society that develops standards for the welding industry, including AWS D1.1 Structural Welding Code-Steel (2015), AWS D1.2 Structural Welding Code-Aluminum (2008), and many others. Site offers ways to access standards and its American Welding Online learning system. Also provides links to SENSE (School Excelling through National Skill Standards Education) and guidelines for welder training.
4. **James F. Lincoln Foundation.** An organization that promotes education and training in welding technology. The site includes technical papers and general information about welding processes, joint design, and guides to welded steel construction. The site includes sections called *Learn It*, *Teach It*, and *Build It* that offer books and digital media.
5. **Miller Electric Company.** A manufacturer of a wide variety of welding equipment and accessories for the professional welding community and occasional welders. Site includes a Resources section containing welding guidelines, and welding videos.
6. **Lincoln Electric Company.** A manufacturer of a wide variety of welding equipment and accessories for the welding industry. The site includes an Education section that provides education support materials and technical articles about welding technology.
7. **Hobart Brothers Company.** A manufacturer of numerous types of filler metal products, electrodes, and steel wire for the welding industry. Site offers a Support tab containing downloadable posters and charts on welding.
8. **Hobart Institute of Welding Technology.** An educational organization that provides instruction in the performance of welding techniques. The site includes a bookstore offering numerous books and DVDs for welding training.
9. **Metal Casting Design.** An online resource for metal casting buyers and design engineers, sponsored by the American Foundry Society. The site includes a metal casting process selector, casting tutorials, and training for buyers and designers. The casting tutorials follow a casting project from design through manufacturing issues and terminology involved in the selection, design, purchase, production, and use of metal castings. See also Internet site 2.
10. **Aluminum Association.** The Association promotes the use of aluminum and represents U.S.- and foreign-based primary producers of aluminum, aluminum recyclers, and producers of fabricated products, as well as industry suppliers. Publishes *Aluminum Standards and Data* in both U.S. and metric units, *Aluminum Design Manual*, and various books on welding aluminum.
11. **North American Die Casting Association.** NADCA promotes industry awareness and the use of and recognition of die castings.

INTERNET SITES FOR MACHINE FRAMES, BOLTED CONNECTIONS, AND WELDED JOINTS

Also refer to the Internet sites from Chapter 19 on Fasteners that includes many related to bolted connections.

1. **Steel Founders' Society of America.** An association of companies providing foundry services. Site includes useful casting information, a glossary of foundry terms, and a framework for designing.

12. **Flow-3D Cast.** Software modeling a wide range of casting processes to guide users with accurate predictions of filling and solidification processes.

PROBLEMS

For Problems 1–6, design a bolted joint to join the two members shown in the appropriate figure. Specify the number of bolts, the pattern, the bolt grade, and the bolt size.

1. Figure P20–1

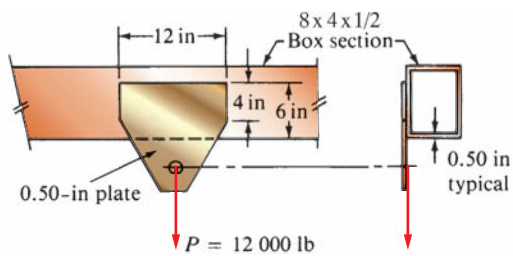


FIGURE P20–1 (Problems 1, 7, and 13)

2. Figure P20–2

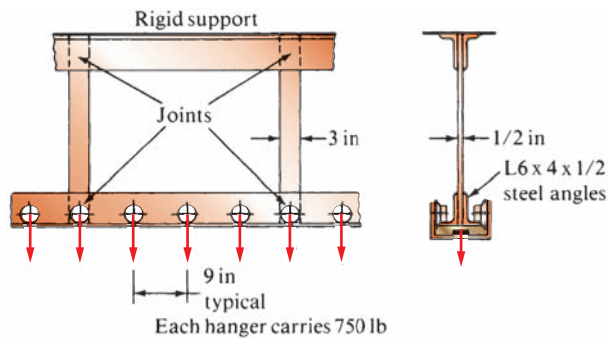


FIGURE P20–2

3. Figure P20–3

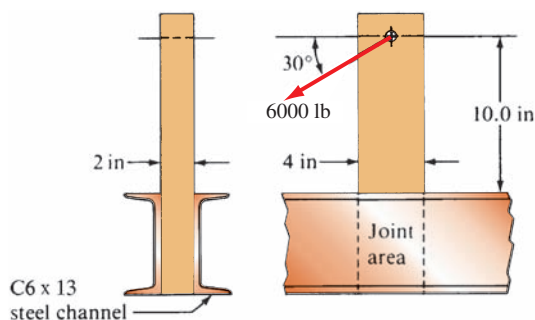


FIGURE P20–3

4. Figure P20–4

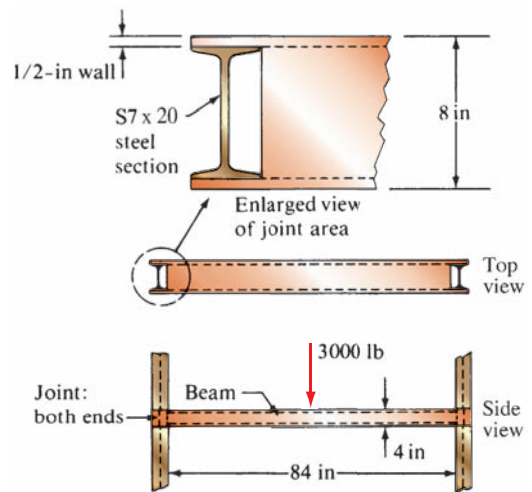


FIGURE P20–4 (Problems 4 and 8)

5. Figure P20–5

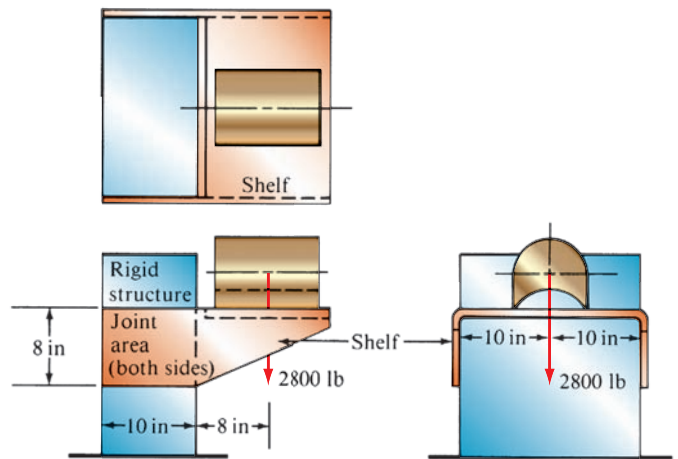
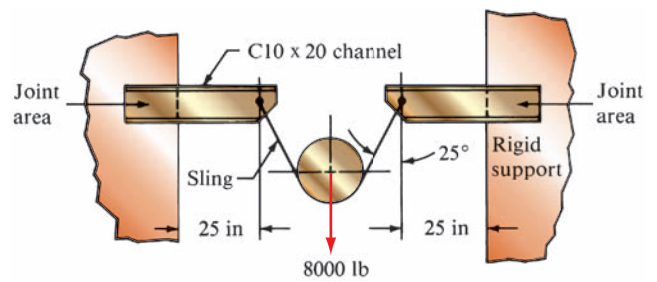


FIGURE P20–5 (Problems 5 and 9)

6. Figure P20–6



Load shared equally by four brackets (only two shown)

FIGURE P20–6 (Problems 6 and 10)

For Problems 7–12, design a welded joint to join the two members shown in the appropriate figure. Specify the weld pattern, the type of electrode to be used, and the size of weld. In Problems 7–9, the members are ASTM A36 steel. In Problems 10–12, the members are ASTM A441 steel. Use the method of treating the joint as a line, and use the allowable forces per inch of leg for building-type structures from Table 20–3.

7. Figure P20–1
8. Figure P20–4
9. Figure P20–5
10. Figure P20–6
11. Figure P20–11

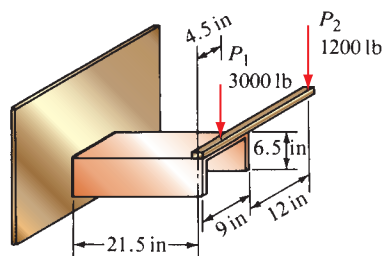


FIGURE P20–11 (Problems 11 and 12)

12. Figure P20–11 (but $P_2 = 0$)

For Problems 13–16, design a welded joint to join the two aluminum members shown in the appropriate figure. Specify the weld pattern, the type of filler alloy, and the size of weld. The types of materials joined are listed in the problems.

13. Figure P20–1: 6061 alloy (but $P = 4000$ lb)
14. Figure P20–14: 6061 alloy

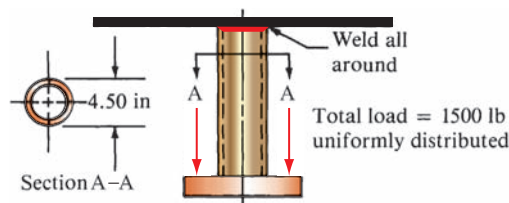


FIGURE P20–14

15. Figure P20–15: 6063 alloy

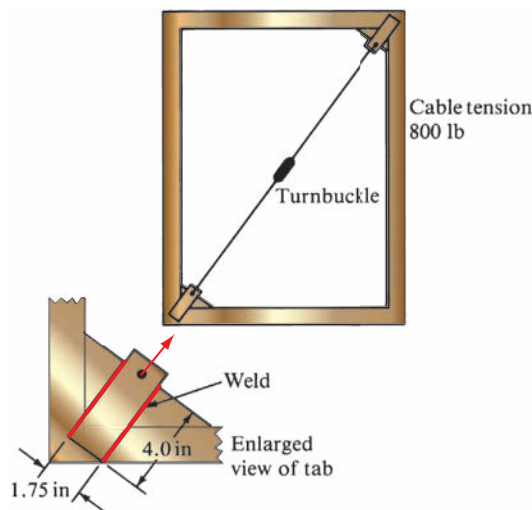


FIGURE P20–15

16. Figure P20–16: 3003 alloy

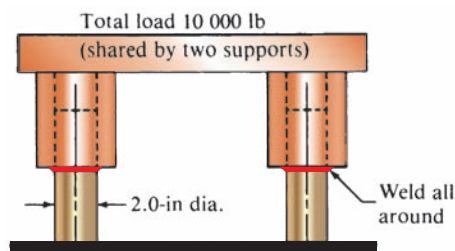


FIGURE P20–16

17. Compare the weight of a tensile rod carrying a dead load of 4800 lb if it is made from (a) SAE 1020 HR steel; (b) SAE 5160 OQT 1300 steel; (c) aluminum 2014-T6; (d) aluminum 7075-T6; (e) titanium 6A1-4V, annealed; and (f) titanium 3A1-13V-11Cr, aged. Use $N = 2$ based on yield strength.

ELECTRIC MOTORS AND CONTROLS

The Big Picture

You Are the Designer

- 21-1 Objectives of This Chapter
- 21-2 Motor Selection Factors
- 21-3 AC Power and General Information about AC Motors
- 21-4 Principles of Operation of AC Induction Motors
- 21-5 AC Motor Performance
- 21-6 Three-Phase, Squirrel-Cage Induction Motors
- 21-7 Single-Phase Motors
- 21-8 AC Motor Frame Types and Enclosures
- 21-9 Controls for AC Motors
- 21-10 DC Power
- 21-11 DC Motors
- 21-12 DC Motor Control
- 21-13 Other Types of Motors

THE BIG PICTURE

Electric Motors and Controls

Discussion Map

- Electric motors provide the power for a huge array of products in homes, factories, schools, commercial facilities, transportation equipment, and many portable devices.
- The two major classifications of motors are *alternating current (AC)* and *direct current (DC)*. Some can operate on either type of power.
- Motors must be connected into complete electrical systems that provide control for the motor for start/stop, speed control, overload protection, and other critical functions.

Discover

Look for a variety of machines and products that are powered by electric motors. Choose both large and small devices; some portable and some that plug into standard electrical outlets. Look around your home, at work, and in a factory.

Try to find the nameplate for each motor, and copy down as much information as you can. How are the nameplate data related to the performance characteristics of the motor? Is it an AC motor or a DC motor? How fast does it run? What is its electrical rating in terms of voltage and current?

Some motors that will operate well in the United States will run differently or will not run at all in other countries. Why? What are the electric voltage and frequency standards in various parts of the world?

This chapter will help you identify many kinds of motors, understand the general performance characteristics of each, and apply them properly.

The electric motor is widely used for providing the prime power to industrial machinery, consumer products, and business equipment. This chapter will describe the various types of motors and will discuss their performance characteristics. The objective is to provide you with the background needed to specify motors and to communicate with vendors to acquire the proper motor for a given application.

Types of motors discussed in this chapter are direct current (DC), alternating current (AC, both single-phase and three-phase), universal motors, stepper motors, and variable-speed AC motors. Motor controls, batteries, and other means of providing DC power are also discussed.

While the appearance of different types of motors and the design of component parts vary significantly,

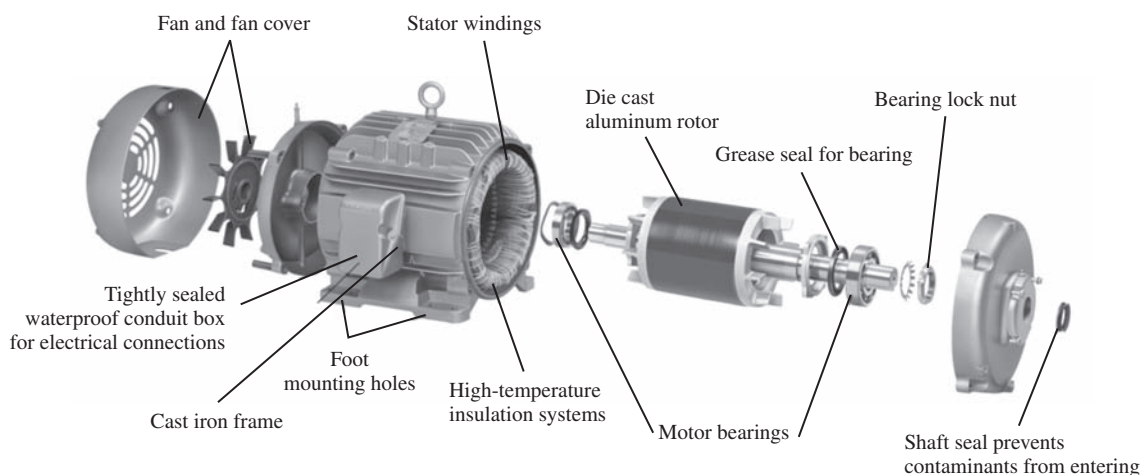


FIGURE 21-1 Exploded view of components for an AC electric induction motor (Baldor Electric Company, Greenville, SC)

it should be helpful at this time to study Figure 21-1 that shows a typical industrial-sized AC electric motor. The exploded view shows the primary components and how they are positioned relative to each other. The delivery of output power from the motor is through the *shaft* that typically runs from the rear bearing through the front of the *frame* extending out to enable connection to the driven machine or to mount a power-transmission element such as a gear, belt sheave, or chain sprocket. The shaft is supported by two precision *bearings*, typically rolling contact type, that accurately locate the axis concentric with that of the *housing*. The *rotor* of the motor is fixed to the shaft between the two bearings. The rotor reacts electromagnetically with the *windings in the stator* that is fitted inside the stationary housing. This particular motor style is called a *totally enclosed, fan cooled AC electric induction motor, TEFC* for short. Note the *fan* on the left side that rotates with the motor shaft and draws air over the housing and out the rear cover to remove heat generated by the induction process.

Now you should look for a variety of machines that are powered by electric motors: some large and some small; some portable, some that plug into standard electrical outlets, or others that are hardwired directly to an electrical circuit. Look around your home, in your car, at work, in gas stations, in shops or movie theaters, or in a factory if possible.

Try to find the nameplate for each motor and copy down as much information as you can. What does each piece of information mean? Are some electrical wiring diagrams included? How are the nameplate data related to the performance characteristics of the motor? Is it an AC motor or a DC motor? How fast does it run? What is its electrical rating in terms of voltage and current? For the motors in portable

devices, what kind of power supply is used? What kinds of batteries are required? How many? How are they connected? In series? In parallel? How do those factors relate to the voltage rating of the motor? What is the relationship between the kind of work done by the motor and the time between charging sessions?

Compare some of the larger motors with the photographs in Section 21-8 of this chapter. Can you identify the frame type? What company manufactured the motor?

See if you can find more information about the motors or their manufacturers from the Internet. Look up some of the company names in this chapter to learn more about the lines of motors that they offer. Do a broader search to find as many different types as you can.

What is the rated power for each motor? What units are used for the power rating? Convert all of the power ratings to watts. (See Appendix 17.) Then convert them all to horsepower to gain an appreciation for the physical size of various motors and to visualize the comparison between the SI unit of watts and the U.S. Customary unit of horsepower.

Separate the motors you find into those that are AC and those that are DC. Do you see any significant differences between the two classes of motors? Within each class, what variations do you see? Do any of the nameplates describe the type of motor, such as *synchronous*, *universal*, *split-phase*, *NEMA design C*, or other designations?

What kinds of controls are connected to the motors? Switches? Starters? Speed controls? Protection devices?

This chapter will help you identify many kinds of motors, understand the general performance characteristics of each, and apply them properly.

YOU ARE THE DESIGNER

Consider a conveyor system that you are to design. One possibility for driving the conveyor system is to use an electric motor. What type should be used? At what speed will it operate? What type of electric power is available to supply the motor? What power is required? What

type of housing and mounting style should be specified? What are the dimensions of the motor? How is the motor connected to the drive pulley of the conveyor system? The information in this chapter will help you to answer these and other questions. ■

21-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Describe the factors that must be specified to select a suitable motor.
2. Describe the principles of operation of AC motors.
3. Identify the typical classifications of AC electric motors according to rated power.
4. Identify the common voltages and frequencies of AC power and the speed of operation of AC motors operating on those systems.
5. Describe single-phase and three-phase AC power.
6. Describe the typical frame designs, sizes, and enclosure styles of AC motors.
7. Describe the general form of a motor performance curve.
8. Describe the comparative performance of *shaded-pole*, *permanent-split capacitor*, *split-phase*, and *capacitor-start*, single-phase AC motors.
9. Describe three-phase, squirrel-cage AC motors.
10. Describe the comparative performance of *NEMA design B*, *NEMA design C*, *NEMA design D*, and *wound-rotor* three-phase AC motors.
11. Describe *synchronous motors*.
12. Describe *universal motors*.
13. Describe three means of producing DC power and the common voltages produced.
14. Describe the advantages and disadvantages of DC motors compared with AC motors.
15. Describe four basic designs of DC motors—*shunt-wound*, *series-wound*, *compound-wound*, and *permanent magnet*—and describe their performance curves.
16. Describe torque motors, servomotors, stepper motors, brushless DC motors, and printed circuit motors.
17. Describe motor control systems for system protection, speed control, starting, stopping, and overload protection.
18. Describe speed control of AC motors.
19. Describe the control of DC motors.

21-2 MOTOR SELECTION FACTORS

As a minimum, the following items must be specified for motors:

- Motor type: DC, AC, single-phase, three-phase, and so on
- Power rating and speed
- Operating voltage and frequency
- Type of enclosure
- Frame size
- Mounting details

In addition, there may be many special requirements that must be communicated to the vendor. The primary factors to be considered in selecting a motor include the following:

- Operating torque, operating speed, and power rating. Note that these are related by the equation

$$\text{Power} = \text{torque} \times \text{speed}$$

- Starting torque.
- Load variations expected and corresponding speed variations that can be tolerated.
- Current limitations during the running and starting phases of operation.
- Duty cycle: how frequently the motor is to be started and stopped.
- Environmental factors: temperature, presence of corrosive or explosive atmospheres, exposure to weather or to liquids, availability of cooling air, and so on.
- Voltage variations expected: Most motors will tolerate up to $\pm 10\%$ variation from the rated voltage. Beyond this, special designs are required.
- Shaft loading, particularly side loads and thrust loads that can affect the life of shaft bearings.

Motor Size

A rough classification of motors by size is used to group motors of similar design. Horsepower (hp) is currently used most frequently, with the metric unit of watts or kilowatts also used at times. The conversion is

$$1.0 \text{ hp} = 0.746 \text{ kW} = 746 \text{ W}$$

The classifications are as follows:

- **Subfractional horsepower:** 1 to 40 millihorsepower (mhp), where 1 mhp = 0.001 hp. Thus, this range includes 0.001 to 0.040 hp (0.75–30 W, approximately).
- **Fractional horsepower:** 1/20 to 1.0 hp (37 to 746 W, approximately).
- **Integral horsepower:** 1.0 hp (0.75 kW) and larger.

References 1–9 provide additional information about the selection and application of electric motors. See also Internet sites 1–4, 14, and 15.

21-3 AC POWER AND GENERAL INFORMATION ABOUT AC MOTORS

Alternating current (AC) power is produced by the electric utility and delivered to the industrial, commercial, or residential consumer in a variety of forms. In the United States, AC power has a frequency of 60 hertz (Hz) or 60 cycles/s. In many other countries, 50 Hz is used. Some aircraft use 400-Hz power from an onboard generator.

AC power is also classified as single phase or three phase. Most residential units and light commercial installations have only single-phase power, carried by two conductors plus ground. The waveform of the power would appear like that in Figure 21-2, a single continuous sine wave at the system frequency whose amplitude is the rated voltage of the power. Three-phase power is carried on a three-wire system and is composed of three distinct waves of the same amplitude and frequency, with each phase offset from the next by 120°, as illustrated in Figure 21-3. Industrial and large commercial installations use three-phase power for the larger electrical loads because smaller motors are possible and there are economies of operation.

AC Voltages

Some of the more popular voltage ratings available in AC power are listed in Table 21-1. Given are the nominal system voltage and the typical motor voltage rating for that system in both single-phase and three-phase. In most cases, the highest voltage available should be used because the current flow for a given power is smaller. This allows smaller conductors to be used.

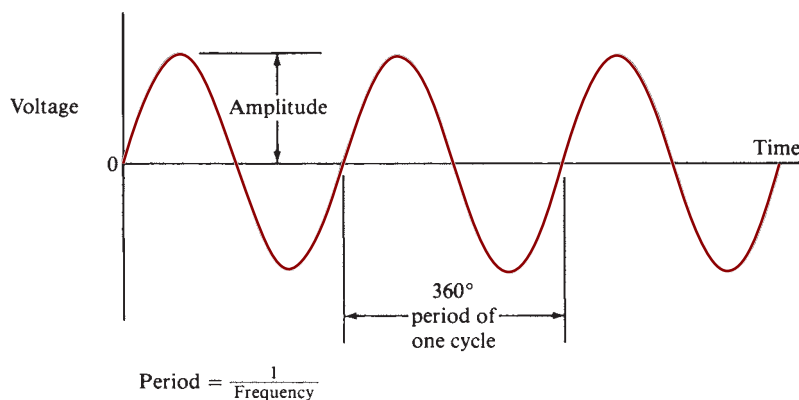


FIGURE 21-2 Single-phase AC power

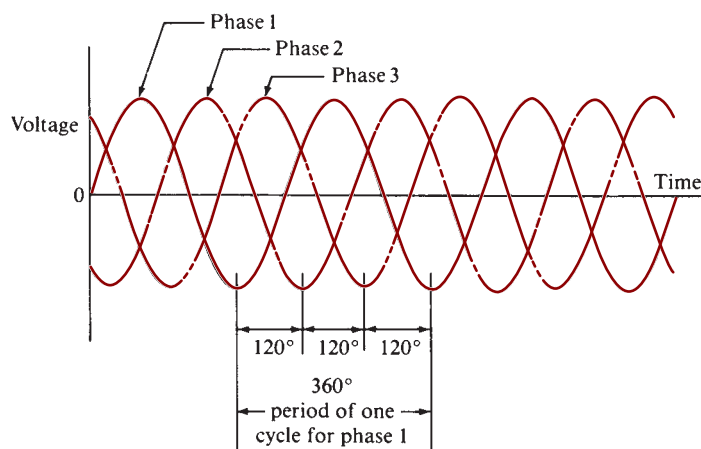


FIGURE 21-3 Three-phase AC power

TABLE 21-1 AC Motor Voltages

System voltage	Motor voltage ratings	
	Single-phase	Three-phase
120	115	115
120/208	115	200
240	230	230
480		460
600		575

TABLE 21-2 AC Motor Speeds for 60-Hz Power

Number of poles	Synchronous speed (rpm)	Full-load ^a speed (rpm)
2	3600	3450
4	1800	1725
6	1200	1140
8	900	850
10	720	690
12	600	575

^a Approximately 95% of synchronous speed (normal slip).

Speeds of AC Motors

An AC motor at zero load would tend to operate at or near its *synchronous speed*, n_s , which is related to the frequency, f , of the AC power and to the number of electrical poles, p , wound into the motor, according to the equation

Synchronous Speed

$$n_s = \frac{120f}{p} \text{ rev/min} \quad (21-1)$$

Motors have an even number of poles, usually from 2 to 12, resulting in the synchronous speeds listed in Table 21-2 for 60-Hz power. But the induction motor, the most widely used type, operates at a speed progressively slower than its synchronous speed as the load (torque) demand increases. When the motor is delivering its rated torque, it will be operating near its rated or full-load speed, also listed in Table 21-2. Note that the full-load speed is not precise and that those listed are for motors with normal slip of approximately 5%. Some motors described later are “high-slip” motors having lower full-load speeds. Some four-pole motors are rated at 1750 rpm at full load, indicating only about 3% slip. It is advisable to consult manufacturers’ literature for the full-load speed of any motor under consideration. *Synchronous motors* operate precisely at the synchronous speed with no slip.

21-4 PRINCIPLES OF OPERATION OF AC INDUCTION MOTORS

Later in this chapter, we will discuss the particular details of several different types of AC motors, but the most common of these is the *induction motor*. Refer again to Figure 21-1 that shows the components in an exploded view. Also look ahead to Figures 21-13 to 21-17 for photos of complete motors. The two active parts of an induction motor are the *stator*, or stationary element, and the *rotor*, or rotating element. Figure 21-4 shows a longitudinal cross section of an induction motor showing the stator, in the form of a hollow cylinder, fixed in the housing. The rotor is positioned inside the stator with a precise, small clearance or *gap*, and is carried on the shaft. The shaft, in turn, is supported by bearings in the housing.

The stator is constructed of many thin, flat discs of steel, called *laminations*, stacked together and insulated from one another. Figure 21-5 shows the shape of the laminations, including a series of slots around the inside. These slots are aligned as the stator laminations are stacked, thus forming channels along the length of the stator core. Several layers of copper wire are passed through the channels and are looped around to form a set of continuous coils,

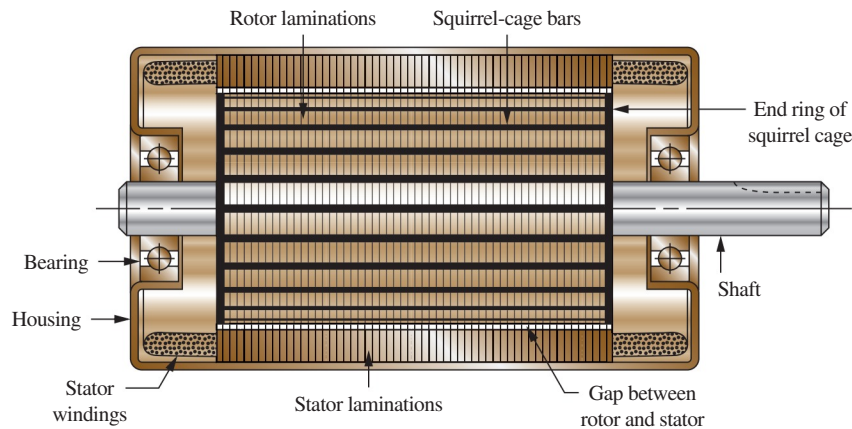


FIGURE 21-4 Longitudinal section through an induction motor

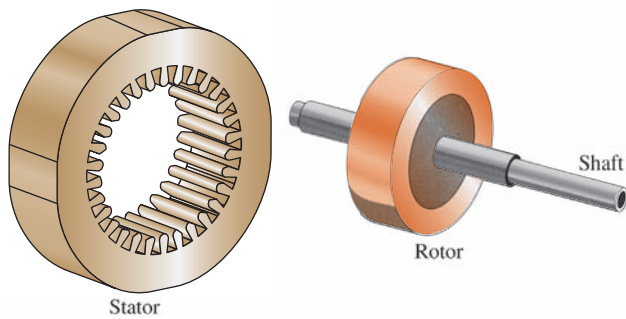


FIGURE 21-5 Induction motor laminations

called *windings*. The pattern of the coils in the stator determines the number of poles for the motor, usually 2, 4, 6, 8, 10, or 12. Table 21-2 shows that the rotational speed of the motor depends on the number of poles.

The rotor also has a lamination stack with channels along the length. The channels are typically filled with solid bars made from a good electrical conductor such as copper or aluminum, with the ends of all bars connected to continuous rings at each end. In some smaller rotors, the complete set of bars and end rings are cast from aluminum as a unit. As shown in Figure 21-6, if this casting were viewed without the laminations, it would appear similar to a squirrel cage. Thus, induction motors are often called squirrel-cage motors. The combination of the squirrel cage and the laminations is fixed onto the motor shaft with good precision to assure concentric alignment with the stator and good dynamic balance while rotating. When the rotor is installed in the supporting bearings and is inserted inside the stator, there is a small gap of approximately 0.020 in (0.50 mm) between the outer surface of the rotor and the inner surface of the stator.

Three-Phase Motors

The principles of operation of AC motors are first discussed for three-phase induction motors. Single-phase motor designs are discussed later. The three-phase electrical power, shown schematically in Figure 21-3, is connected

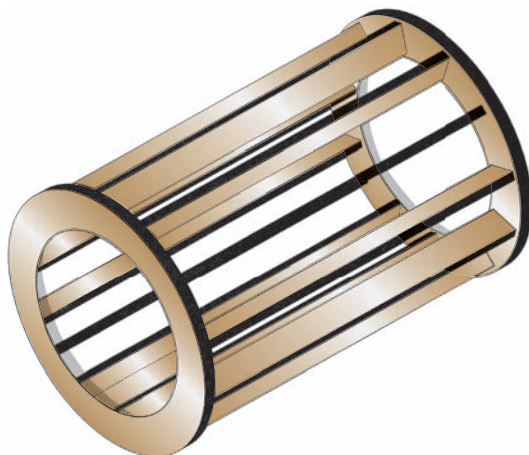


FIGURE 21-6 Squirrel-cage rotor without laminations

to the stator windings. As the current flows in the windings, electromagnetic fields are created that are exposed to the conductors in the rotor. Because the three phases of power are displaced from each other in time, the effect created is a set of fields rotating around the stator. A conductor placed in a moving magnetic field has a current induced in it, and a force is exerted perpendicular to the conductor. The force acts near the periphery of the rotor, thus creating a torque to rotate the rotor and drive the load.

It is the production of the induced current in the rotor that leads to calling such motors as *induction motors*. Note also that there are no direct electrical connections to the rotor, thus greatly simplifying the design and construction of the motor and contributing to its high reliability.

21-5 AC MOTOR PERFORMANCE

The performance of electric motors is usually displayed on a graph of speed versus torque, as shown in Figure 21-7. The *vertical axis* is the rotational speed of the motor as a percentage of synchronous speed. The *horizontal axis* is the torque developed by the motor as a percentage of the full-load or rated torque. When exerting its full-load torque, the motor operates at its full-load speed and delivers the rated torque and power. Table 21-2 lists the synchronous speeds and typical full-load speeds.

The torque at the bottom of the curve where the speed is zero is called the *starting torque* or *locked-rotor torque*. It is the torque available to initially get the load moving and begin its acceleration. This is one of the most important selection parameters for motors, as will be discussed in the descriptions of the individual types of motors.

The “knee” of the curve, called the *breakdown torque*, is the maximum torque developed by the motor during acceleration. The slope of the speed/torque curve in the vicinity of the full-load operating point is an indication of *speed regulation*. A flat curve (a low slope) indicates good speed regulation with little variation in speed as load varies. Conversely, a steep curve (a high slope) indicates poor speed regulation, and the motor will exhibit wide swings

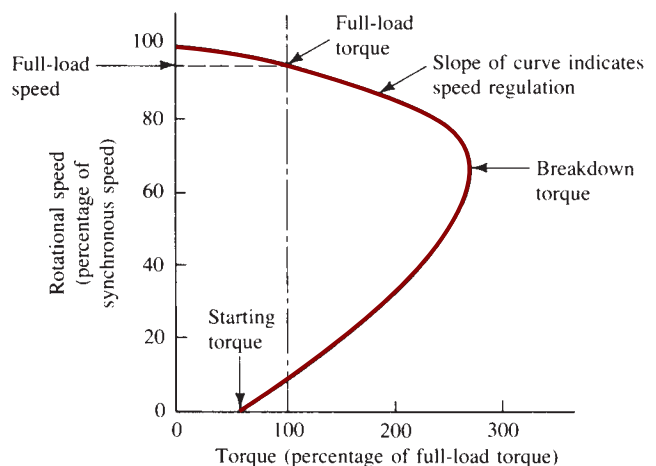


FIGURE 21-7 General form of motor performance curve

in speed as load varies. Such motors produce a “soft” acceleration of a load which may be an advantage in some applications. But where fairly constant speed is desired, a motor with good speed regulation should be selected.

21-6 THREE-PHASE, SQUIRREL-CAGE INDUCTION MOTORS

Three of the most commonly used three-phase AC motors are simply designated as designs B, C, and D by the National Electrical Manufacturers Association (NEMA). They differ primarily in the value of starting torque and in the speed regulation near full load. Figure 21-8 shows examples of the performance curves for these three designs for comparison. Each of these designs employs the solid squirrel-cage type of rotor, and thus there is no electrical connection to the rotor.

The four-pole design with a synchronous speed of 1800 rpm is the most common and is available in virtually all power ratings from 1/4 hp to 500 hp. Certain sizes are available in 2-pole (3600 rpm), 6-pole (1200 rpm), 8-pole (900 rpm), 10-pole (720 rpm), and 12-pole (600 rpm) designs.

NEMA Design B

The performance of the three-phase design B motor is similar to that of the single-phase split-phase motor described later. It has a moderate starting torque (about 150% of full-load torque) and good speed regulation. The breakdown torque is high, usually 200% of full-load torque or more. Starting current is fairly high, at approximately six times the full-load current. The starting circuit must be selected to be able to handle this current for the short time required to bring the motor up to speed.

Typical uses for the design B motor are centrifugal pumps, fans, blowers, and machine tools such as grinders and lathes.

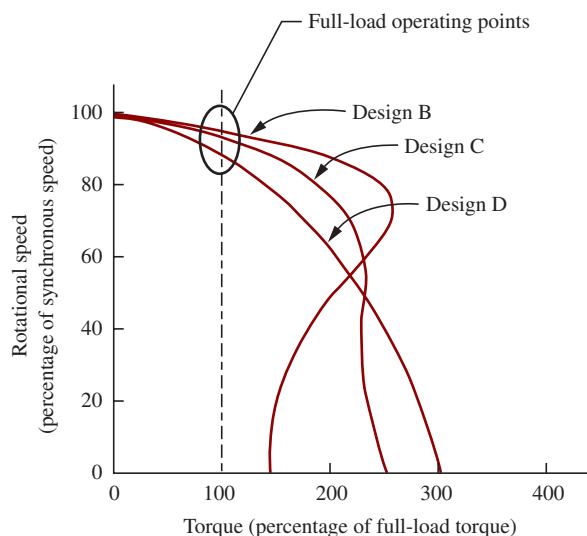


FIGURE 21-8 Examples of typical performance curves for three-phase motors: designs B, C, and D

NEMA Design C

High starting torque is the main advantage of the design C motor. Loads requiring 200% to 300% of full-load torque to start can be driven. Starting current is typically lower than for the design B motor for the same starting torque. Speed regulation is good and is about the same as for the design B motor. Reciprocating compressors, refrigeration systems, heavily loaded conveyors, and ball-and-rod mills are typical uses.

NEMA Design D

The design D motor has a high starting torque, about 300% of full-load torque. But it also has poor speed regulation, which results in large speed changes with varying loads. Sometimes called a *high-slip motor*, it operates at 5% to 13% slip at full load, whereas the designs B and C operate at 3% to 5% slip. Thus, the full-load speed will be lower for the design D motor.

The poor speed regulation is considered an advantage in some applications and is the main reason for selecting the design D motor for such uses as punch presses, shears, sheet-metal press brakes, cranes, elevators, and oil well pumps. Allowing the motor to slow down significantly when loads increase gives the system a “soft” response, reducing shock and jerk felt by the drive system and the driven machine. Consider an elevator: When a heavily loaded elevator cab is started, the acceleration should be smooth and soft, and the cruising speed should be approached without excessive jerk. This comment also applies to a crane. If a large jerk occurs when the crane hook is heavily loaded, the peak acceleration will be high. The resulting high inertia force may break the cable resulting in a dangerous safety problem.

Wound-Rotor Motors

As the name implies, the rotor of the *wound-rotor motor* has electrical windings that are connected through slip rings to the external power circuit. The selective insertion of resistance in the rotor circuit allows the performance of the motor to be tailored to the needs of the system and to be changed with relative ease to accommodate system changes or to actually vary the speed of the motor.

Figure 21-9 shows the results obtained by changing the resistance in the rotor circuit. Note that the four curves are all for the same motor, with curve 0 giving the performance with zero external resistance. This is similar to design B. Curves 1, 2, and 3 show the performance with progressively higher levels of resistance in the rotor circuit. Thus, the starting torque and the speed regulation (softness) can be tuned to the load. Speed adjustment under a given load up to approximately 50% of full-load speed can be obtained.

The wound-rotor design is used in such applications as printing presses, crushing equipment, conveyors, and hoists.

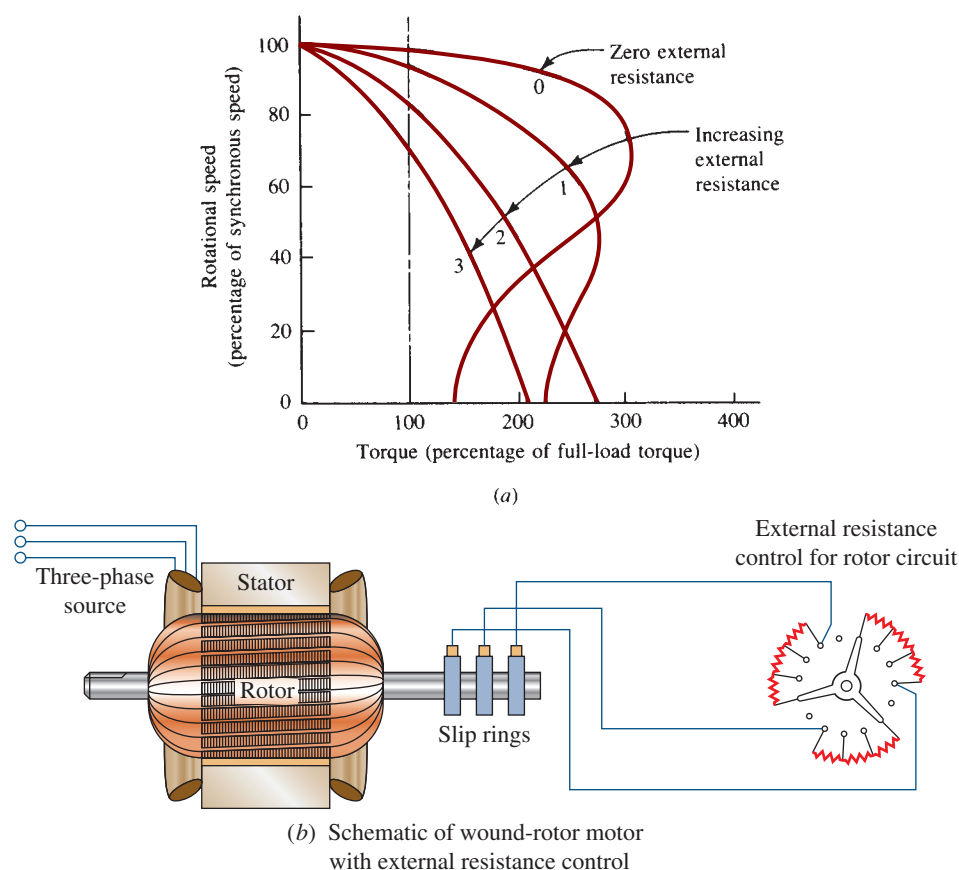


FIGURE 21-9 (a) Examples of typical performance curves for a three-phase, wound-rotor motor with varying external resistance in the rotor circuit (b) Schematic of wound-rotor motor with external resistance control

Electronically Commutated Motors

Systems that apply variable-speed drives, discussed in Section 21-9, can benefit from using electronically commutated (EC) motors. These motors have brushless permanent magnets and are often integrated with a tuned variable frequency motor drive. Such motors, when compared with typical induction motors, exhibit higher efficiency (up to 91%) at design speed, and less decrease in efficiency when lower speeds are used. Applications include fluid pumping systems and conveyors requiring up to about 10 hp (7.5 kW).

Synchronous Motors

Entirely different from the squirrel-cage induction motor or the wound-rotor motor, the *synchronous motor* operates precisely at the synchronous speed with no slip. Such motors are available in sizes from subfractional, used for timers and instruments, to several hundred horsepower to drive large air compressors, pumps, or blowers.

The synchronous motor must be started and accelerated by a means separate from the synchronous motor components themselves because they provide very little torque at zero speed. Typically, there will be a separate squirrel-cage type of winding within the normal rotor which initially accelerates the motor shaft. When the speed

of the rotor is within a few percent of the synchronous speed, the field poles of the motor are excited, and the rotor is pulled into synchronism. At that point the squirrel cage becomes ineffective, and the motor continues to run at speed regardless of load variations, up to a limit called the *pull-out torque*. A load above the pull-out torque will pull the motor out of synchronism and cause it to stop.

Universal Motors

Universal motors operate on either AC or direct current (DC) power. Their construction is similar to that of a series-wound DC motor, which is described later. The rotor has electrical coils that are connected to the external circuit through a commutator on the shaft, a kind of slip ring assembly made of several copper segments on which stationary carbon brushes ride. Contact is maintained with light spring pressure.

Universal motors usually run at high speeds, 3500 to 20 000 rpm. This results in a high power-to-weight and high power-to-size ratio for this type of motor, making it desirable for small appliances and handheld tools such as drills, saws, and food mixers. Vacuum cleaners and sewing machines also frequently use universal motors. Figure 21-10 shows a typical set of speed/torque curves for a high-speed version of the universal motor, showing

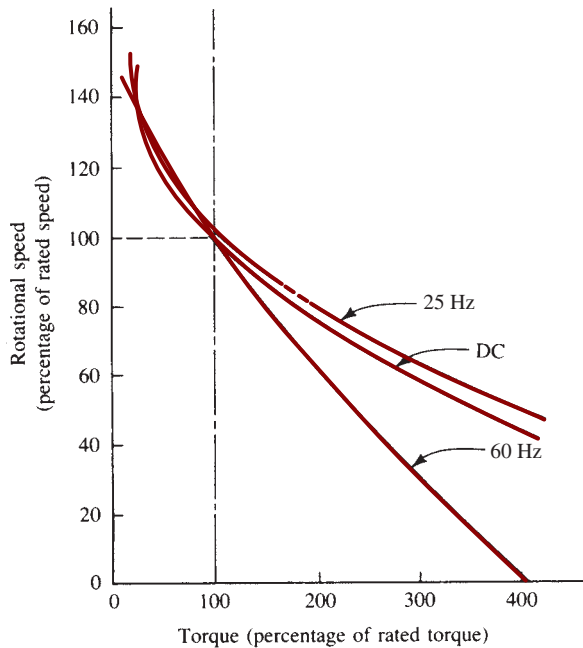


FIGURE 21-10 Examples of typical performance curves for universal motors

the performance for 60-Hz and 25-Hz AC power and DC power. Note that performance near the rated load is similar, regardless of the nature of the input power. Note also that these motors have poor speed regulation, that is, the speed varies greatly with load.

21-7 SINGLE-PHASE MOTORS

The four most common types of single-phase motors are the *split-phase*, *capacitor-start*, *permanent-split capacitor*, and *shaded-pole*. Each is unique in its physical construction and in the manner in which the electrical components are connected to provide for starting and running of the motor. The emphasis here is not on how to design the motors, but rather on the performance, so that a suitable motor can be selected.

Figure 21-11 shows the performance characteristics of these four types of motors so that they can be compared. The special features of the performance curves for the four types of motors are discussed in later sections.

In general, the construction of single-phase motors is similar to that for three-phase motors, consisting of a fixed stator, a solid rotor, and a shaft carried on bearings. The induction principle discussed earlier applies also to single-phase motors. Differences occur because single-phase power does not inherently rotate around the stator to create a moving field. Each type uses a different scheme for initially starting the motor. See Figure 21-12.

Single-phase motors are usually in the subfractional or fractional horsepower range from 1/50 hp (15 W) to 1.0 hp (750 W), although some are available up to 10 hp (7.5 kW).

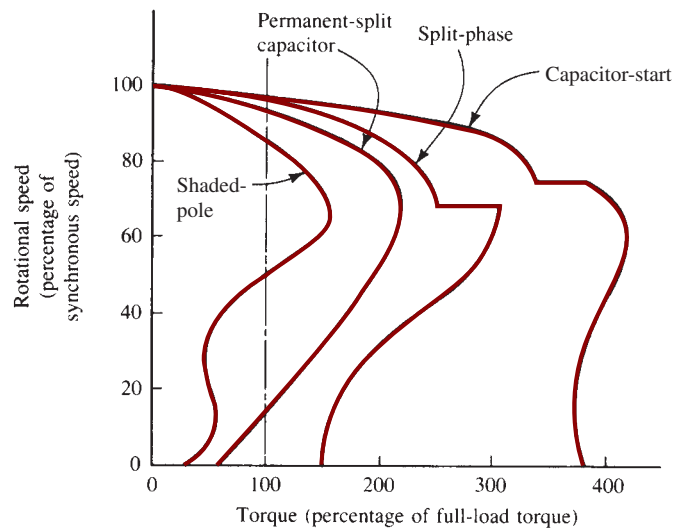


FIGURE 21-11 Performance curves of four types of single-phase electric motors

Split-Phase Motors

The stator of the split-phase motor [Figure 21-12(a) and (b)] has two windings: the *main winding*, which is continuously connected to the power line, and the *starting winding*, which is connected only during the starting of the motor. The starting winding creates a slight phase shift that creates the initial torque to start and accelerate the rotor. After the rotor reaches approximately 75% of its synchronous speed, the starting winding is cut out by a centrifugal switch, and the rotor continues to run on the main winding.

The performance curve for the split-phase motor is shown in Figure 21-11. It has moderate starting torque, approximately 150% of full-load torque. It has good efficiency and is designed for continuous operation. Speed regulation is good. One of the disadvantages is that it requires a centrifugal switch to cut out the starting winding. The step in the speed/torque curve indicates this cutout.

These characteristics make the split-phase motor one of the most popular types, used in business machines, machine tools, centrifugal pumps, AC electric lawn mowers, and similar applications.

Capacitor-Start Motors

Like the split-phase motor, the capacitor-start motor [Figure 21-12(c)] also has two windings: a *main* or *running winding* and a *starting winding*. But in it, a capacitor is connected in series with the starting winding, giving a much higher starting torque than with the split-phase motor. A starting torque of 250% of full load or higher is common. Again a centrifugal switch is used to cut out the starting winding and the capacitor. The running characteristics of the motor are then very similar to those of the split-phase motor: good speed regulation and good efficiency for continuous operation.

Disadvantages include the switch and the relatively bulky capacitor. Frequently the capacitor is mounted

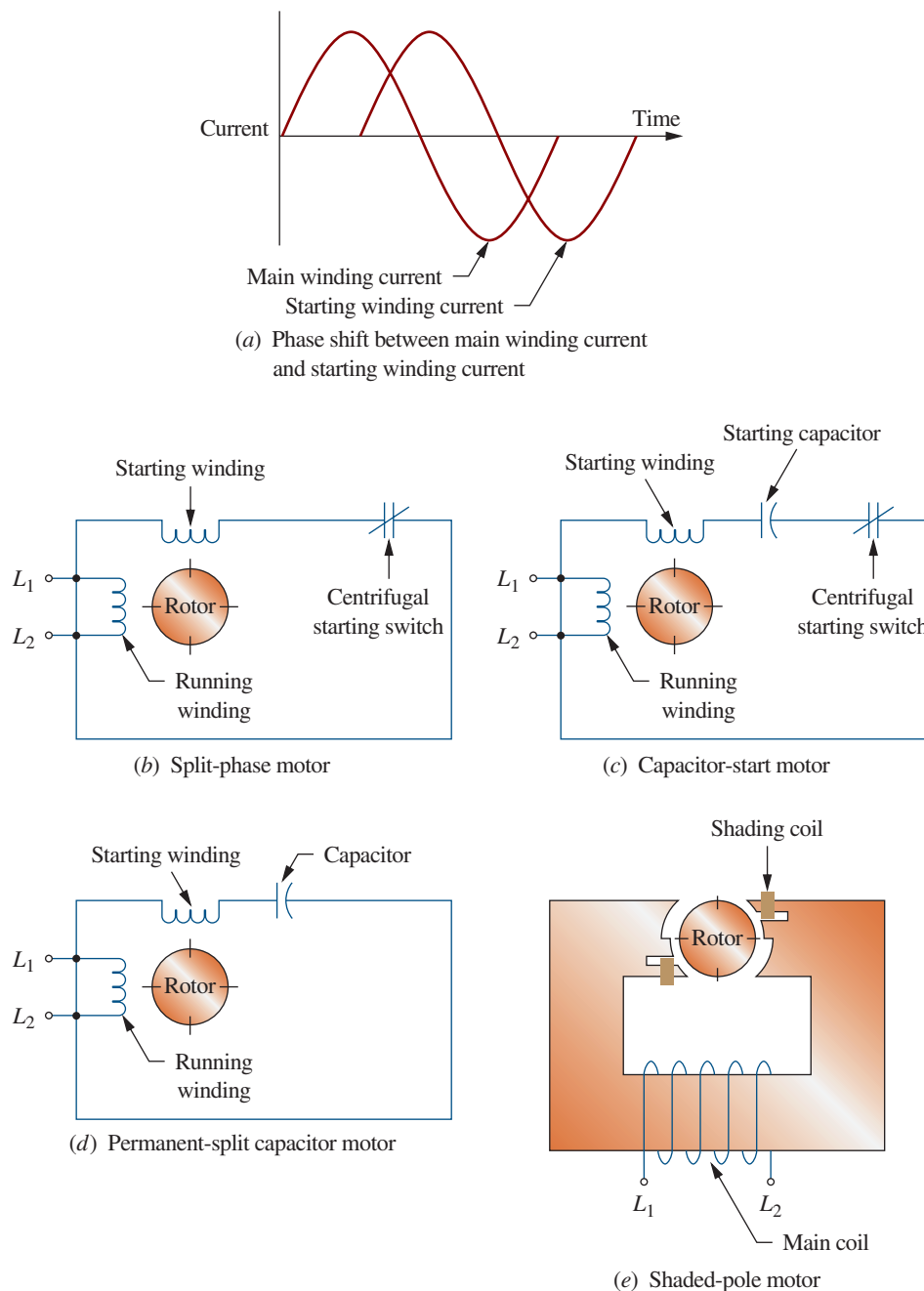


FIGURE 21-12 Schematic diagrams of single-phase motors

conspicuously right on top of the motor. It may also be integrated into a package containing the starting switch, a relay, or other control elements.

Uses for the capacitor-start motor include the many types of machines that need the high starting torque. Examples include heavily loaded conveyors, refrigeration compressors, and pumps and agitators for heavy fluids.

Permanent-Split Capacitor Motors

A capacitor is connected in series with the starting winding at all times. The starting torque of the permanent-split capacitor motor [Figure 21-12(d)] is typically quite low, approximately 40% or less of full-load torque.

Thus, only low-inertia loads such as fans and blowers are usually used. An advantage is that you can tailor the running performance and the speed regulation to match the load by selecting the appropriate capacitor value. Also, no centrifugal switch is required.

Shaded-Pole Motors

The shaded-pole motor [Figure 21-12(e)] has only one winding, the *main* or *running* winding. The starting reaction is created by the presence of a copper band around one side of each pole. The low-resistance band “shades” the pole to produce a rotating magnetic field to start the motor.

The shaded-pole motor is simple and inexpensive, but it has a low efficiency and a very low starting torque. Speed regulation is poor, and it must be fan-cooled during normal operation. Thus, its primary use is in shaft-mounted fans and blowers where the air is drawn over the motor. Some small pumps, toys, and intermittently used household items also employ shaded-pole motors because of their low cost.

21-8 AC MOTOR FRAME TYPES AND ENCLOSURES

Frame Types

The design of the equipment in which the motor is to be mounted determines the type of frame required. Some types are described below.

Foot-mounted. The most widely used type for industrial machinery, the foot-mounted frame has integral feet with a standard hole pattern for bolting the motor to the machine (see Figure 21-13).

Cushion Base. A foot mounting is provided with resilient isolation of the motor from the frame of the machine to reduce vibration and noise. This frame type is often used for fans and appliances where the motor is enclosed in the shell or housing of the product.

C-Face Mounting. A machined face is provided on the shaft end of the motor, which has a standard pattern of tapped holes. Driven equipment is then bolted directly to the motor. The design of the face is standardized by the National Electrical Manufacturers Association (NEMA). See Figure 21-14.

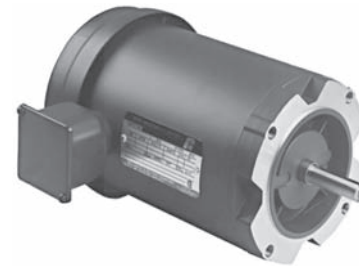


FIGURE 21-14 C-face motor. See Figure 8-25 for an example of a reducer designed to mate with a C-face motor (Baldor/Reliance Electric, Greenville, SC)

D-Flange Mounting. A machined flange is provided on the shaft end of the motor with a standard pattern of through clearance holes for bolts for attaching the motor to the driven equipment. The flange design is controlled by NEMA.

Vertical Mounting. Vertical mounting is a special design because of the effects of the vertical orientation on the bearings of the motor. Attachment to the driven equipment is usually through C-face or D-flange bolt holes as previously described.

Unmounted. Some equipment manufacturers purchase only the bare rotor and stator from the motor manufacturer and then build them into their machine. Compressors for refrigeration equipment are usually built in this manner.

Special-Purpose Mountings. Many special designs are made for fans, pumps, oil burners, and so on.

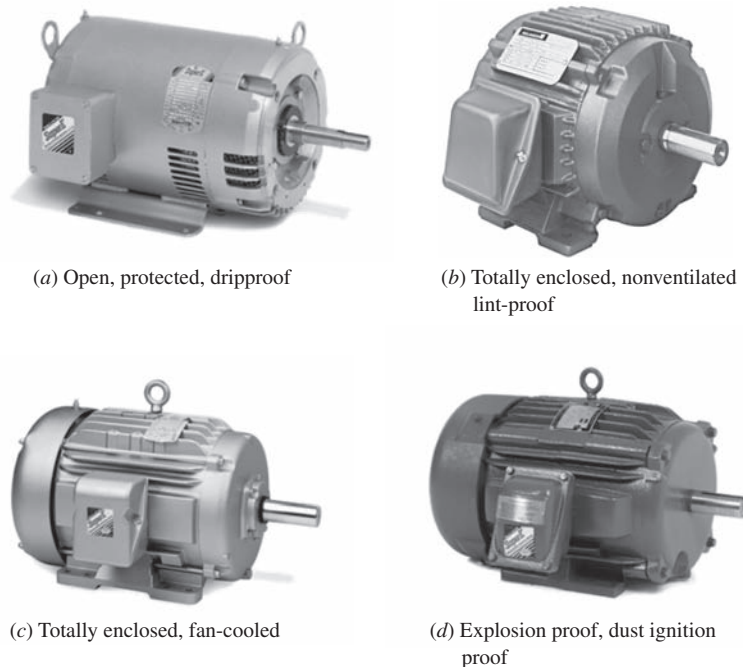


FIGURE 21-13 Foot-mounted motors with various enclosure types (Baldor/Reliance Electric, Greenville, SC)

Enclosures

The housings around the motor that support the active parts and protect them vary with the degree of protection required. Some enclosure types are described next.

Open. Typically a light-gage sheet-metal housing is provided around the stator with end plates to support the shaft bearings. The housing contains several holes or slots that permit cooling air to enter the motor. Such a motor must be protected by the housing of the machine itself.

Protected. Sometimes called *open dripproof*, ventilating openings are provided only on the lower part of the housing so that liquids dripping on the motor from above cannot enter the motor. This is probably the most widely used type [see Figure 21-13(a)].

Totally Enclosed Nonventilated (TENV). No openings at all are provided in the housing, and no special provisions are made for cooling the motor except for fins cast into the frame to promote convective cooling. The design protects the motor from harmful atmospheres [see Figure 21-13(b)].

Totally Enclosed Fan-cooled (TEFC). The TEFC design is similar to the TENV design, except a fan is mounted to one end of the shaft to draw air over the finned housing [see Figure 21-13(c)].

TEFC-XP. The TEFC-XP (explosion-proof) design is similar to the TEFC housing, except special protection is provided for electrical connections to prohibit fire or explosion in hazardous environments [see Figure 21-13(d)].

IEEE 841-2001 Severe Duty Motors. See Figure 21-15. This special type of housing is designed for demanding environments such as in steel mills, petrochemical plants, and pulp and paper mills where

corrosive environments, high temperatures, and high humidity often occur. Extra fin area, some stainless steel components, and corrosion-resistant paints and seals are employed.

Washdown Duty Motors. See Figure 21-16. When motors are used in food processing areas, clean rooms, medical facilities, and similar locations, this special type of housing is designed to be washed down with sanitary cleaning fluid and to withstand direct hosing.

Brake Motors. See Figure 21-17. Note the extension to the left side of the motor that houses a braking system that can be tied into a control system to stop the motor shaft quickly in case of safety issues or to accommodate process cycling. See Chapter 22 for more discussion of brake designs.

Frame Sizes

The critical dimensions of motor frames are controlled by NEMA frame sizes. Included are the overall height and width; the height from the base to the shaft centerline; the shaft diameter, length, and keyway size; and mounting hole pattern dimensions. A few selected motor frame sizes for 1725-rpm, three-phase, induction, foot-mounted dripproof motors are listed in Table 21-3. For the descriptions of the dimensions, refer to Figure 21-18.



FIGURE 21-16 Washdown duty motor (Baldor Electric, Greenville, SC)



FIGURE 21-15 IEEE 841-2001 severe duty motor (Baldor Electric, Greenville, SC)



FIGURE 21-17 Brake motor (Baldor Electric, Greenville, SC)

TABLE 21-3 Motor Frame Sizes

hp	Frame size	Dimensions (in)								Keyway
		A	C	D	E	F	O	U	V	
1/4	48	5.63	9.44	3.00	2.13	1.38	5.88	0.500	1.50	0.05 flat
1/2	56	6.50	10.07	3.50	2.44	1.50	6.75	0.625	1.88	3/16 × 3/32
1	143T	7.00	10.69	3.50	2.75	2.00	7.00	0.875	2.00	3/16 × 3/32
2	145T	7.00	11.69	3.50	2.75	2.50	7.00	0.875	2.00	3/16 × 3/32
5	184T	9.00	13.69	4.50	3.75	2.50	9.00	1.125	2.50	1/4 × 1/8
10	215T	10.50	17.25	5.25	4.25	3.50	10.56	1.375	3.13	5/16 × 5/32
15	254T	12.50	22.25	6.25	5.00	4.13	12.50	1.625	3.75	3/8 × 3/16
20	256T	12.50	22.25	6.25	5.00	5.00	12.50	1.625	3.75	3/8 × 3/16
25	284T	14.00	23.38	7.00	5.50	4.75	14.00	1.875	4.38	1/2 × 1/4
30	286T	14.00	24.88	7.00	5.50	5.50	14.00	1.875	4.38	1/2 × 1/4
40	324T	16.00	26.00	8.00	6.25	5.25	16.00	2.125	5.00	1/2 × 1/4
50	326T	16.00	27.50	8.00	6.25	6.00	16.00	2.125	5.00	1/2 × 1/4

Note: All motors are four-pole, three-phase, 60-Hz, AC induction motors. Refer to Figure 21-18 for description of dimensions.

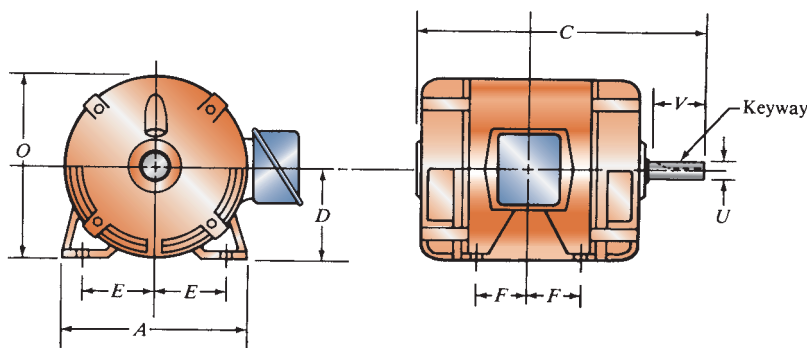


FIGURE 21-18 Key for NEMA standard motor frame dimensions listed in Table 21-3

21-9 CONTROLS FOR AC MOTORS

Motor controls must perform several functions, as outlined in Figure 21-19. The complexity of the control depends on the size and the type of motor involved. Small fractional or subfractional motors may sometimes be started with a simple switch that connects the motor directly to the full line voltage. Larger motors, and some smaller motors on critical equipment, require greater protection. See Internet sites 1-9 and References 2-9.

The functions of motor controls are as follows:

1. To start and stop the motor
2. To protect the motor from overloads that would cause the motor to draw dangerously high current levels
3. To protect the motor from overheating
4. To protect personnel from contact with hazardous parts of the electrical system
5. To protect the controls from the environment
6. To prohibit the controls from causing a fire or an explosion

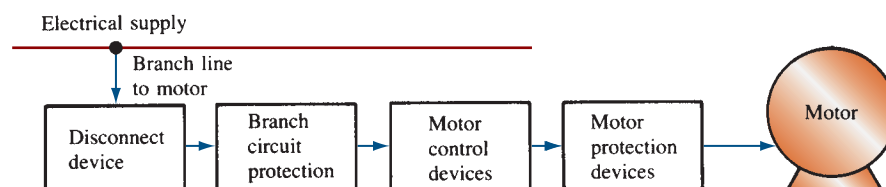


FIGURE 21-19 Motor control block diagram

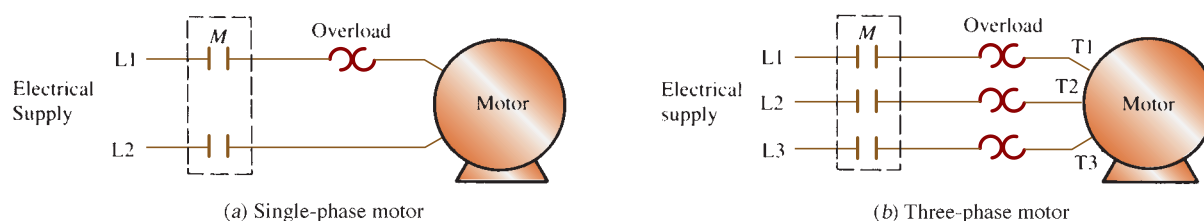


FIGURE 21-20 Manual starters (M = normally open contactors; all actuate together)

7. To provide controlled torque, acceleration, speed, or deceleration of the motor
8. To provide for the sequential starting of a series of motors or other devices
9. To provide for the coordinated operation of different parts of a system, possibly including several motors
10. To protect the conductors of the branch circuit in which the motor is connected

The proper selection of a motor control system requires knowledge of at least the following factors:

1. The type of electrical service: voltage and frequency; single- or three-phase; current limitations
2. The type and size of motor: power and speed ratings; full-load current rating; locked-rotor current rating
3. Operation desired: duty cycle (continuous, start/stop, or intermittent); single or multiple discrete speeds, or variable-speed operation; one-direction or reversing
4. Environment: temperature; water (rain, snow, sleet, sprayed, or splashed water); dust and dirt; corrosive gases or liquids; explosive vapors or dusts; oils or lubricants
5. Space limitations
6. Accessibility of controls
7. Noise or appearance factors

Starters

There are several classifications of motor starters: manual or magnetic; one-direction or reversing; two-wire or

three-wire control; full-voltage or reduced-voltage starting; single-speed or multiple-speed; normal stopping, braking, or plug stopping. All of these typically include some form of overload protection, which will be discussed later.

Manual and Magnetic, Full-Voltage, One-Direction Starting

Figure 21-20 shows the schematic connection diagram for manual starters for single-phase and three-phase motors. The symbol M indicates a normally open contactor (switch) that is actuated manually, for example, by throwing a lever. The contactors are rated according to the motor power that they can safely handle. The power rating indirectly relates to the current drawn by the motor, and the contactor design must (1) safely make contact during the start-up of the motor, considering the high starting current; (2) carry the expected range of operating current without overheating; and (3) break contact without excessive arcing that could burn the contacts. The ratings are established by NEMA. Tables 21-4 and 21-5 show the ratings for selected NEMA starter sizes.

Note in Figure 21-20 that overload protection is required in all three lines for three-phase motors but in only one line of the single-phase motor.

Figure 21-21 shows the schematic connection diagrams for magnetic starters using two-wire and three-wire controls. The “start” button in the three-wire control is a momentary contact type. As it is actuated manually, the coil in parallel with the switch is energized, and it magnetically closes the line contactors marked M . The

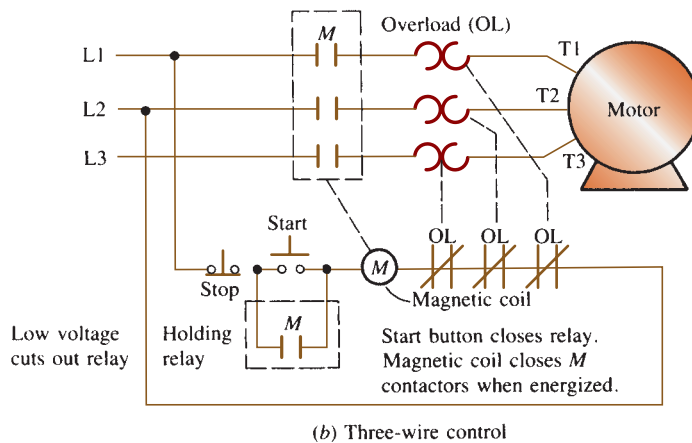
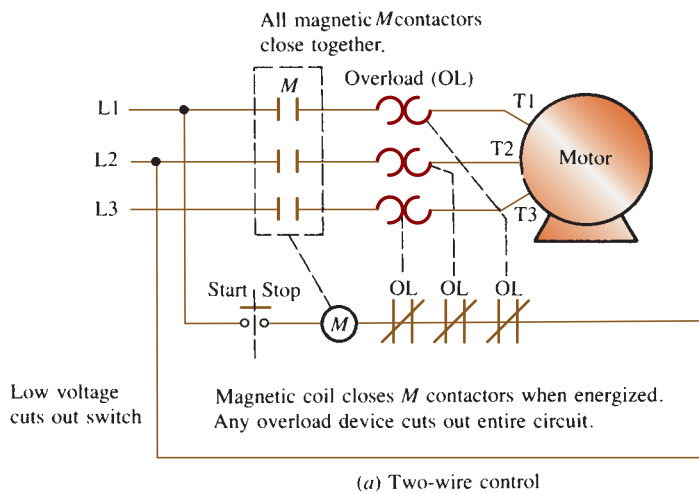
TABLE 21-4 Ratings of AC Full-Voltage Starters for Single-Phase Power

NEMA size number	Current rating (amperes)	Power rating at given voltages					
		110 V		220 V		440 and 550 V	
		(hp)	(kW)	(hp)	(kW)	(hp)	(kW)
00		1/2	0.37	3/4	0.56		
0	15	1	0.75	1 1/2	1.12	1 1/2	1.12
1	25	1 1/2	1.12	3	2.24	5	3.73
2 ^a	50	3	2.24	7 1/2	5.60	10	7.46
3 ^a	100	7 1/2	5.60	15	11.19	25	18.65

^a Applies to magnetically operated starters only.

TABLE 21-5 Ratings of AC Full-Voltage Starters for Three-Phase Power

NEMA size number	Current rating (amperes)	Power rating at given voltages					
		110 V		220 V		440 and 550 V	
		(hp)	(kW)	(hp)	(kW)	(hp)	(kW)
00		3/4	0.56	1	0.75	1	0.75
0	15	1½	1.12	2	1.49	2	1.49
1	25	3	2.24	5	3.73	7½	5.60
2	50	7½	5.60	15	11.19	25	18.65
3	100	15	11.19	30	22.38	50	37.30

**FIGURE 21-21** Magnetic starters for three-phase motors

contacts remain closed until the stop button is pushed or until the line voltage drops to a set low value. (Remember that a low line voltage causes the motor to draw excessive current.) Either case causes the magnetic contactors to open, stopping the motor. The start button must be manually pushed again to restart the motor.

The two-wire control has a manually operated start button that stays engaged after the motor starts. As a safety feature, the switch will open when a low-voltage condition occurs. But when the voltage rises again to an

acceptable level, the contacts close, restarting the motor. You must ensure that this is a safe operating mode.

Reversing Starters

Figure 21-22 shows the connection for a reversing starter for a three-phase motor. You can reverse the direction of rotation of a three-phase motor by interchanging any two of the three power lines. The *F* contactors are used for the forward direction. The *R* contactors would

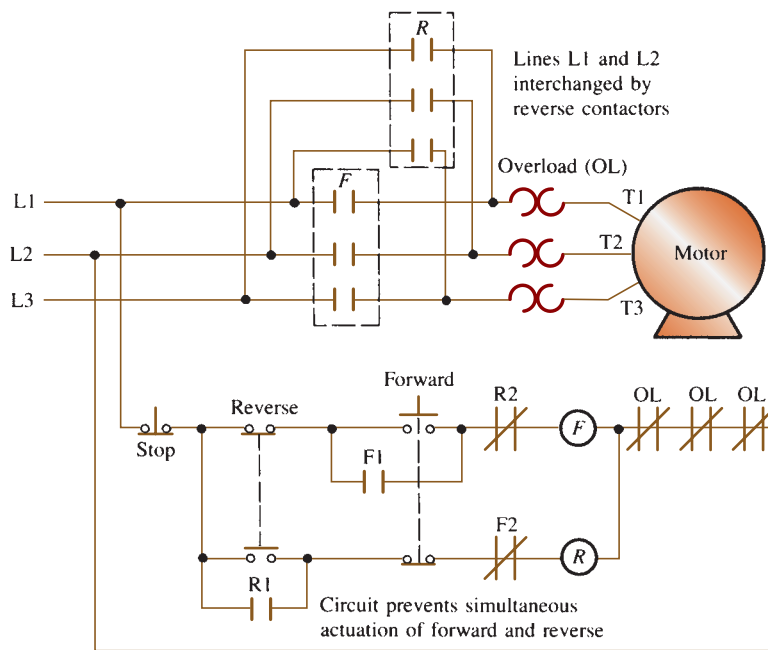


FIGURE 21-22 Reversing control for a three-phase motor

interchange L1 and L3 to reverse the direction. The *Forward* and *Reverse* pushbuttons actuate only one of the sets of contactors.

Reduced-Voltage Starting

The motors discussed in the previous sections and the circuits shown in Figures 21-20 through 21-22 employ full-voltage starting. That is, when the system is actuated, the full line voltage is applied to the motor terminals. This will give the maximum starting effort, but in some cases it is not desirable. To limit jerk, to control the acceleration of a load, and to limit the starting current, reduced-voltage

starting is sometimes used. This gentle start is used on some conveyors, hoists, pumps, and similar loads.

Figure 21-23 shows one method of providing a reduced voltage to the motor when starting. The first action is the closing of the contactors marked A. Thus, the power to the motor passes through a set of resistors that reduces the voltage at each motor terminal. A typical reduction would be to approximately 65% of normal line voltage. The peak line current would be reduced to 65% of normal locked-rotor current, and the starting torque would be 42% of normal locked-rotor torque. (See Reference 4.) After the motor is accelerated, the main contactors M are closed, and full voltage is applied

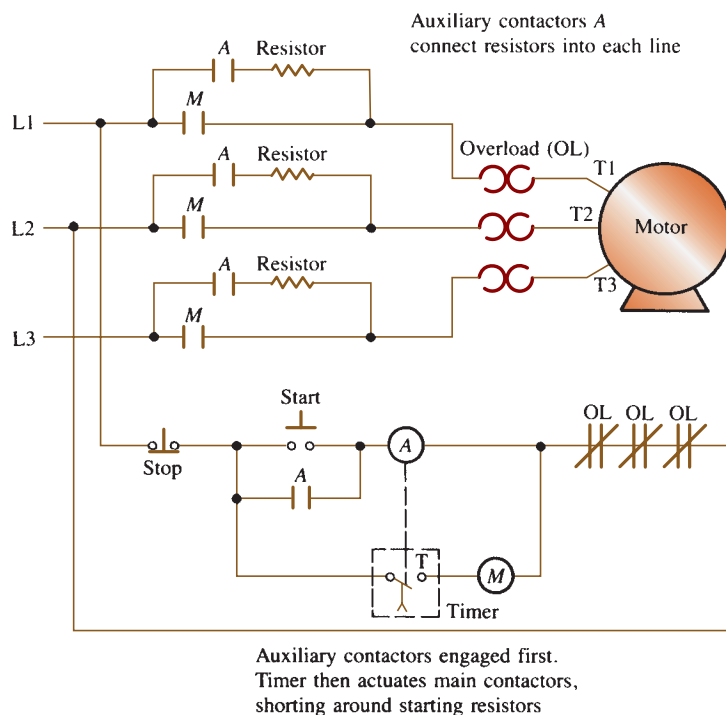


FIGURE 21-23 Reduced-voltage starting by primary resistor method

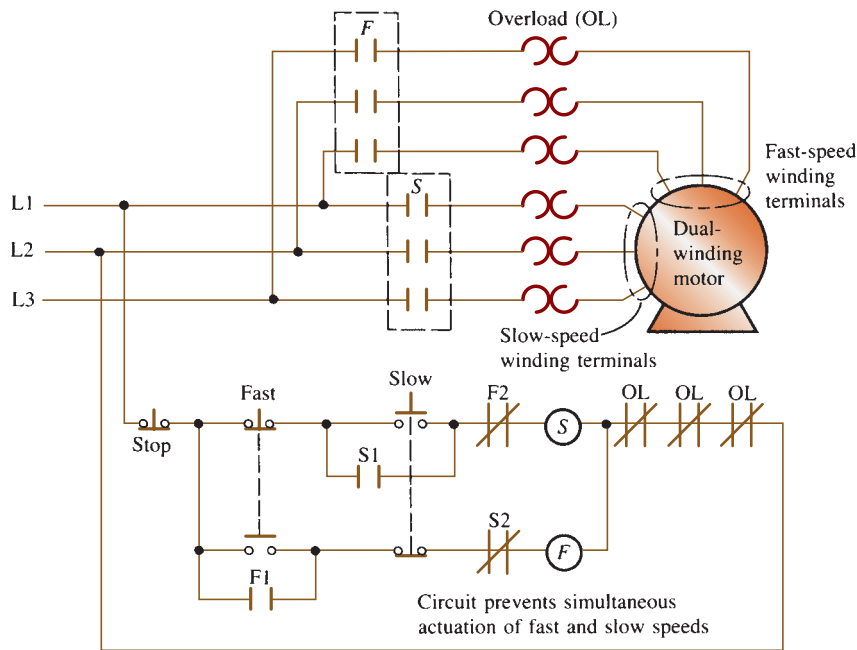


FIGURE 21-24 Speed control for a dual-winding, three-phase motor

to the motor. A timer is typically used to control the sequencing of the A and M contactors.

Dual-Speed Motor Starting

A dual-speed motor with two separate windings to produce different speeds can be started with the circuit shown in Figure 21-24. The operator selectively closes either the *F* (fast) or *S* (slow) contacts to obtain the desired speed. The other features of starting circuits discussed earlier can also be applied to this circuit.

Stopping the Motor

Where no special conditions exist when the system is shut down, the motor can be permitted to coast to a stop after the power is interrupted. The time required to stop will depend on the inertia and the friction in the system. If controlled, rapid stopping is required, external brakes can be used. *Brake motors*, which have a brake integral with the motor, are available. Typically, the design is of the “fail-safe” nature, in which the brake is disengaged by an electromagnetic coil when the motor is energized. When the motor is de-energized, either on purpose or because of power failure, the brake is actuated by mechanical spring force. See Figure 21-17.

On circuits with reversing starters, *plug stopping* can be used. When it is desired to stop the motor running in the forward direction, the control can be switched immediately to reverse. There would then be a decelerating torque applied to the rotor, stopping it quickly. Care must be exercised to cut out the reversing circuit when the motor is at rest to prevent it from continuing in the reverse direction.

Overload Protection

The chief cause of failure in electric motors is overheating of the wound coils due to excessive current. The current

is dependent on the load on the motor. A short circuit, of course, would cause a virtually instantaneously high current of a damaging level.

The protection against a short circuit can be provided by *fuses*, but careful application of fuses to motors is essential. A fuse contains an element that literally melts when a particular level of current flows through it. Thus, the circuit is opened. Reactivating the circuit would require replacing the fuse. Time-delay fuses, or “slow-blowing” fuses, are needed for motor circuits to prevent the fuses from blowing when the motor starts, drawing the relatively high starting current that is normal and not damaging. After the motor starts, the fuse will blow at a set value of overcurrent.

Fuses are inadequate for larger or more critical motors because they provide protection at only one level of overcurrent. Each motor design has a characteristic *overheating curve*, as shown in Figure 21-25. This indicates that the motor could withstand different levels of

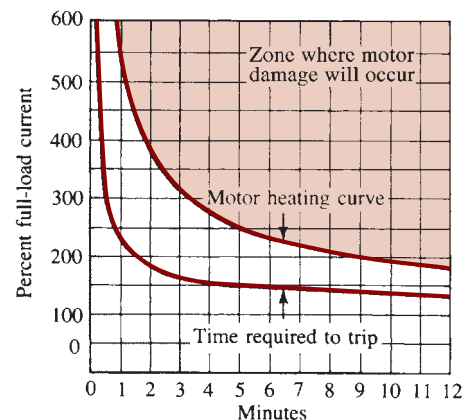


FIGURE 21-25 Motor heating curve and response curve of a typical overload protector

overcurrent for different periods of time. For example, for the motor heating curve of Figure 21–25, a current twice as high as the full-load current (200%) could exist for up to 9 min before a damaging temperature was produced in the windings. But a 400% overload would cause damage in less than 2 min. An ideal overload protection device would parallel the overheating curve of the given motor, always cutting out the motor at a safe current level, as shown in Figure 21–25. Devices are available commercially to provide this protection. Some use special melting alloys, bimetallic strips similar to a thermostat, or magnetic coils that are sensitive to the current flowing in them. Most large motor starters include overload protection integral with the starter. See Internet site 5.

Another type of overload protection uses a temperature-sensitive device inserted in the windings of the motor when it is made. Then it opens the motor circuit when the windings reach a dangerous temperature, regardless of the reason.

Solid-State Overload Relay

Difficulties with thermal element or bimetallic overload devices can be overcome by using a solid-state overload relay. Thermal devices that use a melting element require the replacement of that element after tripping, resulting in extra cost for maintenance supplies and personnel. Both thermal element and bimetallic overload devices are affected by varying ambient temperatures that can change the actual current flow protection level. Temperature compensation devices are available, but they require careful setting and knowledge of expected conditions. Solid-state overload relays overcome these difficulties because only sensed current flow level is used to produce the tripping action. They are inherently insensitive to ambient temperature swings. Furthermore, they can sense the current flow in each of the three windings of three-phase motors and provide protection if any one of the phases experiences a failure or a given increase in current. This provides protection not only for the motor but also for associated equipment that may be damaged if the motor fails suddenly. See Internet site 5 for additional information.

Enclosures for Motor Controls

As stated before, one of the functions of a motor control system is to protect personnel from contact with dangerous parts of the electrical system. Also, protection of the system from the environment must be provided. These functions are accomplished by the enclosure.

NEMA has established standards for enclosures for the variety of environments encountered by motor controls. The most frequent types are described in Table 21–6.

AC Variable-Speed Drives

Standard AC motors operate at a fixed speed for a given load if powered by AC power at a fixed frequency, for example, 60 Hz. Variable-speed operation can be

TABLE 21–6 Motor Control Enclosures

NEMA design number	Description
1	General-purpose: indoor use; not dusttight
3	Dusttight, raintight: outdoor-weather-resistant
3R	Dusttight, rainproof, sleet-resistant
4	Watertight: can withstand a direct spray of water from a hose; used on ships and in food processing plants where washdown is required
4X	Watertight, corrosion-resistant
7	Hazardous locations, class I: can operate in areas where flammable gases or vapors are present
9	Hazardous locations, class II: combustible dust areas
12	Industrial use: resistant to dust, lint, oil, and coolants
13	Oiltight, dusttight

obtained with a control system that produces a variable frequency power. Two such control types are in common use: the *six-step method* and the *pulse-width modulation (PWM) method*. Either system takes the 60-Hz line voltage and first rectifies it to a DC voltage. The six-step method then uses an inverter to produce a series of square waves that provides a voltage to the motor winding which varies both voltage and frequency in six steps per cycle. In the PWM system, the DC voltage is input to an inverter that produces a series of pulses of variable width. The rate of polarity reversals determines the frequency applied to the motor. See Figures 21–26 and 21–27 and Internet sites 5–9.

Reasons for Applying Variable-Speed Drives

It is often desirable to vary the speed of mechanical systems to obtain operating characteristics that are more nearly optimum for the application. For example:

1. Conveyor speed can be varied to match the demand of production.
2. The delivery of bulk materials to a process can be varied continuously.
3. Automatic control can provide synchronization of two or more system components.
4. Dynamic control of system operation can be used during start-up and stopping sequences, for torque control, or for in-process acceleration and deceleration control, often necessary when processing continuous webs such as paper or plastic film.
5. Spindle speeds of machine tools can be varied to produce optimum cutting for given materials, depth of cut, feeds, or cutting tools.
6. The speeds of fans, compressors, and liquid pumps can be varied in response to needs for cooling or for product delivery.

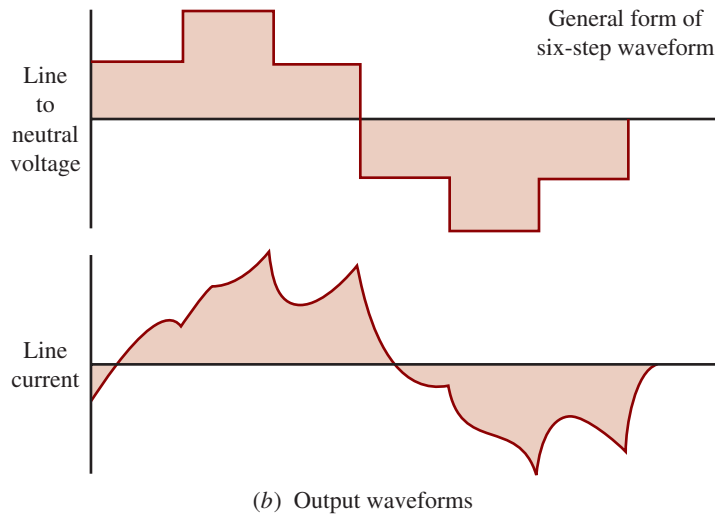
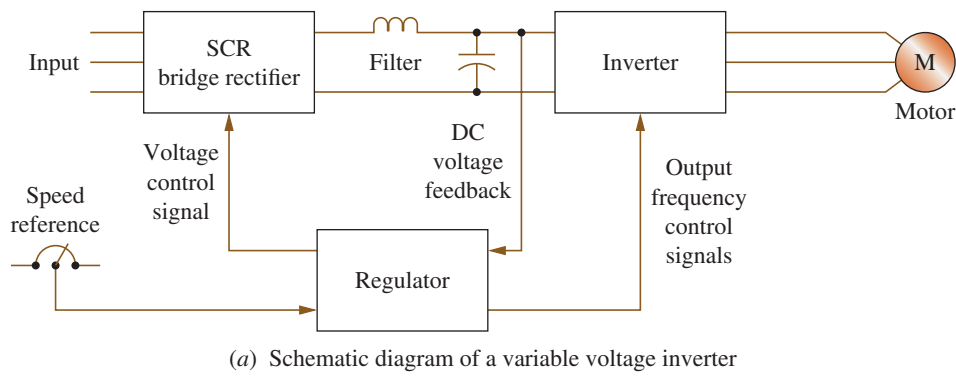


FIGURE 21-26 Six-step method of variable-speed AC motor control (See Internet sites 5-9.)

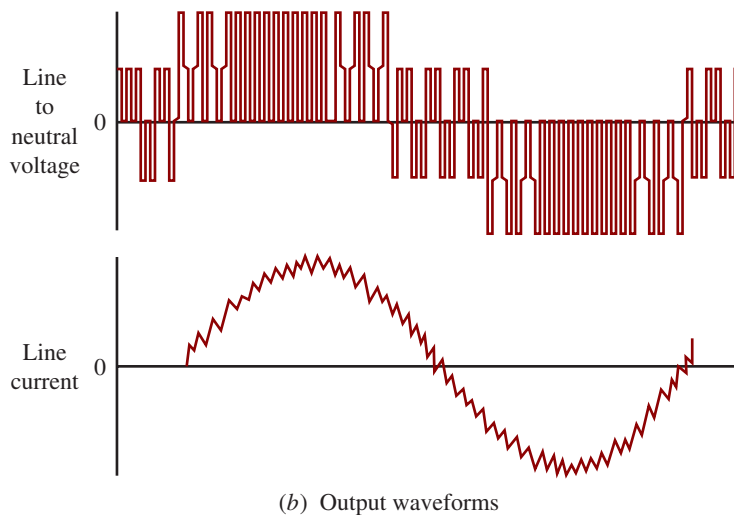
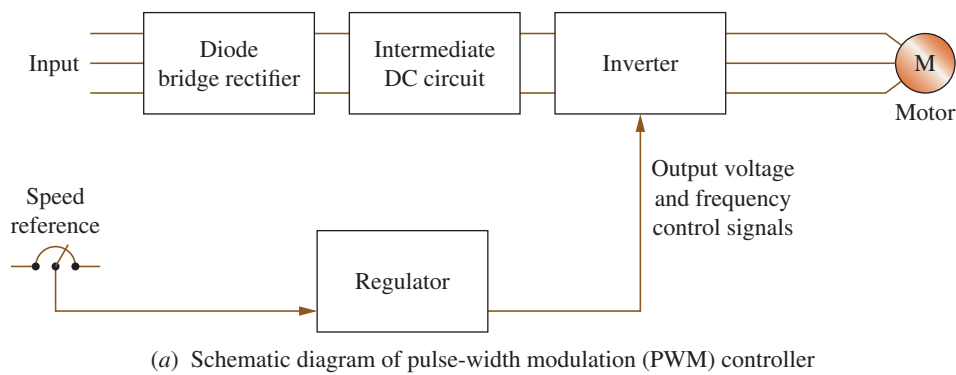


FIGURE 21-27 Pulse-width modulation method of variable-speed AC motor control (See Internet sites 5-9.)

All of these situations allow for more flexible and better process control. Cost savings are also obtained, particularly for item 6. The difference in the power required to operate a pump at two speeds is proportional to the cube of the ratio of the speeds. For example, if the speed of the motor is reduced to one-half of its original speed, the power required to operate the pump is reduced to 1/8 of the original power. Matching the pump speed to the required delivery of the fluid can accumulate significant savings. Similar savings result for fans and compressors.

21-10 DC POWER

DC motors have several inherent advantages over AC motors, as discussed in the next section. A disadvantage of DC motors is that a source of DC power must be available. Most residential, commercial, and industrial locations have only AC power provided by the local utility. Four approaches are used to provide DC power: batteries, hydrogen fuel cells, generators, and rectifiers. See Internet sites 5–9 for additional information on the types of power and control, and Internet sites 10–13 for extended information about batteries.

1. **Batteries:** Typically available battery voltages and their uses are as follows:
 - a. Alkaline and rechargeable nickel metal hydride: 1.5, 3.6, 4.8, 9.0 V
 - Flashlights, portable phones, sensors, small kitchen appliances, small power tools
 - b. Nickel cadmium: 9.6, 12.0, 14.4, 18.0, 24.0 V, power tools
 - c. Sealed lead acid (SLA), flooded valve regulated lead acid (VRLA), absorbed glass mats (AGM): 2.0, 4.0, 6.0, 8.0, 12.0, 24.0, 36.0 V
 - Automotive, marine, wheelchairs, small scooters, forklift trucks, automatic guided vehicles, golf carts, lawn mowers, garden tractors, floor scrubbers, utility switches, network power units, uninterruptible power supplies (UPS), small solar power storage units, telecommunications equipment
 - d. Lithium ion: 12.0, 18.0, 24.0, 28.0, 36.0, 48.0 V
 - Commercial-grade power tools, large capacity standby power, hybrid vehicles, plug-in electric vehicles
2. **Hydrogen fuel cells:** In a single hydrogen fuel cell, hydrogen passing through a plate-type anode, an electrolyte, and a cathode converts chemical energy to electrical energy between two electrodes on either side of the cell at approximately 1.0 V per cell. Only water and heat are discharged from the cells. Multiple cells are stacked together to produce sufficient voltage and power capacity for a given application. Potential uses include small handheld devices and laptop computers, telecommunication

TABLE 21-7 DC Motor Voltage Ratings

Input AC voltage	DC motor rating	NEMA code
115 V AC, one-phase	90 V DC	K
230 V AC, one-phase	180 V DC	K
230 V AC, three-phase	240 V DC	C or D
460 V AC, three-phase	500 V DC or 550 V DC	C or D
460 V AC, three-phase	240 V DC	E

equipment, auxiliary power for residential, commercial, or industrial applications, portable power for remote installations, and all types of transportation. Power capacities range from about 50 W to several hundred kilowatts and higher. Systems rated in the megawatt range are being considered. Still considered an emerging technology, rapid advancements are being made.

3. **Generators:** Powered by AC electric motors, internal combustion engines, turbine engines, wind devices, water turbines, and so on; DC generators produce pure DC. The usual voltages are 115 and 230 V. Some industries maintain such generators to provide DC power throughout the plant.
4. **Rectifiers:** *Rectification* is the process of converting AC power with its sinusoidal variation of voltage with time to DC power, which ideally is nonvarying. A readily available device is the *silicon-controlled rectifier (SCR)*. One difficulty with rectification of AC power to produce DC power is that there is always some amount of “ripple,” a small variation of voltage as a function of time. Excessive ripple can cause overheating of the DC motor. Most commercially available SCR devices produce DC power with an acceptably low ripple. Table 21-7 lists the commonly used DC voltage ratings for motors powered by rectified AC power as defined by NEMA.

21-11 DC MOTORS

The advantages of DC motors are summarized here:

- The speed is adjustable by use of a simple rheostat to adjust the voltage applied to the motor.
- The direction of rotation is reversible by switching the polarity of the voltage applied to the motor.
- Automatic control of speed is simple to provide for matching of the speeds of two or more motors or to program a variation of speed as a function of time.
- Acceleration and deceleration can be controlled to provide the desired response time or to decrease jerk.
- Torque can be controlled by varying the current applied to the motor. This is desirable in tension control applications, such as the winding of film on a spool.

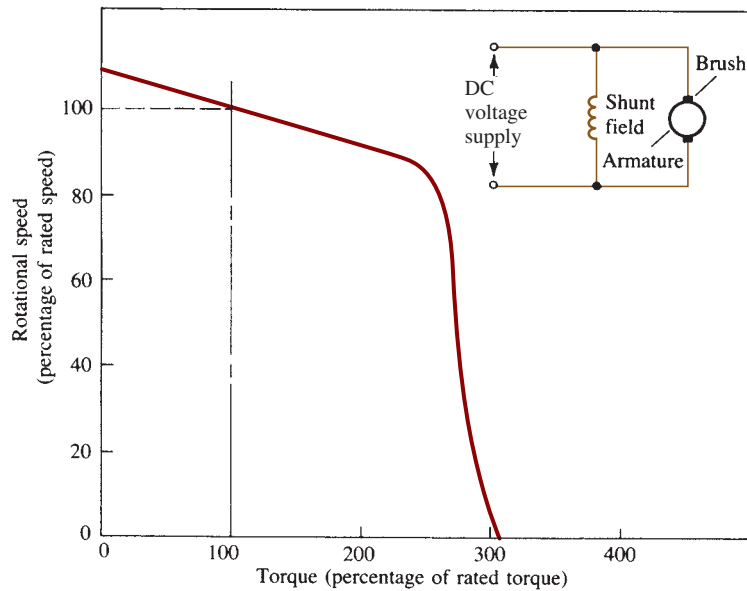


FIGURE 21-28 Example of a performance curve for a shunt-wound DC motor

- Dynamic braking can be obtained by reversing the polarity of the power while the motor is rotating. The reversed effective torque slows the motor without the need for mechanical braking.
- DC motors typically have quick response, accelerating quickly when voltage is changed, because they have a small rotor diameter, giving them a high ratio of torque to inertia.

DC motors have electric windings in the rotor, and each coil has two connections to the commutator on the shaft. The commutator is a series of copper segments through which the electric power is transferred to the rotor. The current path from the stationary part of the motor to the commutator is through a pair of brushes, usually made of carbon, which are held against the commutator by light coil or leaf springs. Maintenance of the brushes is one of the disadvantages of DC motors.

DC Motor Types

Four commonly used DC motor types are the *shunt-wound*, *series-wound*, *compound-wound*, and *permanent magnet* motors. They are described in terms of their speed/torque curves in a manner similar to that used for AC motors. One difference here is that the speed axis is expressed in percentage of *full-load rated speed*, rather than a percentage of synchronous speed, as that term does not apply to DC motors.

Shunt-Wound DC Motor. The electromagnetic field is connected in parallel with the rotating armature, as sketched in Figure 21-28. The speed/torque curve shows relatively good speed regulation up to approximately two times full-load torque, with a rapid drop in speed after that point. The speed at no load is only slightly higher than the full-load speed. Contrast this with the

series-wound motor described next. Shunt-wound motors are used mainly for small fans and blowers.

Series-Wound DC Motor. The electromagnetic field is connected in series with the rotating armature, as shown in Figure 21-29. The speed/torque curve is steep, giving the motor a soft performance that is desirable in cranes, hoists, and traction drives for vehicles. The starting torque is very high, as much as 800% of full-load rated torque. A major difficulty, however, with series-wound motors is that the no-load speed is theoretically unlimited. The motor could reach a dangerous speed if the load were to be accidentally disconnected. Safety devices, such as overspeed detectors that shut off the power, should be used.

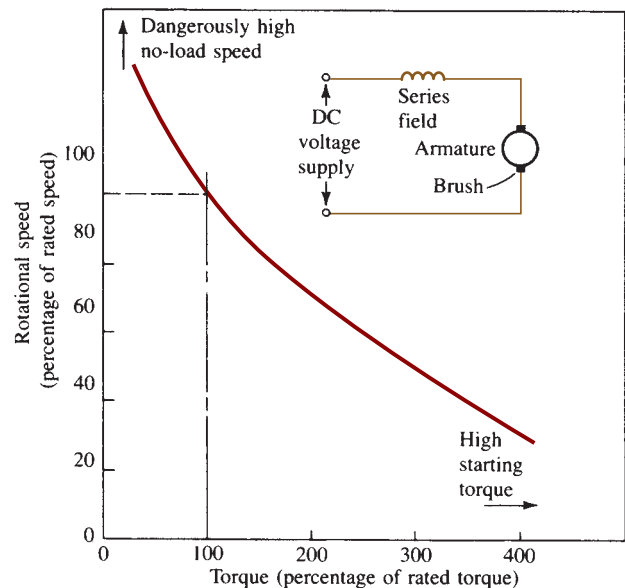


FIGURE 21-29 Example of a performance curve for a series-wound DC motor

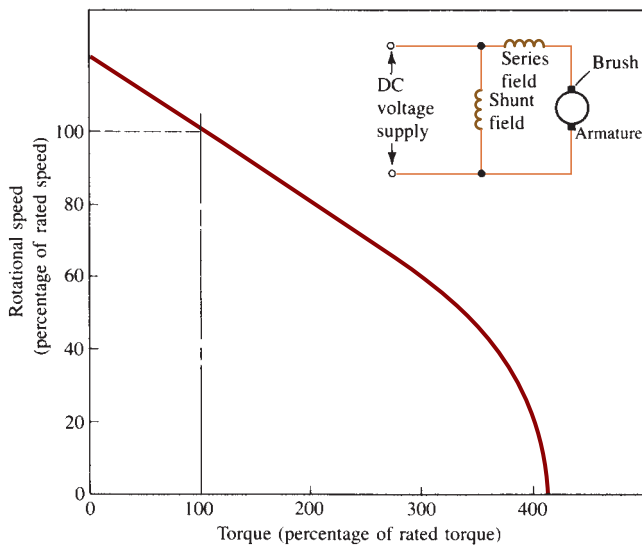


FIGURE 21-30 Example of a performance curve for a compound-wound DC motor

Compound-Wound DC Motor. The compound-wound DC motor employs both a series field and a shunt field, as sketched in Figure 21-30. It has a performance somewhat between that of the series-wound and the shunt-wound motors. It has fairly high starting torque and a soft speed characteristic, but it has an inherently controlled no-load speed. This makes it good for cranes, which may suddenly lose their loads. The motor would run slowly when heavily loaded for safety and control, and fast when lightly loaded to improve productivity.

Permanent Magnet DC Motor. Instead of using electromagnets, the permanent magnet DC motor uses permanent magnets to provide the field for the armature. The direct current passes through the armature, as shown in Figure 21-31. The field is nearly constant at all times and results in a linear speed/torque curve. Current draw also varies linearly with torque. Applications include fans and blowers to cool electronics packages in aircraft, small actuators for control in aircraft, automotive power assists for windows and seats, and fans in automobiles for heating and air conditioning. These motors frequently have built-in gear-type speed reducers to produce a low-speed, high-torque output.

21-12 DC MOTOR CONTROL

Starting of DC motors presents essentially the same problems as discussed for AC motors in terms of limiting the starting current and the provision of switching devices and holding relays of sufficient capacity to handle the operating loads. The situation is made somewhat more severe, however, by the presence of the commutators in the rotor circuit which are more sensitive to overcurrent.

Speed control is provided by variation of the resistance in the lines containing the armature or the field of the motor. The details depend on whether the motor is a

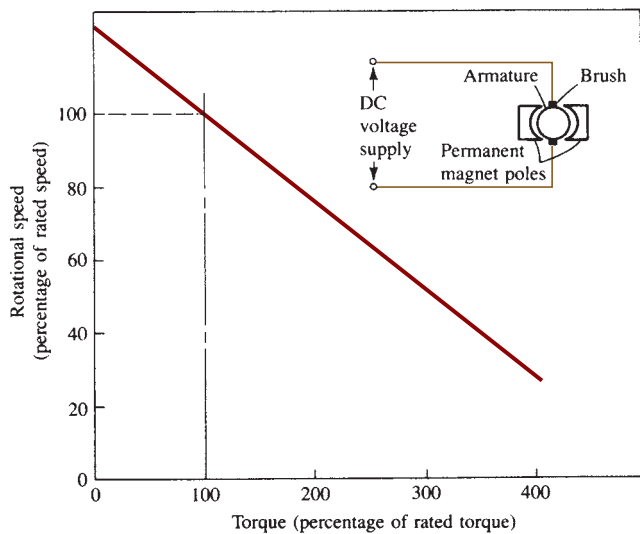


FIGURE 21-31 Example of a performance curve for a permanent magnet DC motor

series, shunt, or compound type. The variable-resistance device, sometimes called a *rheostat*, can provide either stepwise variation in resistance or continuously varying resistance. Figure 21-32 shows the schematic diagrams for several types of DC motor speed controls. See Internet sites 1 and 5-9 and References 2-9.

21-13 OTHER TYPES OF MOTORS

Torque Motors

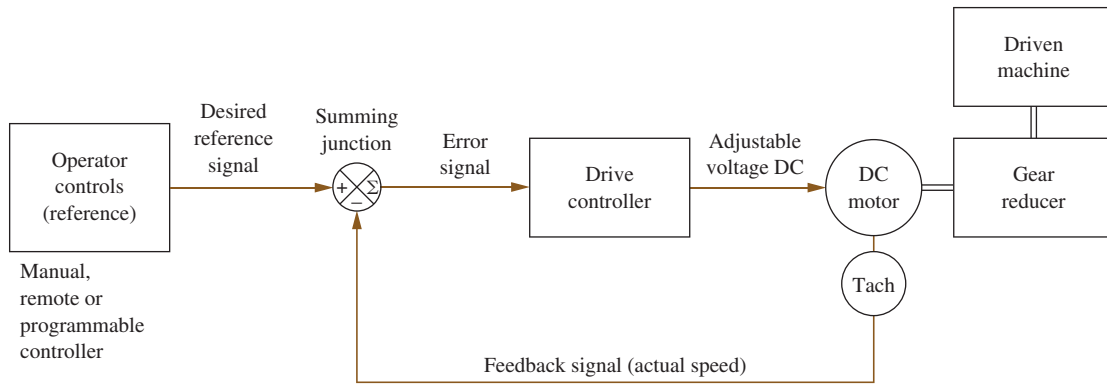
As the name implies, *torque motors* are selected for their ability to exert a certain torque rather than for a rated power. Frequently this type of motor is operated at a stalled condition to maintain a set tension on a load. The continuous operation at slow speed or at zero speed causes heat generation to be a potential problem. In severe cases, external cooling fans may be required.

By special design, several of the AC and DC motor types discussed elsewhere in this chapter can be used as torque motors.

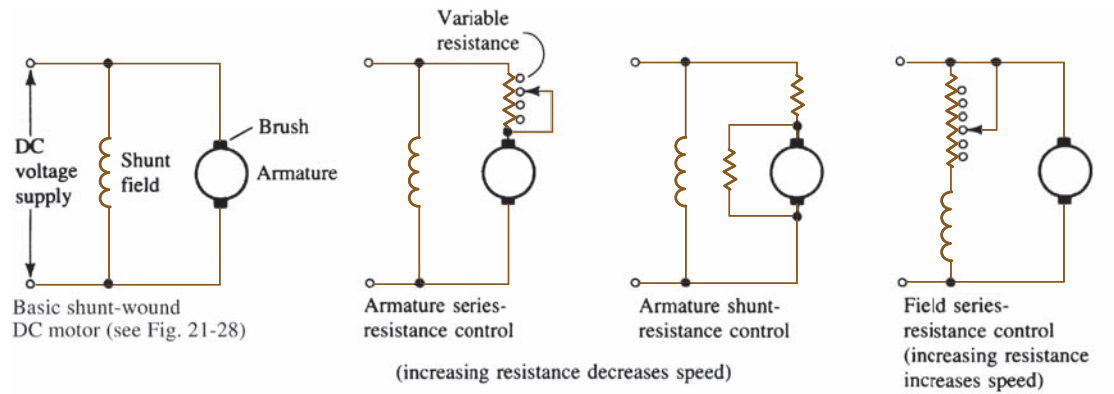
Servomotors

Either AC or DC *servomotors* are available to provide automatic control of position or speed of a mechanism in response to a control signal. Such motors are used in aircraft actuators, instruments, computer printers, and machine tools. Most have rapid response characteristics because of the low inertia of the rotating components and the relatively high torque exerted by the motor. See Internet sites 1, 7-9, and 15.

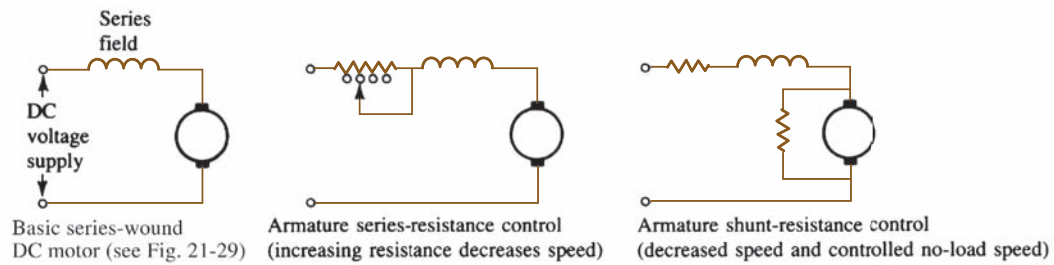
Figure 21-33 shows a schematic diagram of a servomotor controller system. Three control loops are shown: (1) the position loop, (2) the velocity loop, and (3) the current loop. Speed control is effected by sensing the motor speed with a tachometer and feeding the signal



(a) Schematic diagram of DC motor control



(b) Shunt-wound DC motor control



(c) Series-wound DC motor control

FIGURE 21-32 Examples of control circuits for a variety of styles of DC Motors

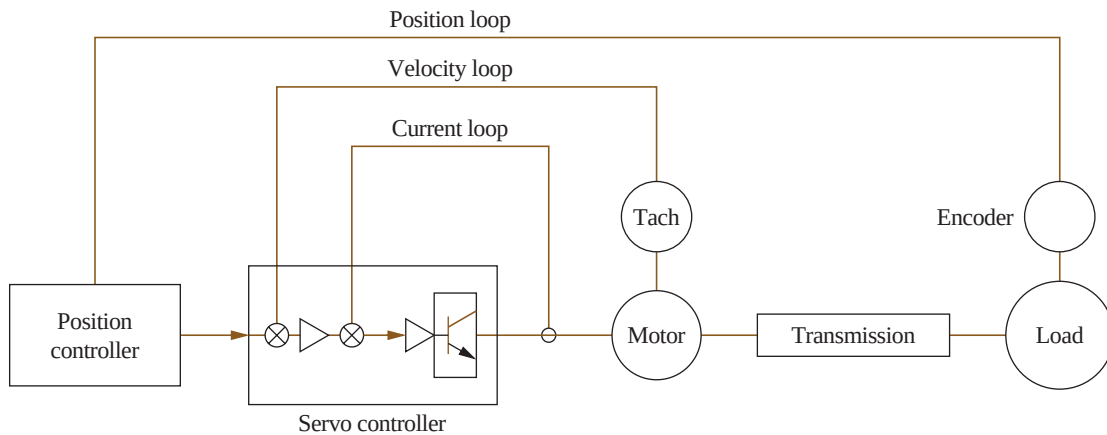


FIGURE 21-33 Example of a control system for a typical servomotor. See Internet sites 7, 9, and 15

back through the velocity loop to the controller. Position is sensed by an optical encoder or a similar device on the driven load, with the signal fed back through the position loop to the controller. The controller sums the inputs, compares them with the desired value set by the control program, and generates a signal to control the motor. Thus, the system is a closed-loop servo control. Typical uses for this type of control would be for numerical control machine tools, special-purpose assembly machines, and aircraft control surface actuators.

Stepper Motors

A stream of electronic pulses is delivered to a *stepper motor*, which then responds with a fixed rotation (step) for each pulse. Thus, a very precise angular position can be obtained by counting and controlling the number of pulses delivered to the motor. Several step angles are available in commercially provided motors, such as 1.8°, 3.6°, 7.5°, 15°, 30°, 45°, and 90°. When the pulses are stopped, the motor stops automatically and is held in position. Because many of these motors are connected through a gear-type speed reducer to the load, very precise positioning is possible to a small fraction of a step. Also, the reducer provides a torque increase. See Internet sites 3 and 14.

Brushless Motors

The typical DC motor requires brushes to make contact with the rotating commutator on the shaft of the motor. This is a major failure mode of such motors. In the *brushless DC motor*, the switching of the rotor coils is accomplished by solid-state electronic devices, resulting in very long life. The emission of electromagnetic interference is likewise reduced compared with brush-type DC motors.

Printed Circuit Motors

The rotor of the *printed circuit motor* is a flat disc that operates between two permanent magnets. The resulting design has a relatively large diameter and small axial length; sometimes it is called a *pancake motor*. The rotor has a very low inertia, so high acceleration rates are possible.

Linear Motors

Linear motors are electrically similar to rotary motors, except the components, stator and rotor, are laid flat instead of being formed into a cylindrical shape. Types include brush-type DC motors, brushless DC motors, stepping motors, and the single-phase type of motor. Capacity is measured in terms of the force that the motor can exert, ranging typically from a few pounds to 2500 lb. Speeds range from 40 to 100 in/s (1.02 to 2.54 m/s). See Internet site 9.

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5. Hughes, Austin, and Bill Drury. *Electric Motors and Drives: Fundamentals, Types and Applications*. 4th ed. Amsterdam, Holland: Newnes-Elsevier Science & Technology Books, 2013.
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INTERNET SITES FOR ELECTRIC MOTORS AND CONTROLS

1. **Baldor-Reliance**. Manufacturer of AC and DC motors, gear motors, servomotors, linear motors, generators, and linear motion products and the associated controls. The site provides a Resource & Support section giving motor performance, applications guidelines, motor construction information, software, and an online catalog.
2. **Regal Beloit EPC, Inc.** Manufacturer of electric motors for driving pumps, conveyors, food processing machinery, machine tools, office equipment, and other commercial and industrial applications. Sizes range from 1/12 to 1250 hp. Brands include Century, Elco, Fasco, and seven others.
3. **OMEGA Engineering, Inc.** Manufacturer of stepper motors and associated controls that can be used for rotary position control, indexing, and linear motion control with a lead screw drive system.
4. **U.S. Electric Motors Company**. Manufacturer of a wide variety of AC and DC electric motors from $\frac{1}{4}$ to 4000 hp for general and specific applications. An online catalog is included. Part of Nidec Motor Corporation.
5. **Schneider Electric**. Manufacturer of AC motor drives, electrical control devices and systems for industry, energy and infrastructure, buildings, residential, and data centers/networks fields. The industry market includes process automation, machine control and monitoring, power supply and distribution, energy monitoring and control, and many others. Some motor control-related products such as starters and contactors are offered under the brand Square D by Schneider Electric.
6. **Eaton/Electrical-USA**. Manufacturer of a wide range of electrical control and power distribution products for industrial, commercial, and residential applications. Select "Products and Solutions" for motor control centers, circuit breakers, power conditioning equipment, switchgear, brakes, and many other products.

7. **Rockwell Automation.** Manufacturer of a wide variety of controls for automation, including motor contactors, AC motor drives, motor control centers, servomotors, programmable logic controllers, sensors, relays, circuit protection devices, network communication, and control systems. Strategic brands include Allen-Bradley®, Rockwell Software®, and Rockwell Automation®.
8. **General Electric Automation & Controls.** Manufacturer of an extensive line of industrial automation systems, motion controls, and process solutions, serving automotive manufacturing, mining & metals, oil & gas, pharmaceuticals, life sciences, power & energy, water & wastewater industries. Includes Alstom-branded products.
9. **Parker Motion Control Systems.** Manufacturer of a wide range of automation devices and systems for industrial applications. Includes the Compumotor line of servomotors and controllers and the Parker Trilogy Systems line of linear electric motors and linear positioners for industrial automation. Part of Parker Hannifin Corporation.
10. **East Penn Manufacturing Co.—DEKA Batteries.** Manufacturer of a wide variety of sizes of lead-acid batteries for power tools, garden equipment, network electric power systems, industrial mobile equipment, golf carts, wheelchairs, railroad diesel engine starting systems, and many others.
11. **Exide Technologies.** Manufacturer of a full line of batteries for automotive, truck, lawn and garden, commercial, marine, industrial forklifts, automated guided vehicles, telecommunications, computer backup and security systems, emergency lighting, and various military equipment.
12. **U.S. Battery.** Manufacturer of 2 V, 6 V, 8 V, and 12 V batteries for golf cars, industrial sweepers and scrubbers, automotive, marine, and multipurpose applications.
13. **Battery Council International.** An organization of manufacturers and users of batteries and battery-powered equipment for training, recommended test procedures, charging, and safety.
14. **Oriental Motor U.S. Corporation.** Manufacturer of rotary and stepper motors with either AC or DC input power and linear actuators for a variety of industrial automation and commercial product applications.
15. **Beckhoff Drive Technology.** Manufacturer of brushless synchronous servomotors for a variety of industrial automation and commercial product applications.
3. Define *duty cycle*.
4. How much variation in voltage will most AC motors tolerate?
5. State the relationship among torque, power, and speed.
6. What does the abbreviation AC stand for?
7. Describe and sketch the form of single-phase AC power.
8. Describe and sketch the form of three-phase AC power.
9. What is the standard frequency of AC power in the United States?
10. What is the standard frequency of AC power in Europe?
11. What type of electrical power is available in a typical American residence?
12. How many conductors are required to carry single-phase power? How many for three-phase power?
13. Assume that you are selecting an electric motor for a machine to be used in an industrial plant. The following types of AC power are available: 120-V, single-phase; 240-V, single-phase; 240-V, three-phase; 480-V, three-phase. In general, for which type of power would you specify your motor?
14. Define *synchronous speed* for an AC motor.
15. Define *full-load speed* for an AC motor.
16. What is the synchronous speed of a four-pole AC motor when running in the United States? In France?
17. A motor nameplate lists the full-load speed to be 3450 rpm. How many poles does the motor have? What would its approximate speed be at zero load?
18. If a four-pole motor operates on 400-Hz AC power, what will its synchronous speed be?
19. If an AC motor is listed as a normal-slip, four-pole/six-pole motor, what will its approximate full-load speeds be?
20. What type of control would you use to make an AC motor run at variable speeds?
21. Describe a C-face motor.
22. Describe a D-flange motor.
23. What does the abbreviation NEMA stand for?
24. Describe a protected motor.
25. Describe a TEFC motor.
26. Describe a TENV motor.
27. What type of motor enclosure would you specify to be used in a plant that manufactures baking flour?
28. What type of motor would you specify for a meat grinder if the motor is to be exposed?
29. Figure P21–29 shows a machine that is to be driven by a 5-hp, protected, foot-mounted AC motor having a 184T frame. The motor is to be aligned with the shaft of the driven machine. Make a complete dimensioned drawing, showing standard side and top views of the machine and the motor. Design a suitable mounting base for the motor showing the motor mounting holes.
30. Define *locked-rotor torque*. What is another term used for this parameter?

PROBLEMS

1. List six items that must be specified for electric motors.
2. List eight factors to be considered in the selection of an electric motor.

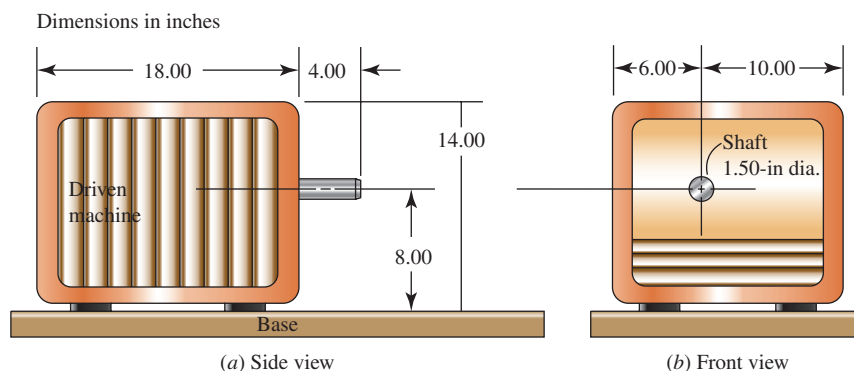


FIGURE P21–29

31. What is meant if one motor has a poorer speed regulation than another?
32. Define *breakdown torque*.
33. Name the four most common types of single-phase AC motors.
34. Refer to the AC motor performance curve in Figure P21–34.
 - a. What type of motor is the curve likely to represent?
 - b. If the motor is a six-pole type, rated at 0.75 hp, how much torque can it exert at the rated load?
 - c. How much torque can the motor develop to start a load?
 - d. What is the breakdown torque for the motor?
35. Repeat Parts (b), (c), and (d) of Problem 34 if the motor is a two-pole type, rated at 1.50 kW.
36. A cooling fan for a computer is to operate at 1725 rpm, direct-driven by an electric motor. The speed/torque curve for the fan is shown in Figure P21–36. Specify a suitable motor, giving the type of motor, horsepower, and number of poles.
37. Figure P21–37 shows the speed/torque curve for a household refrigeration compressor, designed to operate at 3450 rpm. Specify a suitable motor, giving the type, power rating in watts, and number of poles.
38. How is the speed adjusted for a wound-rotor motor?
39. What is the full-load speed of a 10-pole synchronous motor?
40. What is meant by the term *pull-out torque*, as applied to a synchronous motor?
41. Discuss the reasons that universal motors are frequently used for handheld tools and small appliances.
42. Why is the adjective *universal* used to describe a universal motor?
43. Name four ways to produce DC power.
44. List 12 common DC voltages.
45. What is an SCR control? For what is it used?
46. If a DC motor drive advertises that it produces *low ripple*, what does the term mean?
47. If you want to use a DC motor in your home, and your home has only standard 115-V AC, single-phase power, what will you need? What type of motor should you get?
48. List seven advantages of DC motors with respect to AC motors.
49. Discuss two disadvantages of DC motors.
50. Name four types of DC motors.
51. What happens to a series-wound DC motor if the load on the motor falls to nearly zero?
52. Assume that a permanent magnet DC motor can exert a torque of 15.0 N·m when operating at 3000 rpm. What torque could it exert at 2200 rpm?
53. List 10 functions of a motor control.
54. What size motor starter is required for a 10-hp, three-phase motor, operating on 220 V?
55. A single-phase, 110-V AC motor has a nameplate rating of 1.00 kW. What size motor starter is required?
56. What does the term *plug stopping* mean, and how is it accomplished?
57. Why is a fuse an inadequate protection device for an industrial motor?
58. What type of motor control enclosure would you specify for use in a car wash?
59. What could you do to the control circuit for a standard series DC motor to give it a controlled no-load speed?

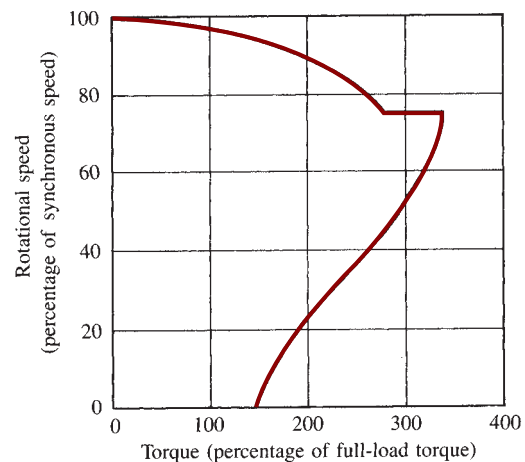


FIGURE P21–34 (Problems 34 and 35)

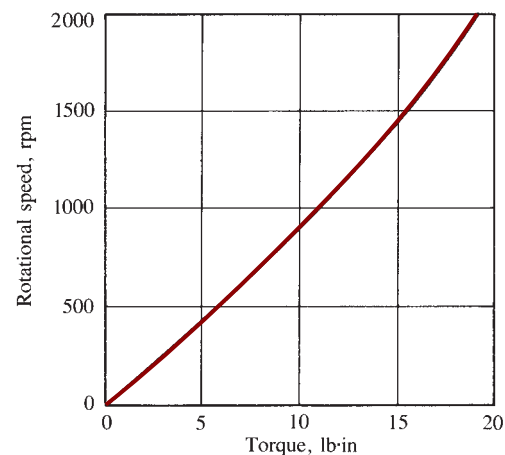


FIGURE P21–36

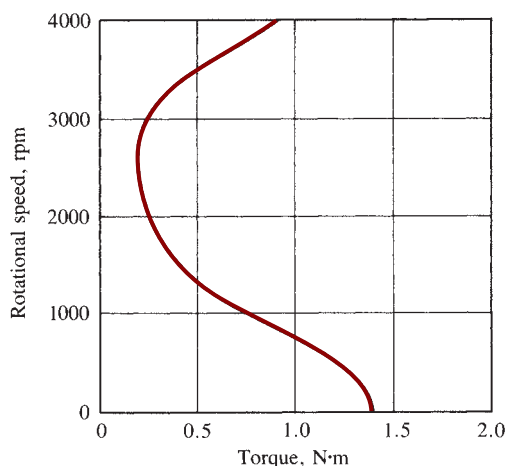


FIGURE P21–37

60. What happens if you connect a resistance in series with the armature of a shunt-wound DC motor?
61. What happens if you connect a resistance in series with the shunt field of a shunt-wound DC motor?

MOTION CONTROL: CLUTCHES AND BRAKES

The Big Picture

You Are the Designer

- 22-1 Objectives of This Chapter
- 22-2 Descriptions of Clutches and Brakes
- 22-3 Types of Friction Clutches and Brakes
- 22-4 Performance Parameters
- 22-5 Time Required to Accelerate or Decelerate a Load
- 22-6 Inertia of a System Referred to the Clutch Shaft Speed
- 22-7 Effective Inertia for Bodies Moving Linearly
- 22-8 Energy Absorption: Heat-Dissipation Requirements
- 22-9 Response Time
- 22-10 Friction Materials and Coefficient of Friction
- 22-11 Plate-Type Clutch or Brake
- 22-12 Caliper Disc Brakes
- 22-13 Cone Clutch or Brake
- 22-14 Drum Brakes
- 22-15 Band Brakes
- 22-16 Other Types of Clutches and Brakes

THE BIG PICTURE

Motion Control: Clutches and Brakes

Discussion Map

- A *brake* is a device used to bring a moving system to rest, to slow its speed, or to control its speed to a certain value under varying conditions.
- A *clutch* is a device used to connect or disconnect a driven component from the prime mover of the system.
- Brakes and clutches either add energy to a moving system or take energy from it. Analyzing the necessary energy changes is an important part of designing effective motion control components.

Discover

Where do you use brakes?

Consider the brakes for a car or a bicycle. Describe the components and the actuation cycle in as much detail as you can. Discuss your findings with your colleagues.

What kinds of equipment other than vehicles use clutches or brakes? Describe some scenarios.

Describe the physics of the operation of a clutch and a brake, considering energy and inertia effects.

This chapter helps you explore all of these issues, and many convenient design and analysis equations are developed. Many kinds of commercially available clutches and brakes are pictured and described.

Machine systems require control whenever the speed or the direction of the motion of one or more components is to be changed. When a device is started initially, it must be accelerated from rest to the operating speed. As an operation is completed, the system must frequently be brought to rest. In continuously operating systems, changing speeds to adjust to different operating conditions is often

necessary. Safety sometimes dictates motion control, as with a load being lowered by a hoist or an elevator.

In this chapter, we will be concerned mostly with control of rotary motion in systems driven by electric motors, engines, turbines, and the like. Ultimately, linear motion may be involved through linkages, conveyors, or other mechanisms.

The machine elements most frequently used for motion control are the clutch and the brake, defined as follows:

- A *clutch* is a device used to connect or disconnect a driven component from the prime mover of the system. For example, in a machine that must cycle frequently, the driving motor is left running continuously, and a clutch is interposed between the motor and the driven machine. Then the clutch is cycled on and off to connect and disconnect the load. This permits the motor to operate at an efficient speed, and it also permits the system to cycle more rapidly because there is no need to accelerate the relatively heavy motor rotor with each cycle.
- A *brake* is a device used to bring a moving system to rest, to slow its speed, or to control its speed to a certain value under varying conditions.

Where do you use brakes? An obvious answer is in a car or a bicycle where safe operation requires fast, smooth stopping when emergency conditions occur or simply when you need to stop at a stop sign or in your driveway. And you do not always need to bring the car or bicycle to a full stop. Slowing down to conform to prevailing traffic flow or to round a curve requires braking to decrease the speed from a higher to a lower speed.

What really happens when you apply the brakes of a bicycle? Can you describe the essential elements of the braking system? With hand brakes, your actuation of the brake lever pulls on a flexible cable that, in turn, pulls on a linkage at the brake assembly mounted above the rim of a wheel. The linkage causes the friction pad to press against the rim. When you pull harder on the lever, a higher force is developed between the pad and the rim. This is called the *normal force*. Recall from physics and statics that a frictional force is created between surfaces moving relative to each other when a normal force presses them together. The frictional force acts in a direction opposite to the relative motion, and thus it tends to slow the motion. With a sufficiently large frictional force applied for a sufficient time, the wheel stops. Notice also that the frictional force acts at a fairly large radius from the center of the wheel. Thus, the force causes a frictional *torque* to be developed, and what is really happening is the deceleration of the angular velocity of the wheel. But since this is directly proportional to the linear speed of the bicycle, you experience the stopping action as a linear deceleration.

But that is not all! Have you ever felt the brake pad after a hard stop? The fact that it gets warm is an indication that the brake absorbs energy during the stopping action. Where does the energy come from? Can you compute the amount of energy that must be

absorbed? What are the parameters involved in this calculation? Compare the amount of energy that must be absorbed in stopping a bicycle that you are riding with that involved in stopping a large airliner landing at 120 mph with a full load of people and baggage in addition to the huge weight of the airplane itself. Imagine what those brakes look like in comparison with the bicycle brake!

What other kinds of equipment besides transportation vehicles require brakes? Consider elevators, escalators, hoists, and winches, which must stop and hold a load after lifting it. Machine tools, conveyors, and other manufacturing equipment must often be brought to a safe, quick stop.

But that equipment must also be *accelerated* to start a new cycle of operation. How can that be achieved? One way is to start and stop the motor or the engine that drives the equipment. However, that is inconvenient and time-consuming, and it may lead to early failure of the system.

How does the automotive drive system allow you to stop and then continue driving without shutting off the engine? Have you ever driven a standard shift car? A “stick shift?” A *clutch* is used to engage and disengage the drive train from the engine. In cars with automatic transmissions clutches are also built into the transmission.

What other kinds of equipment use clutches? Describe some examples from your personal experience, or create a scenario in which it would be desirable to employ a clutch.

Now, consider what the task of a clutch is. Some parts of a machine are rotating continuously, while others are temporarily stationary. Then you engage the clutch. What happens? Consider the physics of the situation. The stationary parts must be accelerated from zero speed to the desired speed according to the design of the drive system. Inertia must be overcome—sometimes rotational inertia; sometimes linear inertia; and sometimes both kinds in the same machine. How long do you want it to take to accelerate the load to its operating speed? You must realize that a shorter time requires a higher value of accelerating torque to be developed by the clutch, and it increases the technical demands on the system in terms of strength of its components, smoothness, and wear life of the frictional materials that actually accomplish the engagement.

This chapter helps you explore all of these issues, and many convenient design and analysis equations are developed. Several different styles of brakes and clutches are discussed, and photographs or detailed drawings of several commercially available designs are shown. It might be desirable to keep this book handy for future projects so that you can look for the details of different clutch and brake designs.

YOU ARE THE DESIGNER

Your company manufactures conveyor systems for warehouses and truck terminals. The conveyors deliver cartons of materials to any of several stations where trucks are to be loaded. To save energy and to decrease the wear on operating parts of the conveyor system, it is decided to operate only the parts of the system that have a demand to deliver a carton. The system is to be operated automatically through a series of sensors, switches, programmable controllers, and an overall supervisory computer control. Your assignment is to recommend the type and the size of clutch and brake units to start and stop the various conveyors.

Some of the design decisions you will have to make are as follows:

1. How much time should be allowed to get the conveyors up to speed after the initial command to start has been given?
2. How fast should the conveyors come to a stop?
3. How many cycles per hour are expected?
4. How much room is available to install the clutch and brake units?
5. What means are available to actuate the clutch and brake units: electric power, compressed air, hydraulic pressure, or some other means?

6. What general type of clutch and brake should be used?
7. What size and model of commercially available unit should be specified?

Along with these decisions, you will need information about the conveyor system itself, such as the following:

1. How much load will be on the conveyors when they are started and stopped?
2. What is the design of the conveyor system itself, and what are the weights, shapes, and dimensions of its components?
3. How is the conveyor driven: electric motor, hydraulic motor, or another means?
4. Are the products moved all on one level, or are there elevation changes in the system?

The information in this chapter will help you design such a system. ■

22-1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Define the terms *clutch* and *brake*.
2. Differentiate between a clutch and a *clutch coupling*.
3. Describe a fail-safe brake.
4. Describe a clutch-brake module.
5. Specify the required capacity of a clutch or a brake to drive a given system reliably.
6. Compute the time required to accelerate a system or to bring it to a stop with the application of a given torque.
7. Define the inertia of a system in terms of its Wk^2 value.
8. Compute the energy-dissipation requirements of a clutch or a brake.
9. Determine the response time for a clutch-brake system.
10. Describe at least five types of clutches and brakes.
11. Name six actuation means used for clutches and brakes.
12. Perform the design and analysis of plate-type, caliper disc, cone, drum and shoe, and band brakes and clutches.
13. Name nine other types of clutches and brakes.

22-2 DESCRIPTIONS OF CLUTCHES AND BRAKES

Several arrangements of clutches and brakes are shown in Figure 22-1. By convention, the term *clutch* is reserved for the application in which the connection is made to a shaft parallel to the motor shaft, as illustrated in Figure 22-1(a). If the connection is to a shaft in-line with the motor, the term *clutch coupling* is used, as shown in Figure 22-1(b).

Also by convention, a brake [Figure 22-1(c)] is actuated by some overt action: the application of a fluid pressure, the switching on of an electric current, or the moving of a lever by hand. A brake that is spring-applied automatically in the absence of an overt action is called a *fail-safe brake* [Figure 22-1(d)]: When the power goes off, the brake goes on.

When the functions of both a clutch and a brake are required in a system, they are frequently arranged in the same unit, the *clutch-brake module*. When the clutch is activated, the brake is deactivated, and vice versa [see Figure 22-1(e)].

A *slip clutch* is a clutch that, by design, transmits only a limited torque; thus, it will slip at any higher torque. It is used to provide a controlled acceleration to a load that is smooth and that requires a smaller motor power. It is also used as a safety device, protecting expensive or sensitive components in the event that a system is jammed.

Most of the discussion in this chapter will concern clutches and brakes that transmit motion through friction at the interface between two rotating parts moving at different speeds. Other types are discussed briefly in the last section.

References 1-7 and Internet sites 1-9 provide additional information on the design, selection, and application of motion control systems, clutches, and brakes.

22-3 TYPES OF FRICTION CLUTCHES AND BRAKES

Clutches and brakes that use friction surfaces as the means of transmitting the torque to start or stop a mechanism can be classified according to the general geometry of the friction surfaces and by the means used to actuate them. In some cases, the same basic geometry can be used as either a clutch or a brake by selectively attaching the friction elements to the driver, the driven machine, or the stationary frame of the machine.

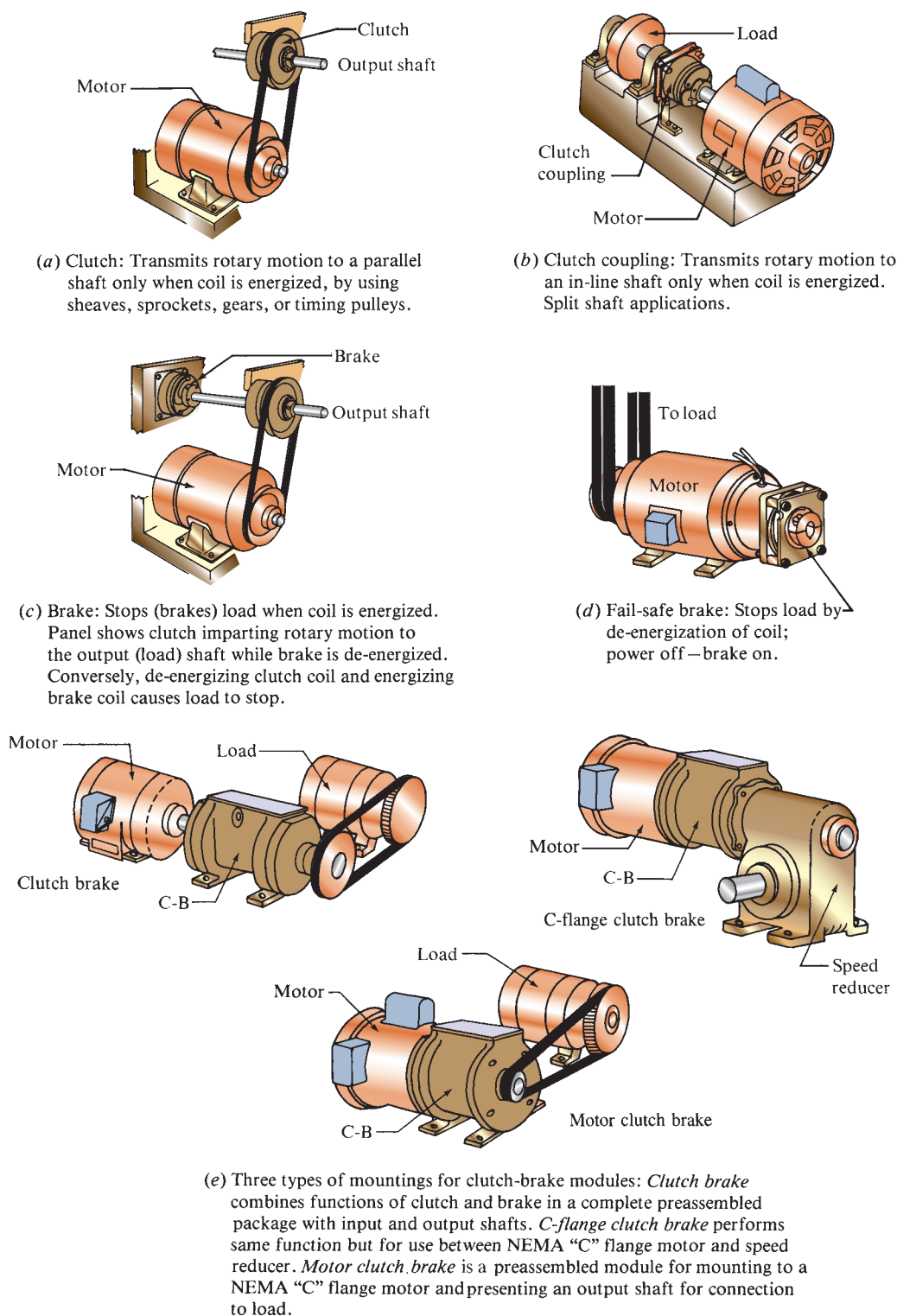


FIGURE 22-1 Typical applications of clutches and brakes

The following types of clutches and brakes are sketched in Figure 22-2:

- (a) **Plate clutch or brake:** Each friction surface is in the shape of an annulus on a flat plate. One or more friction plates move axially to contact a mating smooth plate, usually made of steel, to which the friction torque is transmitted.

- (b) **Caliper disc brake:** A disc-shaped rotor is attached to the machine to be controlled. Friction pads covering only a small portion of the disc are contained in a fixed assembly called a *caliper* and are forced against the disc by air pressure or hydraulic pressure.
- (c) **Cone clutch or brake:** A cone clutch or brake is similar to a plate clutch or brake except that the mating

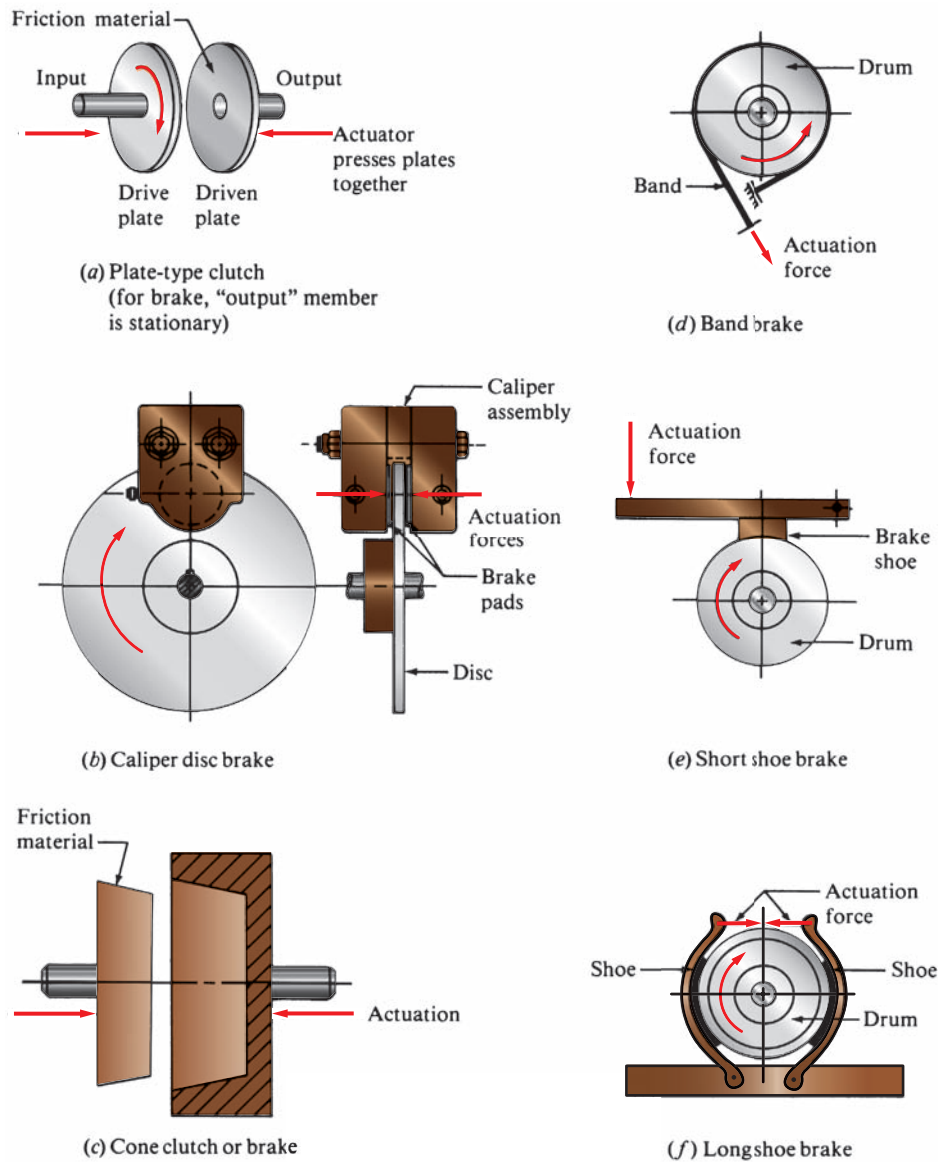


FIGURE 22-2 Types of friction clutches and brakes

surfaces are on a portion of a cone instead of on a flat plate.

- (d) **Band brake:** Used only as a brake, the friction material is on a flexible band that nearly surrounds a cylindrical drum attached to the machine to be controlled. When braking is desired, the band is tightened on the drum, exerting a tangential force to stop the load.
- (e) and (f) **Block or shoe brake:** Curved, rigid pads faced with the friction material are forced against the surface of a drum, from either the outside or the inside, exerting a tangential force to stop the load. Some brake shoes are short and some are long as described later in this chapter.

Actuation

The following are the means used to actuate clutches or brakes. Each may be applied to several of the types described. Figures 22-3 through 22-9 show a variety of commercially available designs.

Manual. The operator provides the force, usually through a lever arrangement to achieve force multiplication. Figure 22-3 shows a power take-off for agricultural or construction equipment where a manually operated clutch engages and disengages the driven shaft from the engine. The driven shaft drives accessory equipment such as a fertilizer spreader, concrete mixer, or wood chipper.

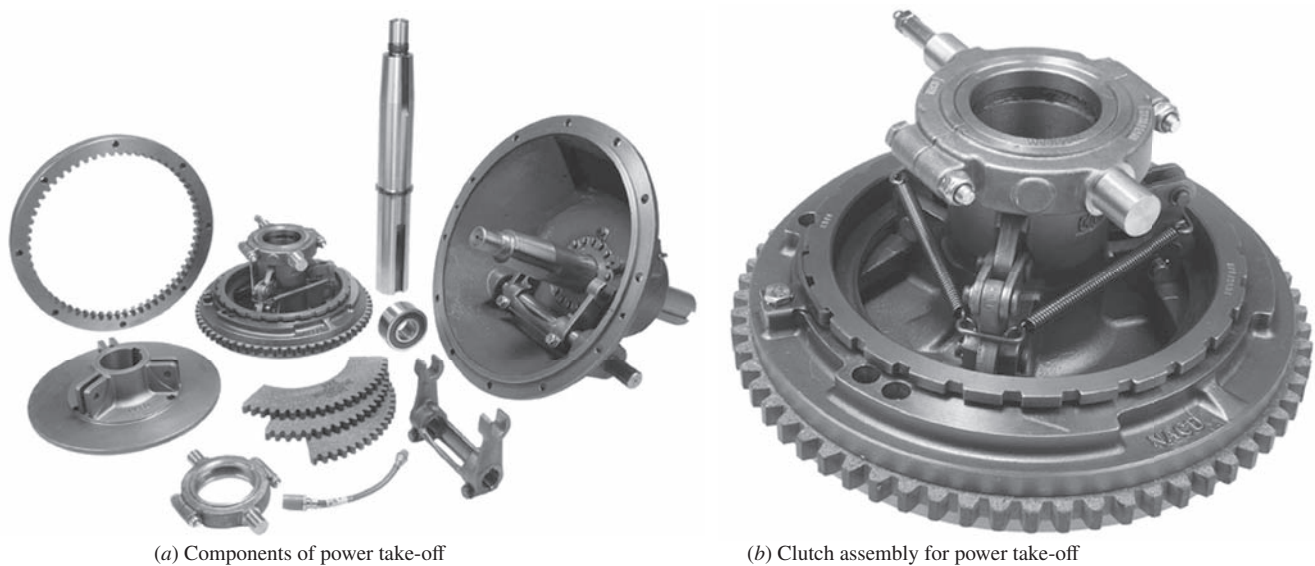


FIGURE 22-3 Power take-off with manual clutch (North American Clutch & Driveline, Inc., Machesney Park, IL)

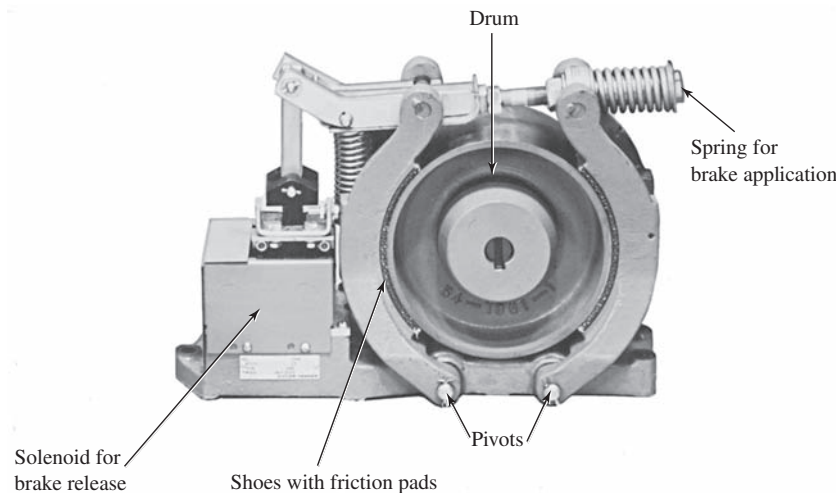


FIGURE 22-4 Spring-applied, electrically released, long shoe brake
(Reprinted with permission and courtesy of Eaton Corporation.)

Spring-applied. Sometimes called a *fail-safe* design when applied to a brake, the brake is applied automatically by springs unless there is some opposing force present. Thus, if power fails, or if air pressure or hydraulic pressure is lost, or if the operator is unable to perform the function, the springs apply the brake and stop the load. The concept may also be used to engage or to disengage a clutch.

Centrifugal. A centrifugal clutch is sometimes employed to permit the driving system to accelerate without a connected load. Then, at a preselected speed, centrifugal force moves the clutch elements into contact

to connect the load. As the system slows, the load will be automatically disconnected.

Pneumatic. Compressed air is introduced into a cylinder or some other chamber. The force produced by the pressure on a piston or a diaphragm brings the friction surfaces into contact with the members connected to the load.

Hydraulic. Hydraulic brakes are similar to the pneumatic type, except that they use oil hydraulic fluids instead of air. The hydraulic actuator is usually applied where high actuation forces are required.

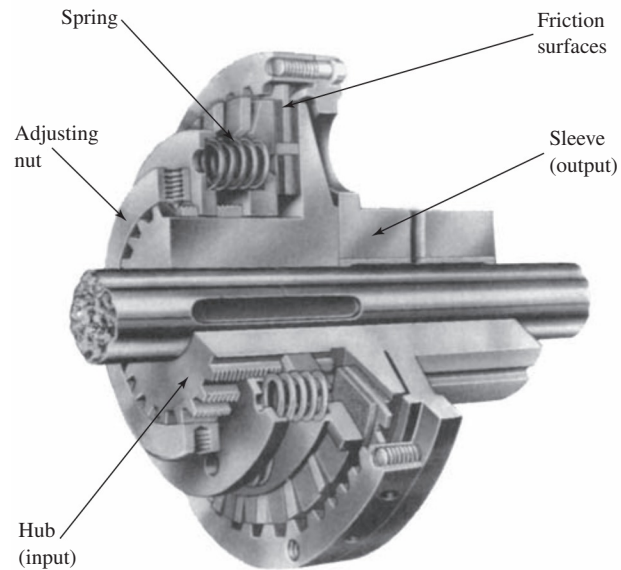
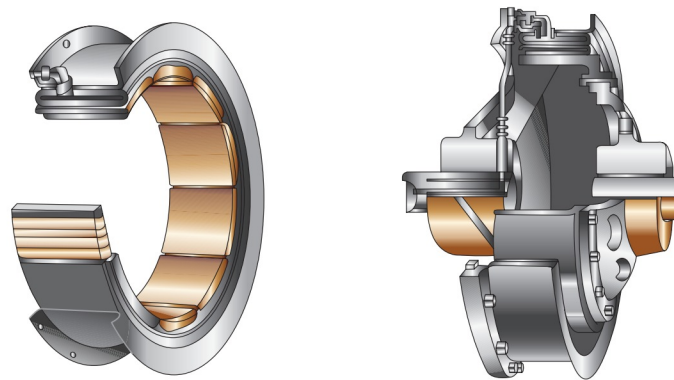
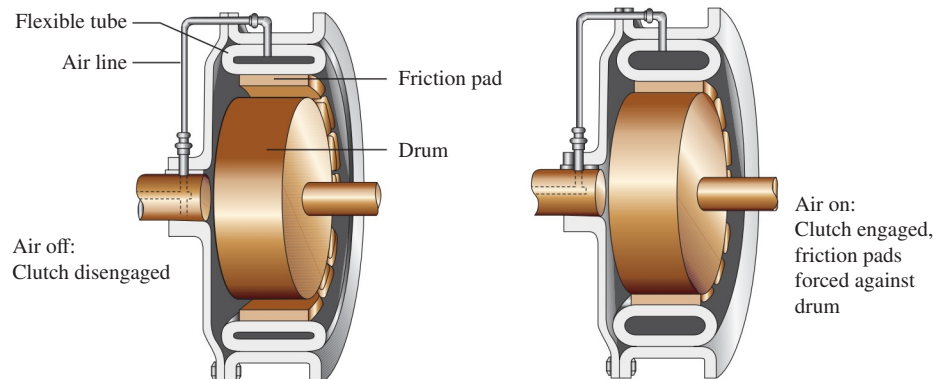


FIGURE 22-5 Slip clutch. Springs apply normal pressure on the friction plates. The spring force is adjustable to vary the torque level at which clutch will slip



(a) Details of clutch design



(b) Clutch activation cycle

FIGURE 22-6 Air-actuated clutch or brake (Eaton Corp., Airflex Division, Cleveland, OH)

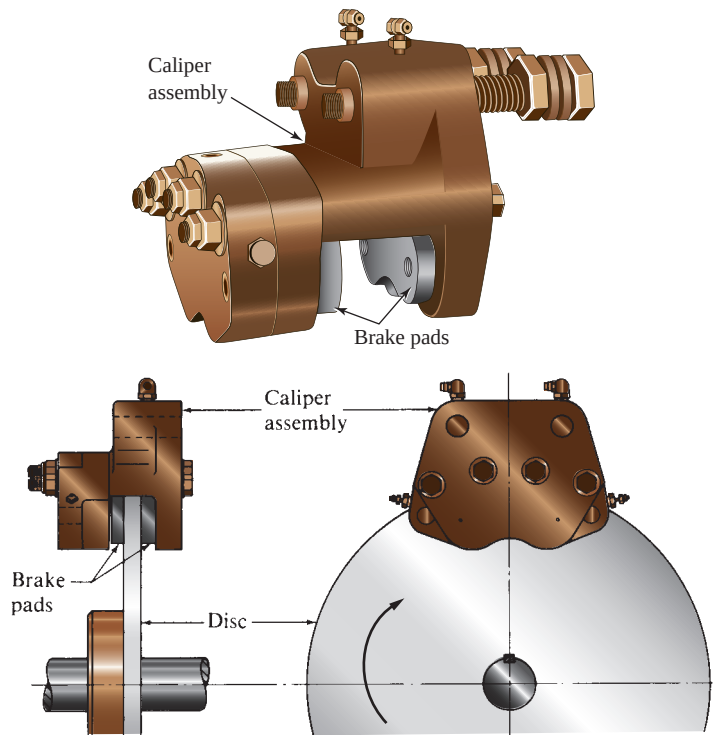
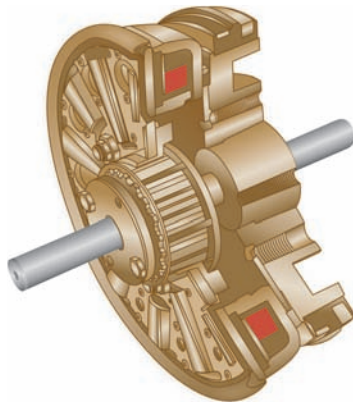
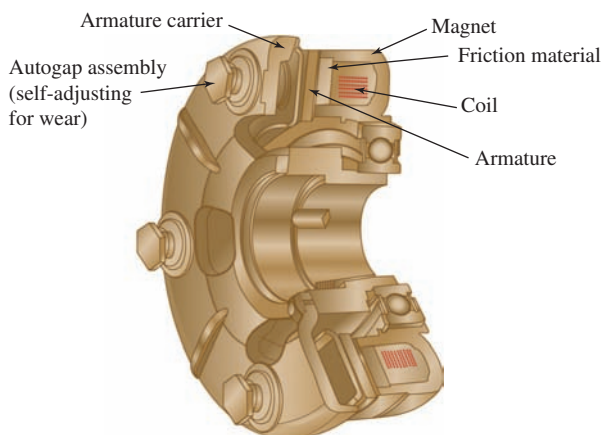


FIGURE 22-7 Hydraulically actuated disc brake assembly



(a) Cutaway view of complete assembly



(b) Components of electromagnet

FIGURE 22-8 Electrically actuated plate-type clutch or brake

Electromagnetic. An electric current is applied to a coil, creating an electromagnetic flux. The magnetic force then attracts an armature attached to the machine that is to be controlled. The armature is usually of the plate type.

22-4 PERFORMANCE PARAMETERS

The principles of physics tell us that whenever the speed or the direction of the motion of a body is changed, there must be a force exerted on the body. If the body is in rotation, a torque must be applied to the system to speed it up or to slow it down. When a change in speed occurs, a corresponding change in the kinetic energy of the system occurs. Thus, motion control inherently involves the control of energy, either adding energy to accelerate a system or absorbing energy to decelerate it.

The parameters involved in the rating of clutches and brakes are as follows:

1. Torque required to accelerate or decelerate the system
2. Time required to accomplish the speed change
3. The cycling rate: number of on/off cycles per unit time
4. The inertia of the rotating or translating parts
5. The environment of the system: temperature, cooling effects, and so on

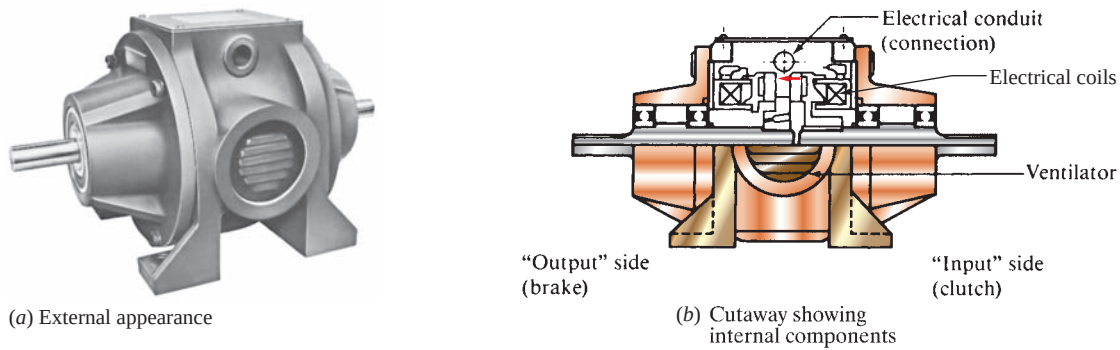


FIGURE 22-9 Electrically operated clutch-brake module

6. Energy-dissipation capability of the clutch or the brake
7. Physical size and configuration
8. Actuation means
9. Life and reliability of the system
10. Cost and availability

Two basic methods are used to determine the torque capacity required of a clutch or a brake. One relates the capacity to the power of the motor driving the system. Recall that, in general, power = torque $\times$ rotational speed ($P = Tn$). The required torque capacity is then usually expressed in the form

⇒ **Required Torque Capacity of Clutch or Brake**

$$T = \frac{CPK}{n} \quad (22-1)$$

where C = conversion factor for units

K = service factor based on the application

More will be said about these later.

Note that the torque required is inversely proportional to the rotational speed. For this reason, it is advisable to locate the clutch or the brake on the highest-speed shaft in the system so that the required torque is a minimum. The size, cost, and response time are all typically lower when the torque is lower. One disadvantage is that the shaft to be accelerated or decelerated must undergo a larger change in speed, and the amount of slipping may be greater. This effect may generate more frictional heat, leading to thermal problems. However, it is offset by the increased cooling effect because of the faster motion of the clutch or brake parts.

The value for the K factor in the torque equation is largely a design decision. Some of the typical guidelines follow:

1. For brakes under average conditions, use $K = 1.0$.
2. For clutches in light duty where the output shaft does not assume its normal load until after it is up to speed, use $K = 1.5$.

3. For clutches in heavy duty where large attached loads must be accelerated, use $K = 3.0$.
4. For clutches in systems having varying loads, use a K factor at least equal to the factor by which the motor breakdown torque exceeds the full-load torque. This was discussed in Chapter 21, but for a typical industrial motor (design B), use $K = 2.75$. For a high starting torque motor (design C or capacitor-start motor), $K = 4.0$ may be required. This ensures that the clutch will be able to transmit at least as much torque as the motor develops and that it will not slip after getting up to speed.
5. For clutches in systems driven by gasoline engines, diesel engines, or other prime movers, consider the peak torque capability of the driver; $K = 5.0$ may be required.

The following list relates the value for C to typically used units for torque, power, and rotational speed. For example, if power is in hp and speed is in rpm, then to obtain torque in lb $\cdot$ ft, use $T = 5252(P/n)$.

Torque	Power	Speed	C
lb $\cdot$ ft	hp	rpm	5252
lb $\cdot$ in	hp	rpm	63 025
N $\cdot$ m	W	rad/s	1
N $\cdot$ m	W	rpm	9.549
N $\cdot$ m	kW	rpm	9549

Although the torque computation method of Equation (22-1) will produce generally satisfactory performance in typical applications, it does not provide a means of estimating the actual time required to accelerate the load with a clutch or to decelerate the load with a brake. The method described next should be used for systems with large inertia such as conveyors or presses having flywheels.

22-5 TIME REQUIRED TO ACCELERATE OR DECELERATE A LOAD

This section uses the acceleration of a load in its examples, wherein a torque must be applied to the load to bring it to a desired rotational speed in a given amount of time. The same principles can be applied in braking situations in which the braking torque either slows or stops a rotating body.

The basic principle involved is taken from dynamics:

$$T = I\alpha$$

where I = mass moment of inertia of the components being accelerated

α (alpha) = angular acceleration—that is, the rate of change of angular velocity

The usual objective of such an analysis is to determine the torque required to produce a change in rotational speed, Δn , of a system in a given amount of time, t . But $\Delta n/t = \alpha$. Also, it is more convenient to express the mass moment of inertia in terms of the *radius of gyration*, k . By definition,

$$k = \sqrt{I/m} \quad \text{or} \quad k^2 = I/m$$

where m = mass

$$m = W/g$$

Then

$$I = mk^2 = Wk^2/g$$

The equation for torque then becomes

✦ Torque Required to Accelerate an Inertia Load

$$T = I\alpha = \frac{Wk^2}{g} \frac{(\Delta n)}{t} \quad (22-2)$$

The term Wk^2 is frequently called simply the *inertia* of the load, although that designation is not strictly correct. A large proportion of the components in a machine system to be accelerated are in the form of cylinders. Figure 22-10 gives the relationships for the radius of gyration and Wk^2 for hollow cylinders. Solid cylinders are simply a special case with an inside radius of zero. We can analyze more complex objects by considering them to be made from a set of simpler cylinders. Example Problem 22-1 illustrates the process.

Now the torque required to accelerate the pulley can be computed. Equation (22-2) can be put into a more convenient form by noting that T is usually expressed in $\text{lb} \cdot \text{ft}$, Wk^2 in $\text{lb} \cdot \text{ft}^2$, n in rpm, and t in s. Using $g = 32.2 \text{ ft/s}^2$ and converting units gives

$$T = \frac{Wk^2(\Delta n)}{308t} \text{ lb} \cdot \text{ft} \quad (22-3)$$

Radius of gyration:

$$k^2 = \frac{1}{2} (R_1^2 + R_2^2)$$

Volume:

$$V = \pi(R_1^2 - R_2^2)L$$

Weight:

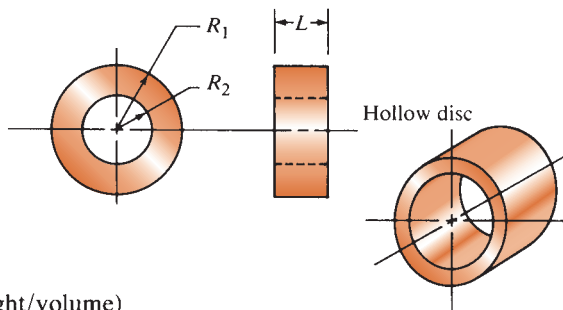
$$W = \delta_w V$$

$$\delta_w = \text{Weight density (weight/volume)}$$

Inertia (Wk^2):

$$Wk^2 = \delta_w V k^2 = \delta_w \pi (R_1^2 - R_2^2) L (R_1^2 + R_2^2) / 2$$

$$Wk^2 = \frac{\pi \delta_w L}{2} (R_1^4 - R_2^4)$$



Typical units: L, R_1, R_2 in inches

$$\delta_w \text{ in lb/in}^3$$

$$Wk^2 \text{ in lb} \cdot \text{ft}^2$$

$$Wk^2 = \frac{\pi}{2} \times \delta_w \frac{\text{lb}}{\text{in}^3} \times L(\text{in}) \times (R_1^4 - R_2^4) \text{ in}^4 \times \frac{1 \text{ ft}^2}{144 \text{ in}^2}$$

$$Wk^2 = \frac{\delta_w L (R_1^4 - R_2^4)}{91.67} \text{ lb} \cdot \text{ft}^2$$

Special case for steel: $\delta_w = 0.283 \text{ lb/in}^3$

$$Wk^2 = \frac{L(R_1^4 - R_2^4)}{323.9} \text{ lb} \cdot \text{ft}^2$$

FIGURE 22-10 Inertia properties of a hollow cylinder

Example Problem 22-1

Compute the value of Wk^2 for the steel flat-belt pulley for which the cross section is shown in Figure 22-11.

Solution

The pulley can be considered to be made up of three components, each of which is a hollow cylinder. The Wk^2 for the total pulley is the sum of that for each component. Use the equation for a steel cylinder from Figure 22-10.

Part 1. Using the formula from Figure 22-10 for a steel cylinder, we have

$$Wk^2 = \frac{(R_1^4 - R_2^4)(L)}{323.9} \text{ lb} \cdot \text{ft}^2 = \frac{[(10.0)^4 - (9.0)^4](6.0)}{323.9}$$

$$Wk^2 = 63.70 \text{ lb} \cdot \text{ft}^2$$

Part 2.

$$Wk^2 = \frac{[(9.0)^4 - (3.0)^4](0.75)}{323.9} = 15.00 \text{ lb} \cdot \text{ft}^2$$

Part 3.

$$Wk^2 = \frac{[(3.0)^4 - (1.5)^4](4.0)}{323.9} = 0.94 \text{ lb} \cdot \text{ft}^2$$

$$\text{Total } Wk^2 = 63.70 + 15.00 + 0.94 = 79.64 \text{ lb} \cdot \text{ft}^2$$

Example Problem 22-2

Compute the torque that a clutch must transmit to accelerate the pulley of Figure 22-11 from rest to 550 rpm in 2.50 s. From Example Problem 22-1, $Wk^2 = 79.64 \text{ lb} \cdot \text{ft}^2$.

Solution

Use Equation (22-3):

$$T = \frac{Wk^2(\Delta n)}{308t} \text{ lb} \cdot \text{ft} = \frac{(79.64)(550)}{308(2.5)} = 56.9 \text{ lb} \cdot \text{ft}$$

In summary, if a clutch that is capable of exerting at least 56.9 lb·ft of torque engages a shaft carrying the pulley shown in Figure 22-11, the pulley can be accelerated from rest to 550 rpm in 2.50 s or less.

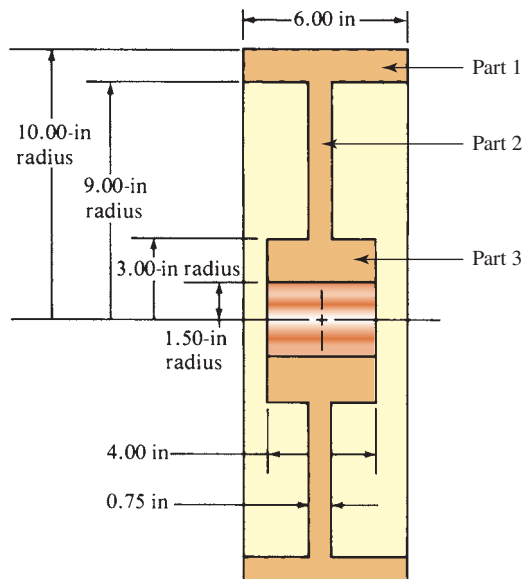


FIGURE 22-11 Steel flat-belt pulley

22-6 INERTIA OF A SYSTEM REFERRED TO THE CLUTCH SHAFT SPEED

In many practical machine systems, there are several elements on several shafts, operating at differing speeds. It is required that the effective inertia of the entire system *as it affects the clutch* be determined. The effective inertia of a connected load operating at a rotational speed different from that of the clutch is

proportional to the square of the ratio of the speeds. That is,

⇒ **Effective Inertia**

$$Wk_e^2 = Wk^2 \left(\frac{n}{n_c} \right)^2 \quad (22-4)$$

where n = rotational speed of the load of interest
 n_c = rotational speed of the clutch

Example Problem 22-3

Compute the total inertia of the system in Figure 22-12 as seen by the clutch. Then compute the time required to accelerate the system from rest to a motor speed of 550 rpm if the clutch exerts a torque of 240 lb·ft. The Wk^2 for the armature of the clutch, which must also be accelerated, is 0.22 lb·ft², including the 1.25-in shaft.

Solution

The clutch and gear A will be rotating at 550 rpm, but because of the gear reduction, gear B, its shaft, and the pulley rotate at

$$n_2 = 550 \text{ rpm} (24/66) = 200 \text{ rpm}$$

Now compute the inertia for each element referred to the clutch speed. Assume that the gears are cylinders having outside diameters equal to the pitch diameter of the gear and inside diameters equal to the shaft diameter. The equation in Figure 22-10 for a steel cylinder will be used to compute Wk^2 .

Gear A

$$Wk^2 = [(2.00)^4 - (0.625)^4](2.50)/323.9 = 0.122 \text{ lb} \cdot \text{ft}^2$$

Gear B

$$Wk^2 = [(5.50)^4 - (1.50)^4](2.50)/323.9 = 7.02 \text{ lb} \cdot \text{ft}^2$$

But, because of the speed difference, the effective inertia is

$$Wk_e^2 = 7.02(200/550)^2 = 0.93 \text{ lb} \cdot \text{ft}^2$$

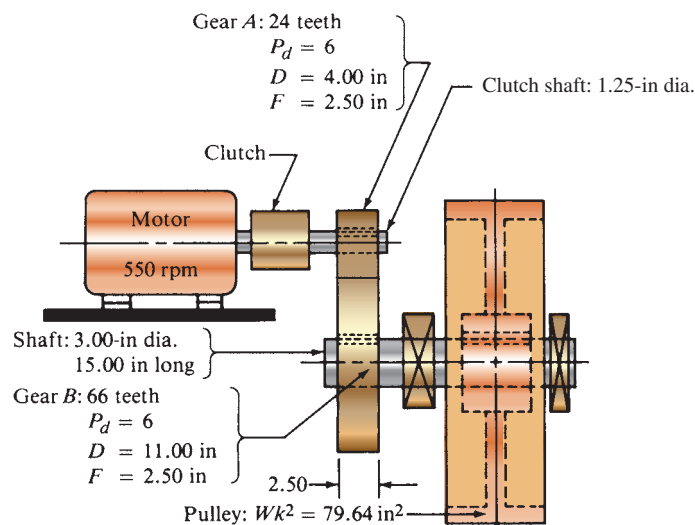


FIGURE 22-12 Given system for Example Problem 22-3.

Pulley

From Example Problem 22–1, $Wk^2 = 79.64 \text{ lb} \cdot \text{ft}^2$. The effective inertia is

$$Wk_e^2 = 79.64(200/550)^2 = 10.53 \text{ lb} \cdot \text{ft}^2$$

Shaft

$$Wk^2 = (1.50)^4(15.0)/323.9 = 0.234 \text{ lb} \cdot \text{ft}^2$$

The effective inertia is

$$Wk_e^2 = 0.234(200/550)^2 = 0.03 \text{ lb} \cdot \text{ft}^2$$

The total effective inertia as seen by the clutch is

$$Wk_e^2 = 0.22 + 0.12 + 0.93 + 10.53 + 0.03 = 11.83 \text{ lb} \cdot \text{ft}^2$$

Solving Equation (22–3) for the time gives

$$t = \frac{Wk_e^2(\Delta n)}{308T} = \frac{(11.83)(550)}{308(240)} = 0.088 \text{ s}$$

22-7 EFFECTIVE INERTIA FOR BODIES MOVING LINEARLY

To this point, we have dealt only with components that rotate. Many systems include linear devices such as conveyors, hoist cables and their loads, or reciprocating racks driven by gears that also have inertia and must be accelerated. It would be convenient to represent these devices with an effective inertia measured by Wk^2 as we have for rotating bodies. We can accomplish this by relating the equations for kinetic energy for linear and rotary motion. The actual kinetic energy of a translating body moving with a linear velocity, v , is

$$KE = \frac{1}{2}mv^2 = \frac{1}{2} \frac{W}{g} v^2 = \frac{Wv^2}{2g}$$

We will use ft/min for the units of velocity.

For a body rotating with an angular velocity in the units of rad/min, the kinetic energy is

$$KE = \frac{1}{2}I\omega^2 = \frac{1}{2} \frac{Wk^2}{g} \omega^2 = \frac{Wk^2\omega^2}{2g}$$

Letting Wk_e^2 be the effective inertia, and equating these two formulas, gives

$$Wk_e^2 = W \left(\frac{v}{\omega} \right)^2$$

We will use ft/min for the units of velocity.

Using n rpm rather than ω rad/min, we must substitute $\omega = 2\pi n$. Then

Effective Inertia for a Load Moving Linearly

$$Wk_e^2 = W \left(\frac{v}{2\pi n} \right)^2 \quad (22-5)$$

Example Problem 22-4

The conveyor in Figure 22–13 moves at 80 ft/min. The combined weight of the belt and the parts on it is 140 lb. Compute the equivalent Wk^2 inertia for the conveyor referred to the shaft driving the belt.

Solution The rotational speed of the shaft is

$$\omega = \frac{v}{R} = \frac{80 \text{ ft}}{\text{min}} \frac{1}{5.0 \text{ in}} \frac{12 \text{ in}}{\text{ft}} = 192 \text{ rad/min}$$

Then the equivalent Wk^2 is

$$Wk_e^2 = W \left(\frac{v}{\omega} \right)^2 = (140 \text{ lb}) \left(\frac{80 \text{ ft/min}}{192 \text{ rad/min}} \right)^2 = 24.3 \text{ lb} \cdot \text{ft}^2$$

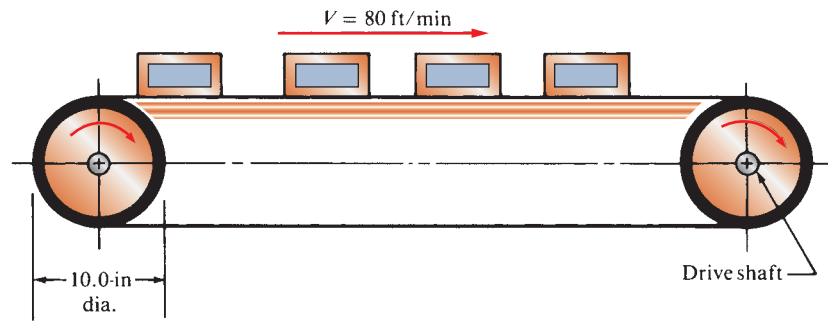


FIGURE 22-13 Conveyor moving at 80 ft/min

22-8 ENERGY ABSORPTION: HEAT-DISSIPATION REQUIREMENTS

When you are using a brake to stop a rotating object or using a clutch to accelerate it, the clutch or brake must transmit energy through the friction surfaces as they slip in relation to each other. Heat is generated at these surfaces, tending to increase the temperature of the unit. Of course, heat is then dissipated from the unit, and for a given set of operating conditions an equilibrium temperature is achieved. That temperature must be sufficiently low to ensure a long life of the friction elements and other operating parts of the unit, such as electric coils, springs, and bearings.

The energy to be absorbed or dissipated by the unit per cycle is equal to the change in kinetic energy of the components being accelerated or stopped. That is,

✧ Energy Absorption by a Brake

$$E = \Delta KE = \frac{1}{2} I \omega^2 = \frac{1}{2} m k^2 \omega^2 = \frac{W k^2 \omega^2}{2g}$$

For typical units in the U.S. Customary System ($\omega = n$ rpm; $W k^2$ in $\text{lb} \cdot \text{ft}^2$; and $g = 32.2 \text{ ft/s}^2$), we get

$$E = \frac{W k^2 (\text{lb} \cdot \text{ft}^2)}{2(32.2 \text{ ft/s}^2)} \frac{n^2 \text{ rev}^2}{\text{min}^2} \frac{(2\pi)^2 \text{ rad}}{\text{rev}^2} \frac{1 \text{ min}^2}{60^2 \text{ s}^2}$$

✧ Energy Absorption in U.S. Customary Units

$$E = 1.7 \times 10^{-4} W k^2 n^2 \text{ lb} \cdot \text{ft} \quad (22-6)$$

In SI units, mass is in kilograms (kg), radius of gyration is in meters (m), and angular velocity is in radians per second (rad/s). Then

$$E = \frac{1}{2} I \omega^2 = \frac{1}{2} m k^2 \omega^2 (\text{kg} \cdot \text{m}^2/\text{s}^2)$$

But the newton unit is equal to the $\text{kg} \cdot \text{m/s}^2$. Then

✧ Energy Absorption in SI Units

$$E = \frac{1}{2} m k^2 \omega^2 \text{ N} \cdot \text{m} \quad (22-7)$$

No further conversion factors are required.

If there are repetitive cycles of operation, the energy from Equation (22-6) or (22-7) must be multiplied by the cycling rate, usually in cycles/min in U.S. Customary units and cycles/s for SI units. The result would be the energy generation per unit time, which must be compared with the heat-dissipation capacity of the clutch or brake being considered for the application.

When the clutch-brake cycles on and off, part of its operation is at the full operating speed of the system, and part is at rest. The combined heat-dissipation capacity is the average of the capacity at each speed, weighted by the proportion of the cycle at each speed (see Example Problem 22-5).

22-9 RESPONSE TIME

The term *response time* refers to the time required for the unit (clutch or brake) to accomplish its function after action is initiated by the application of an electric current, air pressure, spring force, or manual force. Figure 22-14 shows a complete cycle using a clutch-brake module. The straight-line curve is idealized while the curved line gives the general form of system motion.

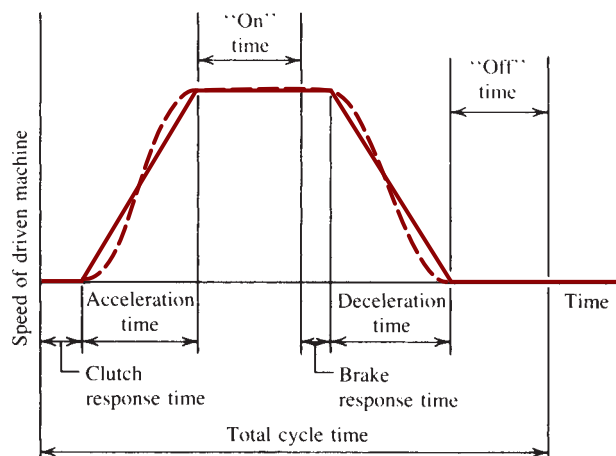


FIGURE 22-14 Typical cycle of engagement and disengagement for a clutch

TABLE 22-1 Clutch-Brake Sample Performance Data

Unit size	Torque capacity (lb · ft)	Inertia, Wk^2 (lb · ft ²)	Heat dissipation (ft · lb/min)		Response time (s)	
			At rest	1800 rpm	Clutch	Brake
A	0.42	0.000 17	750	800	0.022	0.019
B	1.25	0.0014	800	1200	0.032	0.024
C	6.25	0.021	1050	2250	0.042	0.040
D	20.0	0.108	2000	6000	0.090	0.089
E	50.0	0.420	3000	13 000	0.110	0.105
F	150.0	1.17	9000	62 000	0.250	0.243
G	240.0	2.29	18 000	52 000	0.235	0.235
H	465.0	5.54	20 000	90 000	0.350	0.350
I	700.0	13.82	26 000	190 000	0.512	0.512

Note: Torque ratings are static. The torque capacity decreases as the speed difference between the parts being engaged increases. Interpolation may be used on heat-dissipation data.

The actual response time will vary, even for a given unit, with variations in the load, environment, or other operating conditions.

Commercially available clutches and brakes for typical machine applications have response times from a few milliseconds (1/1000 s) for a small device, such

as a paper transport, to approximately 1.0 s for larger machines, such as an assembly conveyor. Manufacturers' literature should be consulted. To give you a feel for the capabilities of commercially available clutches and brakes, Table 22-1 gives sample data for electrically powered units.

Example Problem 22-5

For the system in Figure 22-12, and using the data from Example Problem 22-3, estimate the total cycle time if the system is controlled by unit G from Table 22-1 and the system must stay on (at steady speed) for 1.50 s and off (at rest) for 0.75 s. Also estimate the response time of the clutch-brake and the acceleration and deceleration times. If the system cycles continuously, compute the heat-dissipation rate, and compare it with the capacity of the unit.

Solution

Figure 22-15 shows the estimated total cycle time for the system to be 2.896 s. From Table 22-1, we find that the clutch-brake exerts 240 lb · ft of torque and has a response time of 0.235 s for both the clutch and the brake.

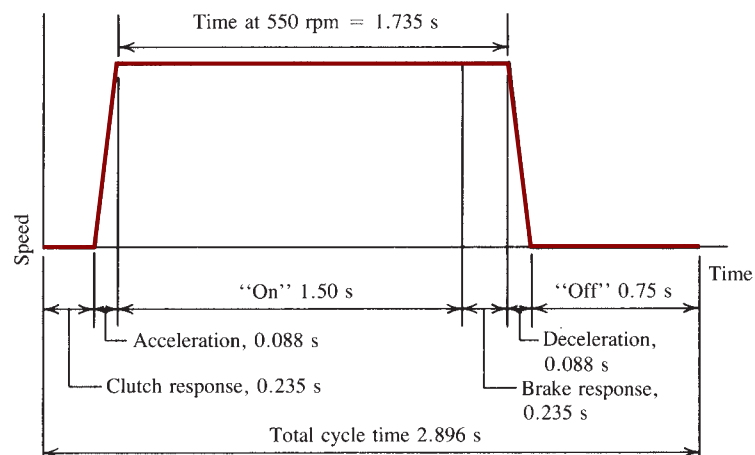


FIGURE 22-15 Estimated total cycle time